

Spaghet Boi's Handbook of Handbuilding

November 16, 2024

Contents

1	Tile Efficiency	2
1.1	Complex Taatsu	2
1.1.1	Complex Taatsus with Pairs	3
1.1.2	Ryankanchan and Its Derived Shapes	9
1.1.3	134 Shape	17
1.1.4	When the 5 of 578 Shape is an Akadora	19
1.1.5	3568 Shape	19
1.1.6	35689 Shape	20
1.1.7	Sankanchan and Yonkanchan	21
1.1.8	Shapes with Overlapping Acceptance	22
1.2	4-Tile Shapes	23
1.2.1	Yonrenkei	23
1.2.2	Nakabukure	25
1.2.3	Aryanmen	28
1.2.4	Confirming a Mentsu or Confirming a Head	30
1.3	6-Tile Shapes	33
1.4	Other Shapes	38
1.4.1	1345 Shape	38
1.4.2	13457 Shape	38
1.4.3	Entotsu	39
1.5	6 Block Theory	39
1.5.1	Criteria of Keeping 6 Blocks	40
1.5.2	Order of Dropping a Block	42
1.5.3	Dealing with Multiple Pairs	45
2	Building Yaku	46
2.1	Yakuhai	46
2.2	Tanyao	51
2.3	Pinfu	57
2.4	Ipeiko	58
2.5	Honitsu and Chinitsu	59
2.6	Yakuyaku	63
2.7	Sanshoku Doujun	64

2.8	Ikkitsuukan	69
2.9	Toitoiho	72
2.10	Chiitoitsu	74
3	Call Judgement	78
3.1	Tanyao	79
3.2	Yakuhai	84
3.3	Honitsu	86
3.4	Toitoiho	87
3.5	Kan	88
3.6	Canceling Ippatsu and Shifting Haitei	90
4	Bibliography	90

1 Tile Efficiency

Tile efficiency is a set of policies and techniques that allows us to reach an optimal point when taking speed, value, win rate, and many other factors into consideration. There is also the concept of *pure* tile efficiency which concerns of only the speed of reaching *tenpai*, one tile away from completing our hand.

In most rulesets such as M. League ruleset where there are yakus such as dora, akadora, ippatsu, and uradora to inflate the expected value of riichi, the best course of action is often to become the first player to call riichi, and the speed and opportunity of reaching tenpai depend on how wide the acceptance of our hand is. Therefore, pure efficiency is very often prioritized and preached. Indeed, mahjong is not a math contest where the player who can calculate the most acceptance automatically wins the game; there are many other factors that would require us to sacrifice efficiency for a better overall outcome, and we will see how those factors play in a bigger picture of hand building.

The following text assumes that the reader has a basic understanding of mahjong rules such as suits, tiles, win conditions, call mechanisms, and riichi. There are, however, many terms that I cannot find a proper translation in English, so I will use Japanese terms as is. Some of them are listed in Table 1. Many other terms such as *ryanmenkanchan* and *agaritoppu* will be introduced later in the text.

1.1 Complex Taatsu

Complex taatsus are shapes involving simple taatsus (penchans, kanchans, and ryanmens) and an additional tile. Table 2 lists complex taatsus that uses three tiles.

Indeed, there are other kinds of complex shapes such as 12234m, 2345m, 334568m, and so on. Shapes with mentsus involved have more tiles and thus less frequent than shapes with less tiles, but their complexity can still easily confuse us. In Section 1.1, we will deal with complex shapes that has no mentsu inside. Shapes with mentsus will be covered later.

Term	Definition
Ryanmen	The shape consisting of two neighboring number tiles such as 23p (23p)
Kanchan	The shape consisting of tiles whose numbers have difference of 2 such as 24p (24p)
Penchan	The shape consisting of number tiles of 1 or 9 and a neighboring tile such as 12p (12p)
Toitsu	The shape consisting of two identical copies such as 11p (11p)
Taatsu	The set of ryanmen, kanchan, and penchan
Shuntsu	The shape consisting of three consecutive number tiles such as 123p (123p)
Koutsu	The shape consisting of three identical copies such as 111p (111p)
Minko	A koutsu resulting from pon call and thus open to all players
Anko	A koutsu in our hand invisible to other players
Mentsu	The set of koutsu and shuntsu, completed shapes
Tenpai	One tile away from completing our hand
1-away	One tile away from tenpai, also called <i>iishanten</i>
2-away	Two tiles away from tenpai, also called <i>ryanshanten</i>
Shanpon	A wait pattern where we have two pairs in our hand

Table 1: Terminology

Name	Shape	Acceptance
Penchan-toitsu	112m	13m 6 tiles
Kanchan-toitsu	113m	12m 6 tiles
Ryankanchan	135m	24m 8 tiles
Ryanmen-toitsu	223m	124m 10 tiles

Table 2: Types of complex taatsus using three tiles

1.1.1 Complex Taatsus with Pairs

Since complex taatsus from Table 2 (except for ryankanchan) contain a pair, they must serve as head when there is no other head candidate in our hand. On the other hand, when we have more than two complex taatsus containing pairs, the pairs will impair our acceptance, so in most cases we opt to break those pairs and reduce them to single taatsus. In order to increase the probability of reaching good waits, as a rule of thumb, we drop one tile from the strongest complex taatsu (ネマタ, 2014m,n). Table 3 lists our actions based on the number of pairs, complex taatsus (excluding ryankanchans), and ryanmen-toitsus in our hand.

Pair	Complex taatsu	Ryanmen-toitsu	Action
2	2		Reduce the weakest complex taatsu to a pair
3	2	0	Break the pair in the inner kanchan-toitsu
			Reduce outer penchan-toitsu to a pair
3	2	1	Reduce the ryanmen-toitsu to a ryanmen
3	3	0	Reduce the weakest complex taatsu to a pair
3	3	1	Reduce the ryanmen-toitsu to a ryanmen
3	3	2	Reduce the weakest complex taatsu to pair

Table 3: Action based on the number of pairs, complex taatsus, and ryanmen-toitsus.

Note that Table 3 is just a rule of thumb. Depending on many other factors such as the existence

of yakuhai pairs or table situations, we may take alternatives.

Two Pairs

We must drop a tile from either 233p or 466s from (1) since we will not be touching completed groups and simple ryanmens. To keep two pairs within our hand, our choices narrow down to 2p and 4s. Their acceptances are listed in Table 4, which suggests that dropping 4s has 4 more tiles than dropping 2p in terms of acceptance. This is a case of reducing the weaker complex taatsu to a pair and keeping the stronger complex taatsu, and this particular shape of 1-away is called *perfect 1-away* or *kanzen iishanten* in Japanese.

(1) 12334578m 233p 466s

Drop	4s	2p
Acceptance	69m 134p 6s 20 tiles	69m 3p 56s 16 tiles

Table 4: Acceptance of (1).

(2) has 112345p which makes dropping 1p and reducing it to 45p ryanmen attractive. But by doing so, we lost flexibility in manzu shape and we must make either 66m or 77m our head, thus reducing our acceptance. If we see (2) as choosing between 667m¹ and 112p, it becomes clear that dropping 2p here gives us the widest acceptance. Now, dropping 5p also gives the same acceptance as 2p, but considering the case when we draw 47s and wait on 69m1p, dropping 2p here makes the outside tile 1p more likely to come out. Hence drop 2p (渋川難波, 2024b). Table 5 lists our acceptance.

(2) East-1, East seat, turn 6, dora 4m
666778m 112345p 06s

Drop	1p	2p
Acceptance	36p47s 15 tiles	56789m1p47s 24 tiles

Table 5: Acceptance of (2).

If our choices are between two ryanmen-toitsus, reduce the one on the inside into a pair to make sure that when we reach tenpai our waits are outside, which has higher win rate. Therefore, between 4m and 5s from (3), we drop 5s and keep the possibility of 25m wait (お知らせ, 2020, pp. 208–209).

(3) 334789m 34p 111566s

(4) also has two ryanmen-toitsus 778m and 556s, and one is connected to a completed mentsu 345s. Upon seeing this complex shape, one may be reluctant to touch complex souzu shape and tempted to drop 7m to fix 78m ryanmen or drop 8m. The 7m drop forces us to treat 55s as our head and renders 36s surplus, thus having less acceptance than 8m drop, while 8m drop leads us

¹Strictly speaking,  can be treated as either  +  or  + , depending on which mentsu we extract from it. Either way, manzu shape is made of a ryanmen-toitsu and a mentsu; it does not change the fact that we are comparing between a ryanmen-toitsu and a penchan-toitsu.

to final wait on either 47p or 47s. Our previous conclusion applies here as well: drop 6s since the ryanmen-toitsu 556s is inside (渋川難波, 2024b). Table 6 lists our acceptance.

- (4) East-1, East seat, turn 6, dora 6p
778m 12356p 345556s

Drop	7m	8m	6s
Acceptance	69m47p 16 tiles	7m47p2457s 22 tiles	679m47p25s 23 tiles

Table 6: Acceptance of (4).

It should be noted that in all cases with only two pairs, reducing complex taatsus to simple taatsus will not lead to maximal acceptance. When we have two pairs in our hand, we can accept tiles of the same kind as these pairs; any tiles that follows these pairs also increases acceptance. If we only have one pair, say 23p466s, then the tile that follows the pair becomes useless, since drawing 5s destroys the pair and produces a mentsu, which renders our hand headless and unready. This means that sometimes we have to counterintuitively destroy ryanmens.

Three Pairs, Two Complex Taatsus

Since we have more than two pairs in (5), we opt to break one, either from 233p or 466s². Breaking the pair in either complex taatsus will give us the same number of acceptance, but the numbers of draws that lead to ryanmen tenpai are different and listed in Table 7. This is a case of reducing stronger complex taatsu to a simple taatsu (ネマタ, 2014m).

- (5) 12334588m 233p 466s

Drop	3p	6s
Acceptance	8m14p56s 16 tiles	8m134p5s 16 tiles
Ryanmen-tenpai draws	8m 56s 8 tiles	5s 4 tiles

Table 7: Acceptance of (5).

If our choices are between complex taatsus without ryanmens, there is no difference in direct acceptance and ryanmen-tenpai draws. If our complex taatsus contains kanchans, there is a chance for those to upgrade into ryanmens. It is therefore optimal to break the pair from inner kanchan-toitsus, since they have 8 tiles of ryanmen improvement as opposed to 4 tiles for outside kanchans. This means dropping 6s from 466s in (6). Drawing 37s improves our 46s kanchan into ryanmen, while only 4p can produce a ryanmen taatsu out of 13p (ネマタ, 2014m).

- (6) 12334588m 133p 466s

For complex taatsus with few ryanmen-improvement draws, if the remaining pair is in the middle of a suit, we can expect its ryanmen improvement. We need to reduce a complex taatsu into a pair for that outcome, and it then becomes optimal to fix the weakest complex taatsu into a pair. (7) has 899s that, when turned into a penchan (simple taatsu), has no ryanmen improvement,

²Note that we do not drop 𠄎 here since that is equal to dropping entire block and increasing our shanten number.

compared to 113p which produces a ryanmen when we draw 4p, so we drop 8s and make it a pair (ネマタ, 2014m).

(7) 123345m 11366p 899s

(8) has a more complex shape 11333445p that can be treated as either 11p + 345p + 334p or 11p + 333p + 445p, which leads to its wide acceptance of 123456p. We will lose some of its acceptance if we drop any tile from this shape, so instead we reduce 778s to a ryanmen, since technically we still have three pairs in our hand. Table 8 lists the acceptance after different drops. This leads to the rule of thumb that we keep the more complex shape with more acceptance intact (G・ウザク, 2016, p. 24).

(8) East-1, North seat, turn 7, dora 3p
234m11333445p778s

Drop	7s	4p	3p
Acceptance	123456p69s 24 tiles	1236p679s 21 tiles	125p679s 19 tiles

Table 8: Acceptance of (8).

(9) is also choosing between two ryanmen-taatsus 556m and 566s, and reducing either to a ryanmen does not make any difference in terms of acceptance. However, if we keep 345566s and draw another dora 3s, we can drop 22z and go for tanyao. In that regard, it is best that we drop 5m here (G・ウザク, 2016, p. 24).

(9) East-1, East seat, turn 6, dora 3s
556m 678p 345566s 22z

There is no difference in terms of immediate acceptance between 8m drop and 2s drop from (10). However, in case of drawing 7p and thus the potential 789 sanshoku, we must fix 78m ryanmen. With 8m drop and 7p draw, we can proceed to drop 3s from the shape 78m788p223s (ネマタ, 2015c).

(10) Dora 4z
788m 23488p 223789s

When there is a yakuhai pair, keep 3 pairs (G・ウザク, 2019, p. 72). Yakuhai pairs contribute almost³ no value unless it becomes a triplet or at least is part of our final waits. Therefore, we do not expect it to be our head, and we must find our head somewhere else. Dropping either of 8p2s from (11) gives us 2 less tiles of acceptance than maximal number, but it keeps the option of 7z pon. Since pinzu shape has draws like 2p that potentially improve our shapes and souzu shape only has 5s improvement, we drop 8p here and keep the follow-up tile for 4s pair.

(11) 789m 345668p 244s 77z

³Well, it does give 2 fu, but that is negligible compared to the value an addition han brings.

Similarly for (12), under the premise that 2z will be ponned, we must drop either of 8m and 7s. If we drop 7s and draw 7m, 05s can only serve as head and cannot accept dora 4s. Therefore, drop 8m.

- (12) South-3, East seat, turn 5, dora 4s
135899m 123p 057s 22z

Three Pairs, Three Complex Taatsus, No Ryanmen-Toitsus

If there are three complex taatsus, all of which contain no ryanmen, then it is optimal to reduce the one with least ryanmen improvement into a pair, so that when the remaining complex taatsus becomes a mentsu, there will be no surplus tiles. (14) is the hand derived from (13) by dropping 8s and completing 668p. (15) is derived from (13) by dropping 9s instead and completing 668p. We can see, in Table 9, that dropping 8s loses 7s acceptance and we have 2 tiles of acceptance less than dropping 9s, but once 668p becomes a mentsu, (14) has 4 tiles of acceptance more than (15). We therefore prioritize the width of 1-away acceptance and sacrifice a little bit at 2-away. Drop 8s (ネマタ, 2014m).

- (13) 12378m 113668p 899s

- (14) 12378m 113678p 99s

- (15) 12378m 113678p 89s

Drop	8s	9s
Acceptance	69m1267p9s 22 tiles	69m1267p7s 24 tiles
Hand after completing 668p	(14)	(15)
Acceptance after completing 668p	69m12p9s 16 tiles	69m7s 12 tiles

Table 9: Acceptance of (13), (14), and (15)

Three Pairs, Three Complex Taatsus, One Ryanmen-Toitsu

If there is one ryanmen-toitsu among all three complex taatsus, it is optimal to reduce the ryanmen-toitsu to a ryanmen. There will be no difference in immediate acceptance between the drop candidates 26p9s from (16), but there will be huge differences when we draw any of the acceptances, as shown in Table 10. If we drop 6p, completing the remaining two complex taatsus 223p and 899s will leave us only 3 kinds 12 tiles of acceptance. Likewise for 9s drop. On the other hand, if we drop 2p, all effective draws will guarantee us 16 tiles of acceptance. One may be tempted by the 7p draw after dropping 6p which gives us perfect 1-away, but also keep in mind that it is more likely to draw ourselves into 3 kinds 12 tiles of acceptance.

- (16) 12378m 223668p 899s

Since it is difficult to calculate all the acceptance of all possible effective draws after dropping each possible tiles, we can just follow the principle to reduce the strongest complex taatsu to a ryanmen.

Drop	Next draw	Max acceptance
2p	69m14p	drop 8p $\rightarrow$ 146p79s 16 tiles
		drop 8s $\rightarrow$ 1467p9s 16 tiles
	6p	drop 8p8s $\rightarrow$ 69m14p 16 tiles
	7p	drop 6p8s $\rightarrow$ 69m14p 16 tiles
	7s	drop 8p9s $\rightarrow$ 69m14p 16 tiles
	9s	drop 8p8s $\rightarrow$ 69m14p 16 tiles
6p	69m	drop 8s $\rightarrow$ 1247p9s 16 tiles
	14p	drop 2p8s $\rightarrow$ 69m7p 12 tiles
	2p	drop 3p8s $\rightarrow$ 69m7p 12 tiles
	7p	drop 8s $\rightarrow$ 69m124p9s 20 tiles
	7s	drop 3p9s $\rightarrow$ 69m7p 12 tiles
	9s	drop 3p8s $\rightarrow$ 69m7p 12 tiles

Table 10: Differences between 2p drop and 6p drop from (16)

Three Pairs, Three Complex Taatsus, Two Ryanmen-Toitsus

If there are two ryanmen-toitsus, then we reduce the remaining complex taatsu to a pair. After that, since we have three pairs in our hand, we will have 2 tiles of acceptance less than reducing the other taatsus to a ryanmen. Considering the acceptance when we reach 1-away, however, it becomes clear that this sacrifice of 2 tiles at 2-away is worth. If we choose to fix any ryanmen in (17), completing any complex taatsu leaves us 1 surplus tile. On the contrary, by making the weakest complex taatsu our head, we are guaranteed perfect 1-away no matter what we draw, as shown in Table 11. Drop 8s, even though it may be a little bit counterintuitive because we have been hearing that three pairs are the weakest.

(17) 12378m223677p899s

Drop	Next draw	Max acceptance
2p	69m14p	drop 8s $\rightarrow$ 14578p9s 20 tiles
	58p	drop 7p8s $\rightarrow$ 69m14p 16 tiles
	7p9s	drop 6p8s $\rightarrow$ 69m14p 16 tiles
	7s	drop 6p9s $\rightarrow$ 69m14p 16 tiles
8s	69m	drop 2p $\rightarrow$ 14578p9s 20 tiles
		drop 7p $\rightarrow$ 12458p9s 20 tiles
	14p	drop 2p $\rightarrow$ 69m578p9s 20 tiles
	2p	drop 3p $\rightarrow$ 69m578p9s 20 tiles
	58p	drop 7p $\rightarrow$ 69m124p9s 20 tiles
	7p	drop 6p $\rightarrow$ 69m124p9s 20 tiles
	9s	drop 3p $\rightarrow$ 69m2578p 20 tiles
		drop 6p $\rightarrow$ 69m1247p 20 tiles

Table 11: Differences between dropping 2p and 8s from (17)

1.1.2 Ryankanchan and Its Derived Shapes

The shape 246p is called *ryankanchan* in Japanese, sometimes *ryankan* for short. The shape contains two kanchans 24p and 46p, and *ryan* in Japanese means “two”, hence the name. It takes up the space of 3 tiles and accepts at most 8 tiles, and we are left to choose any of its two kanchans when other parts of our hands get filled up, so ryankanchan is weaker than ryanmen in terms of space efficiency and final waits.

Comparing Ryankanchans

Even between different kinds of ryankanchans, there are differences other than the tiles used. Table 12 lists three kinds of ryankanchan 135p, 246p, and 357p. Note that we omit 468p and 579p because they are identical to 246p and 135p due to symmetry of mahjong tiles. We can see that 357p has the most improvement draws, and dropping 5p from 135p makes our suji 2p wait better than non-suji 2p kanchan wait. 246p and the ryanmen 67p developed from it are all tanyao tiles, so it is a shape we would like to keep for tanyao.

Shape	Ryanmen draw	Comment
135p	6p	Dropping 5p makes 2p wait suji
246p	7p	Dropping 2p still keeps the acceptance of 0p
357p	28p	

Table 12: Comparison between ryankanchans

Comparing Ryankanchan with Other Taatsus

In the case where there are no floating tiles to drop and we must choose between ryankanchan and other taatsus, when we choose to drop reduce the ryankanchan to a simple kanchan, we lose 4 tiles of acceptance. Therefore, to destroy ryankanchan or not depends on the trade-off of the loss of 4 tiles of acceptance:

1. If destroying ryankanchan still gives us the maximal acceptance, it is the right choice
2. If there is no difference in terms of acceptance,
 - (a) When pinfu is feasible, keep ryankanchan
 - (b) When pinfu is impossible, drop ryankanchan and set up suji trap

We must drop a tile from either 135m or 233345p (18). Notice that the complex pinzu shape has an acceptance of 1346p 12 tiles. If we drop 3p, we lose the acceptance of 369p 7 tiles; if we drop 2p, we lose the acceptance of 14p 7 tiles. Either way, we lose more than dropping 5m (which loses 4m 4 tiles.) Therefore, we destroy ryankanchan in this case. Dropping 1m or 5m leaves us the same number of acceptance, but a 5m in our river⁴ makes 2m more easily to come out, so 5m is the best choice here (福地誠, 2017, pp. 44–45).

(18) Dora 9p
135m 23334599p 789s

⁴River, or *kawa* in Japanese, refers to the discards of a player, neatly placed in front of them.

Drop	5m	3p	2p
Acceptance	2m13469p 18 tiles	24m14p 15 tiles	24m369p 15 tiles

Table 13: Acceptance of (18).

(19) is a hand similar to (18), but pinzu shape is not complex in a way that would lead to more loss when we drop a tile from it. Notice that, if we keep 135m ryankanchan, we have a 50-50 chance of getting pinfu by drawing 24m which makes our hand value 7700 (riichi pinfu dora dora.) If we drop 5m, only 9p draw (2 tiles out of the acceptance of 16 tiles total) gives us mangan tenpai; in other cases, we are stuck with 5200 bad wait. Therefore, although there is no difference in acceptance, in terms of expected value, 3p drop is optimal (福地誠, 2017, pp. 46–47).

(19) Dora 9p
135m 23356799p 789s

Drop	5m	3p
Acceptance	2m1349p 16 tiles	24m14p 16 tiles

Table 14: Acceptance of (19).

(20) has a 8s anko and thus pinfu is impossible. With 9m toitsu tanyao is also far away. Therefore, no matter what we drop, there is no difference in terms of acceptance and value. In this case, we drop 5p and make our 2p wait a suji trap (石橋伸洋, 2015, pp.23–25).

(20) South seat, turn 4, dora 4s
23399m 135p 406888s

(21) has a ryankanchan and an anko. If we drop 8s and make 88s our head, we will have only 16 tiles of acceptance at 1-away, as opposed to 24 tiles if we drop 26m while still maintaining 1-away, as shown in Table 15. Besides the number of acceptance, when we keep 456888s, any draw of 367s improves our souzu shape, and any draw of manzu tile that gives us head leaves us ryanmen tenpai. The problem becomes which of 26m to drop. Considering the acceptance of 0m and the improvement draw 7m, drop 2m (G・ウザク, 2019, p. 84).

(21) 246m 23467p 456888s

Drop	2m	6m	8s
Acceptance	456m5678p 24 tiles	234m5678p 24 tiles	35m58p 16 tiles

Table 15: Acceptance of (21).

Since (22) has two ryanmens with overlapping acceptance, dropping ryanmens and dropping any of 268s from 24668s will not lead to any difference in terms of the number of acceptance. However, considering the shape of our final waits, we can't touch ryanmens. We must keep 6s pair in case we draw any of 469m which gives us perfect 1-away. We also cannot overlook the possibility of 678 sanshoku, so we are incentivized to keep 8s. In this regard, 2s becomes the most unwanted tile. Drop 2s (G・ウザク, 2016, p. 14).

- (22) East-1, East seat, turn 5, dora 7p
55678m 3467p 24668s

Drop	2s	6s	8s
Acceptance	5m258p567s 24 tiles	258p357s 24 tiles	5m258p356s 24 tiles

Table 16: Acceptance of (22).

When the 5 of 135 Ryankanchan is Akadora

The draws that are related to the akadora of 130p ryankanchan are 4567p, which makes 1p seemingly useless. That said, when we draw 2p, 0p becomes floating and 34567p draws make 0p useful. Besides, when we draw another 1p, it produces a head and a kanchan; drawing 6p further develops it into a ryanmen. Therefore, we would like to keep it intact, unless we have enough blocks to aim for tenpai (ゆ 5 せー, 2024b).

For example, in (23) since we only have $345m + 688m + 130p + 1335s$ four blocks, we must drop 1z and aim for draws that give us another block. If we draw 2p, as in (24), then we can drop 1s and expect blocks around floating 0p and 5s.

- (23) East-1, turn 1, dora 5z
345688m 130p 1335s 1z

- (24) East-1, turn 2, dora 5z
345688m 1230p 1335s

(25) is a similar hand, but since we already have 5 blocks, we can drop 1p and go for tanyao.

- (25) East-1, turn 2, dora 5z
23445688m 130p 335s

The 10m shape in (26) is the predecessor of 130m ryankanchan, so it will be a loss if we drop 1m over otakaze 2z. However, this possibility will not make 1m stronger than seat wind or yakuhai 46z. So we drop 2z for now, and drop 1m later on.

- (26) East-2, North seat, dora 4m
10m 1389p 3666s 246z

Vertically Developed Ryankanchans such as 2246 and 2466

When we have 1135p and draw another copy of 5p, the result shape 11355p is a strong shape that accepts 1245p 4 kinds 12 tiles. Likewise for 2246p with 6p draw, and 1355p with 1p draw. If we draw both 46p, we end up the shape 134556p, which is called *ryanmenkanchan*⁵ (chirno, 2008). Despite all these improvements, we still have to consider the cases where we have more stronger taatsus and must drop one from 1135p or 2246p.

⁵See Section 1.3 for more details on ryanmenkanchan.

We can drop either 4p or 8p from (27), which will lead to the same number of 16 tiles of acceptance, as shown in Table 17. Keeping two pairs 99m44p in our head does not give us more ryanmen improvement draws than keeping 468p shape (both having only 3p), and drawing any of these pairs into an anko leaves us rii-nomi⁶. In that regard, it is slightly more advantageous to keep 468p shape and hope for pinfu. Drop 4p (G・ウザク, 2019, p. 86).

(27) 12399m 4468p 34789s

Drop	4p	8p
Acceptance	57p25s 16 tiles	9m45p25s 16 tiles
Pinfu draws	57p 8 tiles	5p 4 tiles

Table 17: Acceptance of (27).

(28) has 77m instead of 99m compared to (27), which increases the draws that give us perfect 1-away from 3p to 68m3p. While it is true that 4p drop still gives us the maximal 8 tiles of pinfu draw, having 12 tiles of perfect 1-away draw means that it is more likely to get good waits with 8p drop. Therefore, the best choice here is to drop 8p (G・ウザク, 2019, p. 88).

(28) 12377m 4468p 34789s

Dropping 3s and 7s from (29) leaves us the same number of acceptance. However, if we drop 3s and have 99m077s, drawing 9m7s means that 0s is useless. Since we have a 68p kanchan in our hand, there is no pinfu or even perfect 1-away either. If we are going to riichi anyway, it is best to make full use of 0s to achieve maximal expected value. Thus drop 7s (G・ウザク, 2016, p. 26).

(29) East-1, East seat, turn 7, dora 2p
99m 23468p 3077s 444z

Since we have a 77z pair in (30), when we call pon on 7z we must have a pair elsewhere. In that regard, we must keep 77s pair and drop 3s. It is likely that 0s becomes useless, but the difference is only 1000 points. After we open our hand, we go full steam ahead (G・ウザク, 2016, p. 26).

(30) East-1, West seat, turn 6, dora 2m
666789m 34p 3077s 77z

We must drop either 4s or 8s from the shape 4468s from (31). Since we already have pairs 77m77p, 8s drop loses 2 more tiles than 4s drop. When our hand becomes 1-away with, say, 6m draw, there is no difference in terms of acceptance (ryanmen + pair + kanchan-toitsu versus ryanmen + head + ryankanchan), but if we have 468s shape in our hand, we have a higher chance of getting pinfu than 446s shape. Therefore, based on the acceptance at 2-away and pinfu chance at 1-away, the best choice is 4s drop (ずんだもんの麻雀何切るチャンネル, 2023).

⁶*Rii-nomi* refers to a state where we have no yaku other than riichi. The term is a portmanteau of riichi and *nomi*, a Japanese word for “only.” Rii-nomi itself is not inherently bad; ryanmen rii-nomi is still better than not calling riichi especially when we are the first player to do so. Bad wait rii-nomi would require us to think twice before calling riichi and taking an amount of risk that is not justified by the cheap value. A part of hand building is about how to avoid rii-nomi without sacrificing efficiency too much.

- (31) East-1, West seat, turn 6, dora 1p
23577m 12377p 4468s

Bulging Ryankanchans such as 1335 and 2446

To handle shapes like 1335p and 2446p:

- 2-away
 - Drop the outside tile, aim for ryanmen improvement
- 1-away
 - If pinfu is within sight, drop the bulging part
 - If pinfu is infeasible, drop the inside tile

Suppose that a part of our hand is 77m1335p and the rest of our hand being completed blocks and a ryanmen. Table 18 lists the pinfu draws, ryanmen improvement draws, and final waits (when the ryanmen is completed) of different drops from 1335p. When we reduce 1335p to a regular ryankanchan, it has the most draws that give us pinfu; if we cannot draw 24p in time, we can still cut 5p and riichi and make our 2p wait a suji. When pinfu is not feasible, we should cut 5p in advance to make our suji wait 2p more likely to come out⁷. When our hand is callable, the 3p pair becomes more valuable.

Drop	1p	3p	5p
Result shape	335p	135p	133p
Pinfu draws	4p 4 tiles	24p 8 tiles	2p 4 tiles
Ryanmen improvement draws	68m6p 12 tiles	6p 4 tiles	68m 8 tiles
Final waits	7m3p shanpon	2p kanchan	2p kanchan or 7m3p shanpon

Table 18: Comparing drops from the shape 77m1335p.

(32) has pinfu chance if we drop 3p and draw 24p. Meanwhile, (33) cannot have pinfu no matter what we drop; in this case, drop 5p and make 2p more likely to come out. (34) has no pinfu either, and it has 555z anko, which means that, after dropping 5p, calling pon on both 7m3p will give us tenpai (ゆ ー せー, 2021d).

- (32) East-1, East seat, turn 5, dora 9s
12377m 1335p 45789s

- (33) East-1, East seat, turn 5, dora 9s
77m 1335p 45789s 333z

- (34) East-1, East seat, turn 5, dora 9s
77m 1335p 45789s 555z

⁷The argument that the suji of a tile cut in advance is more likely to come out than the suji of a riichi declaration tile is based on the deal-in rate difference of sujis. Sujis of riichi declaration tiles have higher deal-in rates than other kinds of sujis (みーにん, 2017, pp. 152–153). If a player's policy is to cut the suji with lower deal-in rate, they will tend not to cut riichi declaration tile suji.

Horizontally Developed Ryankanchans such as 2357

Drop the kanchan part; generally, drop 5 first; when we have no head, drop 7 (ゆうせー, 2022a). Even if tanyao is within reach, keeping ryankanchan takes up the space which we could use for safe tiles instead. It is also possible that our ryankanchan ends up a kanchan bad wait, which leaves us no reason to not destroy it when we have a ryanmen.

When we add 4m to 2357m, since 23m also accepts 4m, we actually end up with a mentsu and a kanchan. Compared to other inside kanchans⁸, this 57m kanchan is inferior. In this regard, drop 57m and keep 46p in (35).

(35) East-1, turn 3, dora 4z
2357m 46p 22256788s

In addition to ryanmen improvement draws, considering final waits, the kanchan part of 2357m is inferior to outside kanchans⁹. In this regard, drop 57m and keep 79s in (36).

(36) East-1, turn 3, dora 4z
2357m 2225699p 79s

When we drop 7p from 2357p, the 235p shape produces a head if we draw any of 25p. Drawing 4p gives us a yonrenkei shape¹⁰, which also produces a head easily. Since (37) has no head, drop 7m.

(37) East-1, turn 3, dora 4z
2357m 234678p 3456s

Ryankanchans with Following Ryanmens such as 34468

The 34468p shape produces a mentsu if we draw any of 2457p, as shown in Table 19, and therefore is a strong shape (クリアレイン, 2024; ゆうせー, 2022c).

Draw	Result shape	Composition
2p	234468p	Shuntsu 234p + ryankanchan 468p
4p	344468p	Anko 444p + kanchan 68p or ryanmen-toitsu 344p + ryankanchan 468p
5p	344568p	Ryanmenkanchan (see 1.3)
7p	344678p	Ryanmen-toitsu 344p + shuntsu 678p

Table 19: Draws that produce a mentsu out of 34468p shape.

Since dropping any tile from 34468p annihilates the benefits mentioned earlier, when compared to yonrenkei or nakabukure¹¹, we tend to keep 34468p and reduce other 4-tile shapes to a mentsu.

⁸Inside kanchans are kanchans with 8 tiles of ryanmen improvement draws. For pinzu suit, for example, that would be , , and .

⁹Outside kanchans are kanchans with 4 tiles of ryanmen improvement draws. For pinzu suit, for example, that would be , ,  and .

¹⁰See Section 1.2.1.

¹¹See Section 1.2.2.

As shown in Table 20, dropping either of 36p also gives us far more acceptance than dropping any tile from 34468m. Therefore, drop 3p from (38).

(38) 34468m 3456p 11789s

Drop	4m	8m	3p
Acceptance	257m 12 tiles	25m 8 tiles	2457m1s 16 tiles

Table 20: Acceptance of (38).

When we have 6 blocks, we still tend to keep 34468p shape. When there are only ryanmens in the rest of our hand, drop 8p. In this regard, since there is a weaker 79s kanchan in (39), we drop 79s; since there is no weaker taatsu in (40), drop 8m.

(39) 34468m 45789p 2279s

(40) 34468m 45789p 2399s

When we have 5 blocks, we still tend to keep 34468p shape. When there is no head in our hand, how we handle 34468p shape will largely depend on table situation and turn number. When we are still early in the game, we can drop 8p and focus on 34p ryanmen. When we are late and there is no room for improvement, make 44p our head and drop 3p. Therefore, even for the same hand such as (41), our decision changes depending on other information.

(41) 34468m 123789p 357s

Dropping 5m from 13556m in advance at 1-away is not recommended (ゆうせー, 2021a). When we drop 5m from (42), our ryanmen-tenpai draw reduces to only 2m, opposed to 24m if we keep it, while both options have the same number of bad-wait draws, as shown in Table 21. Keeping a safe tile here worsens our wait quality too much.

(42) East-1, West seat, turn 6, dora 8p
13556m 33789p 123s 4z

Drop	5m	4z
Result shape	1356m33p4z	13556m33p
Ryanmen-tenpai draws	2m 4 tiles	24m 8 tiles
Bad-wait draws	47m 8 tiles	57m3p 8 tiles

Table 21: Acceptance of (42).

That said, we can drop the ryankanchan part when we draw 6m. Even though the shape 5566m has overlapping acceptance 47m, our bad-wait draws reduce to 56m3p 3 kinds 6 tiles, 2 tiles less than keeping 13556m, as shown in Table 22. We also gain space for safe tiles. Therefore, drop 3m and then 1m from (43).

(43) East-1, West seat, turn 6, dora 8p
135566m 33789p 123s

Drop	3m	6m
Result shape	15566m33p	13556m33p
Ryanmen-tenpai draw	47m 8 tiles	24m 8 tiles
Bad-wait draw	56m3p 6 tiles	57m3p 8 tiles

Table 22: Acceptance of (43).

The same principle applies to inner 34468s shape (ゆ う せー, 2022c). Drop 6s from (44) and (45).

(44) East-1, West seat, turn 4, dora 9p
444m 33p 678p 334468s

(45) East-1, West seat, turn 4, dora 9p
11 789p 334468s 444z

When we have yakuhai anko such as (46) and draw 1m with 13556m shape in our hand, we can drop 5m and keep 113m so that when we pon either 1m1p or draw 2m, we have a good ryanmen wait. Drop 5m from (46).

(46) East-1, West seat, turn 6, dora 8p
113556m 11789p 555z

Detached Ryankanchan such as 245679

The 245679p shape in (47) is dubbed “detached ryankanchan” since it accepts 38p far apart. If we draw 56p, we can further improve it into a ryanmenkanchan. On the other hand, dropping either 29p will leave us a mentsu and kanchan; which to drop largely depends on the rest of our hand.

In (47), dropping either of 29p from detached ryankanchan or dropping 6m from 566m will leave us the same number of acceptance at 1-away. However, considering the draws that will give us pinfu and the sanshoku chance, fixing 56m ryanmen here is better, as shown in Table 23 (G・ウザク, 2019, p. 89).

(47) 566m 245679p 45699s

Drop	6m	2p	9p
Acceptance	47m38p 16 tiles	467m8p9s 16 tiles	467m3p9s 16 tiles
Pinfu draws	38p 8 tiles	8p 4 tiles	3p 4 tiles
Sanshoku-chance draws	4m8p	4m8p	None

Table 23: Acceptance of (47).

Choosing between two ryankanchans

(48) has two ryankanchans, one bulging and one vertically developed on one end. Since there is no pair in the rest of our hand, we will focus on how we can make tiles following 55p and 66s pair

function. Dropping 2s makes 4s useless since completing 466s shape leaves us no pair¹². Dropping either of 357p will make full use of all tiles while giving us the same number of acceptance. Taking ryanmen improvement into consideration, 5p drop becomes optimal, as shown in Table 24.

(48) 123789m 3557p 2466s

Drop	3p	5p	2s
Acceptance	56p36s 12 tiles	46p3s 12 tiles	46p 8 tiles
Ryanmen-improvement draws	8p5s 8 tiles	28p5s 12 tiles	28p 8 tiles

Table 24: Acceptance and improvement of (48).

When we must choose between two simple ryankanchans, drop the tile that will make our wait better (Feng, 2018, p. 26) Thus, we drop 5p from (49) and make our wait suji. Note that while 6m drop also makes 3m wait suji, 3m is more inside than 8p and thus still has an overall lower win rate.

(49) 246999m 33567s 579p

1.1.3 134 Shape

It is a common pitfall to only see the ryanmen part of a 134p shape and drop 1p early. The possibilities of 134p can help us tremendously especially when it is still early and our hand lacks shapes (ヨートル, 2024b). In short,

- Makes our hand *kuttsuki*¹³
- Easy to produce a pair

Draw	Shape	Comment
1m	1134m	A pair and a ryanmen.
2m	1234m	A yonrenkei, which produces a pair easily or functions as nobetan when we reach tenpai.
3m	1334m	A kanchan and a ryanmen.
4m	1344m	A kanchan and a pair. Decent as our final wait.
5m	1345m	A floating 1m and a mentsu. However, drawing 2m gives us 45m ryanmen, drawing 3m also gives us a decent kanchan. Drawing 6m and we have the 13456m shape which can easily further develop into a mentsu and a ryanmen.
6m	1346m	Two kanchans.

Table 25: Development of 134m.

Therefore, when our hand does not have 5 blocks ready, we keep 134p over single yakuhais. When there is no head in our hand, we also prioritize 134p. Otherwise, dropping 1p is fine.

¹²Technically, there is, if we treat it as 45s + 66s, but we are then required to drop either of 37p and we are still not tenpai yet despite (48) already at 1-away.

¹³*Kuttsuki* is a Japanese term whose literal meaning is “sticky,” In mahjong context, it refers to a state where tiles next to or 1 tile away from floating tiles all become accepted.

(50) does not have enough blocks, so we keep 134m and expect it to develop. Drop 56z from (50).

(50) East-1, West seat
134m 340899p 248s 56z

(51) already has 5 blocks to aim for tenpai, so we can drop 1m and keep 56z as safe tiles.

(51) East-1, West seat
13488m 3468p 345s 56z

(52) has no head confirmed, so we keep 134m to increase our chances of getting a head. Drop 2z.

(52) East-1, West seat, dora 8s
134m 1236789p 888s 2z

Play Defensively when Our Hand is Too Messed Up

(53) is 5-away and only has one ryanmen 34m. No matter how good 134m can develop, it is more likely that some one will slam riichi earlier than us. In this case, we drop 1m and keep safe tiles 567z.

(53) 1348m 4899p 136s 567z

Avoid Bad-wait Rii-Nomi

(54) has the same tiles as (52) but dora is 7z instead of 8s. The possible kanchan that 134m brings us becomes unfavorable, so we drop 1m and fix ryanmen.

(54) East-1, West seat, dora 7z
134m 1236789p 888s 2z

When the 1 is Also a Dora

As shown in Table 26, when we have no akadora in (55), keeping dora 1p or taking 34p ryanmen does not make too much difference in terms of hand value. Hence drop 1p and take ryanmen to increase win rate. When we have one or more akadoras, keeping dora 1p, and possibly calling pon 1s and pivoting to 1p tanki or drawing another 1p and waiting on shanpon greatly increase hand value. When it is still early, with akadoras in our hand, it is better to drop 4p (ゆ ー せー, 2022j).

(55) East-1, West seat, dora 1p
?67m 134p 11?67s -7?77z

?	Drop 1p	Drop 4p
5m + 5s	1000 points: 25p 7 tiles; 2000 points: 0p 1 tile	2000 points
5m + 0s	2000 points: 25p 7 tiles; 3900 points: 0p 1 tile	3900 points
0m + 0s	3900 points: 25p 7 tiles; 7700 points: 0p 1 tile	7700 points

Table 26: Value of (55) depending on drop and draw.

1.1.4 When the 5 of 578 Shape is an Akadora

Hand state	Action
Far away from tenpai	Keep 078m
1-away	Drop 0m if tanyao is infeasible
Tenpai	Generally drop 0m

Table 27: Handing 078m.

When we are still far away from tenpai, we can expect to draw tiles around 0m. Since (56) has the risk of ending up no dora bad-wait rii-nomi, we drop 7s from 778s. (57) also has the same risk, and 557p has more ryanmen-improvement draws than 224s, so we drop 4s (ゆ う せ ー, 2022f).

(56) 078m 23557p 224 778s

(57) 078m 123557p 22478s

When we are 1-away, acceptance and final wait become more important than the value the dora brings. (58) already has 5z anko and we do not have to cling to 0m for value. Drop 0m

(58) 078m 22456p 688s 555z

However, as an exception, when keeping 07m gives us sanshoku or tanyao which enables us to call and speed up our hand, we drop 8m. When reaching tenpai, if dropping 8m guarantees sanshoku and tanyao, we can also dama our hand instead of dropping 0m and riichi. In this regard, (59) is at tenpai but does not have sanshoku confirmed, so we drop 0m and riichi; on the other hand, (60) has confirmed sanshoku, and its value is already mangan, we drop 8m without riichi.

(59) 078m 22456p 22256s -7s

(60) 078m 22567p 22256s -7s

1.1.5 3568 Shape

It is easy to get a head out of 3568p shape (ゆ う せ ー, 2022b). Not just stacking 38p, but drawing 47p also gives us yonrenkei. 3p can also function as a floater. Thus, it is generally treated as 2 blocks. Table 28 lists our policy when taking the rest of our hand into consideration.

(61) already has 5z anko, and directly accepting 3m using 12m penchan is valuable. Thus, we drop 8p first, and then drop 3p if it does not give us ryanmen.

(61) East-1, West seat, turn 3, dora 1s
12m 3568p 22788s 555z

(62), however, gains tanyao if 12m is removed. Therefore, we drop 1m and expect to draw 8p or tiles around 3p.

(62) East-1, West seat, turn 3, 1s
12m 3568p 22455788s

Number of block	Compared with	Action
4		Keep 3568p
5	Floater (no head in hand)	Keep 3568p
	Floater tiles other than 34567	Keep 3568p
	Penchan (bottleneck of tanyao)	Keep 3568p
	Penchan (not bottleneck of tanyao)	Drop 3568p
	Floater tiles of 34567	Drop 3568p
6		Drop 3568p

Table 28: Handling 3568p or 2457p (ゆうせー, 2022b; クリアレイン, 2023c)

Compared to 3–7 floaters, generally, we drop 8p from 3568p since it has 4 less tiles of acceptance as a floater. However, for hands like (63) where there is no confirmed head, we drop 7m and expect to find a head in shapes developed from 3568p.

(63) East-1, West seat, turn 3, dora 1s
237m 3568p 1236778s

There are 6 blocks in (64), which makes us choose between 2m89p to drop. If we simply treat 2457m as two kanchans, we are tempted to drop the weakest taatsu, which is 89p penchan. However, dropping penchan has the following demerits:

- 45m cannot function as ryanmen (as long as we do not draw 278m)
- We lose 7p acceptance

On the other hand, if we keep 457m89p shape, we will not lose 3m acceptance because we still have 45m ryanmen, and drawing 7p completes a block. The merits of keeping 2m here, which is to increase the probability of confirming a head, becomes irrelevant due to 11p pair, so we can drop 2m without much loss.

(64) 2457m 1189p 667s 444z

1.1.6 35689 Shape

In situations where we are choosing between a 3–7 floater and 35689p shape, drop the latter, but not its penchan part. **Generally, drop 3p and keep penchan at 1-away** (ゆうせー, 2022i). With 35689p, riichi chance on 7p draw is more valuable than forming a taatsu around 3p. Besides, this floater 3p is inferior to an isolated floater 3 because:

- Drawing 2p produces a ryanmen with overlapping acceptance
- Drawing 4p does not give us a ryanmen; it produces a mentsu 345p and a kanchan 68p

While it is true that 34568p has 245p draws that give us ryanmen, from dropping 9p to drawing any of 245p upgrade will inevitably take a couple of turns.

That said, there are two cases where we can drop penchan 89p and go back to 2-away after dropping 3p:

1. Penchan is the bottleneck of tanyao
2. We have other strong shapes such as yonrenkei and nakabukure

When we are comparing between floaters in (65) at 2-away¹⁴, 3s from 35689s is inferior to isolated 6p. Therefore, drop 3s.

(65) East-1, West seat, turn 5, dora 8m
22378m 1136p 35689s

Likewise, for (66) which is 1-away, we keep 89s penchan and cut 3s floater.

(66) East-1, West seat, turn 5, dora 8m
234789m 115p 35689s

If we later draw 6m and get yonrenkei 6789m in (67), we can drop 89s penchan. Since this penchan still has overlapping acceptance, when we have very strong shapes in the rest of our hand, we can opt to drop penchan at 1-away.

(67) East-1, West seat, turn 6, dora 8m
2346789m 115p 5689s

If this 89s penchan is the only bottleneck of tanyao, we can drop it in exchange of value and flexibility even if it increases our shanten number. Drop 9s from (68).

(68) East-1, West seat, turn 5, dora 8m
22367m 2247p 35689s

1.1.7 Sankanchan and Yonkanchan

Shapes like 1357p are dubbed *sankanchan* and 13579p *yonkanchan* in Japanese, because they have three kanchans and four kanchans respectively. In general, the order of importance is “even-numbered sankanchan < floaters < odd-numbered sankanchan < outside kanchan < yonkanchan < inside kanchan” (chirno, 2008).

For odd-numbered sankanchan like 1357p, drawing 17p produces a ryankanchan and a head, drawing 9p produces a yonkanchan, drawing 35p produces a ryankanchan and a kanchan, drawing 8p produces a ryanmen, and drawing 246p completes a mentsu. Therefore, the acceptance of 1357p is 123456789p. However, if we must choose between an outside kanchan and a sankanchan, dropping 1p from 1357p while keeping the kanchan gives us the same number of acceptance but more ryanmen-improvement draws, as shown in Table 29. Hence the conclusion that sankanchans are weaker than outside kanchans.

¹⁴While generally we take penchan over floaters at 1-away, when we are 2-away instead, we have some leeway to seek taatsu around floaters and hope for better shapes. Indeed ゆうせー (2020b) states that we can drop penchan if we have two 3–7 floaters at 2-away. However, as stated previously, the  floater is actually weaker than an isolated 3–7 floater, so, in the case of , we do not really meet the standard of dropping penchan.

¹⁵Since we just cut , this could make us end up furiten.

Drop	1p	13s
Acceptance	46p2s 12 tiles	246p 12 tiles
Ryanmen-improvement draws	(2p) ¹⁵ 8p4s	8p

Table 29: Acceptance of 1357p13s.

Yonkanchans like 13579p accept 2468p but lack any ryanmen improvement. When we choose between yonkanchan and an inside kanchan, we can reduce yonkanchan to a sankanchan without losing acceptance while increasing our ryanmen improvement draws.

Even-numbered sankanchans such as 2468m also lack any ryanmen improvement. In most cases, we reduce it to a ryankanchan. The only case where we would keep it as is is when we are aiming for tanyao and treat it as 24m and 68m two kanchans. Therefore, we drop 8s from 2468s in (69) for the faint possibility of 234 sanshoku, and we drop 11m pair from (70) to aim for tanyao (𠂔うせー, 2022n).

(69) 24m 44699p 2468s 777z

(70) 11357m 46688p 2468s

1.1.8 Shapes with Overlapping Acceptance

When two taatsus share the same tiles as their acceptance, our effective number of acceptance reduces. This is known as *nidouke* in Japanese, which literally means “to accept twice.” Shapes like 3467p shares 5p acceptance, so only 258p 3 kinds 12 tiles complete a mentsu out of this shape, whereas 2569s 4 kinds 16 tiles complete a mentsu out of 3478s shape. Shapes with overlapping acceptance are not always bad or weak; there are cases where we still keep them over other weak taatsus.

Neighboring Pairs

Shapes like 3344p can be treated as two neighboring pairs 33p and 44p or two ryanmens 34p and 34p, therefore shapes with overlapping acceptance. We can easily produce a ryanmen out of this shape, so when pinfu is within sight, its priority is higher than weak taatsus such as kanchans, but lower than other pairs and ryanmens. Therefore, we drop one tile from 3344p in (71), and we drop 68s kanchan from (72) (G・ウザク, 2019, pp. 112–113).

(71) 33678m 3344p 23478s

(72) 33678m 3344p 23468s

Since (73) has no possibility of pinfu, and dropping 99p gives us tanyao, it becomes optimal to drop 9p.

(73) 33678m 8899p 23468s

Skipping Pairs

Shapes like 7799p can be treated as two pairs 77p and 99p with a gap in between or two kanchans 79p and 79p, therefore shapes with overlapping acceptance. Since 8p draw gives us iipeikou chance, and tiles around the pair on the inside gives us perfect 1-away, we still keep it over other kanchans and penchans, and destroy it when the rest of our hand are completed mentsus, pairs, and ryanmens (G・ウザク, 2019, pp. 114–117).

We can drop either of 79m from (74). Their respective merits are:

- 7m drop:
 - When 9m becomes our final wait, it has higher win rate
- 9m drop:
 - 6m draw gives us perfect 1-away and higher chance of pinfu

Especially for (74) where 47p draw gives us tanyao potential, in terms of expected value, 9m drop is the best.

(74) 7799m 123456p 5578s

Drop 68s from (75) for the possibility of perfect 1-away with 6m draw. Drop 8s first so that we still keep the possibility of 5s which gives us a ryanmen.

(75) 237799m 123456p 68s

Shapes like 224466p consists of three skipping pairs can be treated in a similar fashion. However, since 35p draw give iipeiko chance, we will discuss it in the context of building iipeiko yaku in Section 2.4.

1.2 4-Tile Shapes

1.2.1 Yonrenkei

Shapes like 3456p are called *yonrenkei* in Japanese, which means “the shape of 4 consecutive numbers.” If we draw 36p, we get a pair and a mentsu; if we draw 2457p, we get a ryanmen and a mentsu. It is therefore a strong shape. Depending on the rest of our hand, we may have to sacrifice this shape and reduce it to a mentsu.

(76) has two yonrenkei shapes. If we drop either of 13s, we get the most acceptance, but it is also very likely that we end up kanchan or shanpon bad-wait rii-nomi. The 122233s consists of 123s + 223s, which gives us iipeikou possibility. Therefore we will not touch 122233s and we opt to drop one tile from the weakest yonrenkei. Drop 1m (G・ウザク, 2016, p. 14).

(76) East-1, West seat, turn 7, dora 9s
1234m 5678p 122233s

(77) has tanyao ready, and with 07m kanchan we have tanyao dora 2, a very valuable hand. On top of that, we already have 6 blocks in our hand, the merit of producing head or ryanmen out of

5678s yonrenkei becomes very little. Considering also the possibility of 567 sanshoku, we drop 8s here (G・ウザク, 2016, p. 16).

(77) East-1, East seat, turn 6, dora 7m
2207m 3466p 225678s

Reaching Tenpai with 3456 Shape

Generally, drop the outermost tile (ヨーテル, 2023b). If we have 3456p and we must dropping one to reach tenpai, dropping 3p works in general cases. Merits of cutting the outermost tile are:

- Outer tiles generally have less deal-in rate
- Not giving other players easy time calling
- Strength of discards: a 6p signals that 39p are relatively safe, while a 3p gives way less information about our hand.

Exceptions are:

- Doras and sanshoku are involved
- To keep the possibility of catching kamichi's akadora
- To keep possibilities of improvements
- Inner tiles are safer according to table situation
- Early in a round

Cases involving doras and sanshoku

(78) is an open hand with 3456s. Our general policy is to drop 3s. However, since dora is 2s and, if we keep 345s shape and drawing one 2s, we can drop 5s instead. This is called sliding and is something one must keep in mind when they have a shuntsu close to dora. Drop 6s from (78).

(78) Dora 2s
3456s 2256p -2'13m 77'7z

If we drop 2s from (79), we will not be able to call chi on kamicha's 0s discard. If we instead keep 234456s, we can chi 0s with 46s and discard 2s. If 0s is already discarded, follow the general policy and drop 2s.

(79) 2344556s 3367p -55'5z

(80) has a slight possibility of 123 or 234 sanshoku. In this case, drop 5p.

(80) 123m 2345p 2399s -55'5z

Iipeikou-Like Shapes Has Less Improvements

Dropping either of 36p from (81) will give us the same waits 47p. If we drop 6p and keep 34556p, drawing 2p and dropping 5p gives us 23456p sanmenchan. Dropping 3p means that we will lose this upgrade. Therefore, drop 6p here.

(81) 345566p 23488s -6'66z

Note that this is for open hands only since iipeikou shapes gives us no additional value when our hand is open. If this is a closed hand, definitely go for iipeikou since we cannot change our tiles after we riichi.

Half Suji

When another player already declared riichi and we must drop either 3p or 6p in order to take tenpai, and 9p is genbutsu, half suji 6p is safer than musuji 3p (みーにん, 2017, pp. 140–141). Likewise, when 8p is genbutsu, half suji 5p is safer than musuji 2p.

The main reason why half suji is safer than musuji is because, in terms of possibility of the incomplete shape we deal in, musuji 3p deals into 12p penchan, which 6p does not deal into. The number of possible combinations of kanchan waits and ryanmen waits are the same for both 36p, so 3p has more possible shapes to deal into than 6p, hence higher deal-in rate.

Early in a Round

To make sure that, when we draw 6p back later in the game we can drop 3p, we can drop 3p when it is very early in a round.

Cases of 1234 and 2345

To be able to slide when we draw middle tiles, in most cases, we still drop outermost tiles from 1234p or 2345p. Other general rules such as keeping sanshoku chance or dora slide still apply. 川村晃裕 (2017, pp. 171–172) also gives similar merits and demerits and suggests to drop innermost tile instead in the following situations:

- Hand is cheap
- We prioritize defense over winning, and 1 (or 2) are generally much safer than 4 (or 5)
- No aka dora acceptance, or aka dora can still be picked up by other blocks
- No tanyao chance
- Shimocha still stays menzen
- We are in huge lead, and we don't mind shimocha opening their hand

For 112344m however, we should drop 4m and use 234m mentsu to catch possible 0m draw.

1.2.2 Nakabukure

The shape 3445p is called *nakabukure* in Japanese due to the duplicate of the middle tile of the sequence making it look like it is bulging (and hence nicknamed “the bulging shape.”)

Nakabukure works the best with the head of our hand confirmed. 3445p nakabukure accepts 23456p, which, except for 4p draw, make a mentsu and a ryanmen. If we are 1-away and have the head ready, upon drawing any of 2356p, we are ready for ryanmen tenpai. Contrarily, if 3445p becomes our final wait, we are forced to wait on 4p tanki, a very *very* weak wait. Therefore,

in cases where we have nakabukure and a ryanmen-toitsu, reducing the latter to a pair often is the best choice.

Dropping 3p from (82) and (83) is the best choice (G・ウザク, 2019, pp. 96–97). Table 30 lists the acceptance of (83) and it shows that confirming our head is better than destroying nakabukure or aryanmen.

(82) 5667m 344678p 4556s

(83) 5667m 344678p 4456s

Drop	Acceptance	Good-wait draw
3p	45678m4p234567s 38 tiles	4578m345s 23 tiles
6m	12345p234567s 37 tiles	245p35s 18 tiles
4s	45678m12345p 33 tiles	45678m2p 20 tiles

Table 30: Acceptance of (83).

4-tile shapes such as *ryantan*¹⁶ and *kantan*¹⁷ work the best without a head, and thus are incompatible with nakabukure. Since the number of waits of ryanmen and kantan are close to or greater than ryanmens, we destroy nakabukure for better final waits. Therefore, dropping 3s from (84) and 6s from (85) are the best choices here (G・ウザク, 2019, pp. 98–99). Especially in the case of (85), even if 0m is not double dora, knowing that dropping any of 5m4p6s gives us the same number of 34 tiles of acceptance, reducing nakabukure to a mentsu is still slightly better as suggested by Table 31.

(84) 6777m 234788p 2334s

(85) Dora 5m
0777m 234455p 5667s

Drop	Good-wait (waiting on ≥ 6 tiles) draw
5m	245p4578s 21 tiles
4p	46m5p4578s 24 tiles
6s	456m23456p 25 tiles

Table 31: Acceptance of (85).

Nakabukure and a Tile One Tile Away

Shapes such as 23346p can be treated as a 246p ryankanchan and 33p pair, or 2334p nakabukure and a floating 6p. Therefore, when there are no heads confirmed in our hand, we keep this shape and hope that we get a head with 356p draws.

¹⁶Shapes such as  which can be treated as a  ryanmen and  pair, or a  tanki and  anko, hence the name – ryan(men and)tan(ki). As a final wait, it waits on .

¹⁷Shapes such as  which can be treated as a  kanchan and  pair, or a  tanki and  anko, hence the name – kan(chan and)tan(ki). As a final wait, it waits on .

Drop candidates of (86) are 6m3p6p, but dropping 6m gives us the widest acceptance as is suggested by Table 32 (G・ウザク, 2019, p. 100).

(86) 2346m 23346p 45678s

Drop	Acceptance
6m	356p369s 20 tiles
3p	6m6p369s 17 tiles
6p	6m3p369s 16 tiles

Table 32: Acceptance of (86). All the candidates share the same acceptance of 369s 11 tiles, which unfortunately give us tanki wait.

In the case of 233468p in (87), as is the case of taking headless 1-away, dropping 3p gives us the widest 1-away. In terms of the number of good wait draws, as shown in Table 33, there is only 1 tile of difference between 8p drop and 3p drop. Considering the possibility of improvements with 124p draw, dropping 8p seems better. As long as no yakus or doras are involved, when there is no head confirmed, we prioritize nakabukure and a tile one tile away (G・ウザク, 2019, p. 101).

(87) 234m 233468p 45678s

Drop	Acceptance	Draws giving us ≥ 6 tiles of tenpai
2p	57p369s 19 tiles	57p 8 tiles
3p	678p3456789s 33 tiles	678p 10 tiles
8p	356p369s 20 tiles	356p 9 tiles

Table 33: Acceptance of (87).

Cases where We Drop Yonrenkei or Nakabukure Before Tenpai

We have learned that both yonrenkei and nakabukure are good at making good shapes. This, however, forces us to treat it as 2 blocks, and to complete 2 mentsus out of these 4-tile shapes, we must draw at least 2 turns. Therefore, when we have enough blocks and they are all sufficient for advancing our hand towards tenpai, we can drop yonrenkei or nakabukure without sacrificing too much acceptance. According to クリアレイン (2023d), when there are no other floating tiles to dispose of, the conditions to drop yonrenkei or nakabukure are,

- 2-away
 - Other 4 blocks are all ryanmens and pairs.
- 1-away
 - There is at least 1 ryanmen block.

Hand (88) already has 5 blocks (treating yonrenkei as 1 here.) All blocks are good shapes, so we drop either 3p or 6p from yonrenkei shape.

(88) 2256m 3456p 334788s

On the other hand, there is a 68m kanchan in (89). In such cases, we keep yonrenkei or nakabukure in hope for better shapes, while also keeping the bad block so as to not lose its direct acceptance.

Therefore, we turn to ryanmen-toitsus and drop 3s. We keep 788s so that we have a better chance of completing this block with tanyao tiles.

(89) 2268m 3456p 334788s

While it is tempting to drop bad blocks and rely on yonrenkei or nakabukure to give us good shapes, when we are at 1-away, we have to compromise and accept the chance of bad waits. Therefore, we do not drop 799m from (90); we drop 5p instead.

(90) 11799m 4556p 34s 333z

There is also a 34m ryanmen in (91), so we drop 2s from 2345s (多井隆晴・郡道美玲, 2021, pp. 99–100). Dropping 9p loses the direct tenpai chance on 8p draw.

(91) West seat, turn 7, dora 8m
34m 66779p 2345888s

However, when the remaining blocks are so bad that there is just no chance for good wait, we can opt to drop one of the bad blocks and fall back on yonrenkei or nakabukure. Therefore, we drop 2s from (92) and hope for improvement with 5s draw. We do not touch 799m since it accepts 89m 6 tiles, which is more than a simple kanchan like 24s (クリアレイン, 2023d).

(92) 11799m 4556p 24s 333z

1.2.3 Aryanmen

Shapes like 3345p are called *aryanmen* in Japanese. The *a-* prefix in *aryanmen* means “next to,” and considering that, as a final wait, the shape has a ryanmen wait but 2 tiles less than an ordinary ryanmen, we can see how this shape is treated as something inferior than a ryanmen.

1123p is also an aryanmen. Since it has no ryanmen improvements, nor is it a strong wait (since we already have two copies of the outermost wait 1p), it is the weakest aryanmen of all shapes in this family. In most cases, it is simply treated as 123s sequence and a floating 1s.

Aryanmen is the Weakest 4-Tile Shape

In terms of good shape improvement, aryanmens have the least number of draws, as shown in Table 34. That is why it is often treated as a weak shape and, in literatures such as Uzaku’s efficiency book, you will find advices such as “when in doubt, drop aryanmen, and more often than not you’ll be right.”

Shape	Acceptance	Good shape draw
Yonrenkei 3456p	12345678p 28 tiles	2457p 14 tiles
Nakabukure 3445p	23456p 16 tiles	2356p 14 tiles
Aryanmen 3455p	234567p 20 tiles	456p 9 tiles

Table 34: Acceptance of 4-tile shapes.

Hand (93) has 3 kinds of 4-tile shapes. We must drop one of them, and we opt to drop the weakest of them all. Drop 4s (G・ウザク, 2019, p. 102).

(93) East round, South seat
3456m 6778p 4456s 44z

Dropping 6p from (94) and having both sanmenchan and the acceptance of dora 2p confirmed seem good. However, as is shown in Table 35, dropping 5s actually gives us the widest acceptance and the most number of good wait draws.

(94) East-1, East seat, turn 4, dora 2p
4567m 345667p 3455s

Drop	5s	6p
Acceptance	23456789m2356789p 51 tiles	47m258p25s 23 tiles
Good-wait draws	345678m2358p 26 tiles	47m258p25s 23 tiles

Table 35: Acceptance of (94).

Let Go of the “Ryanmen” in Aryanmens

Aryanmens does not work well with headless hands. In a 14-tile hand with 2 uncompleted taatsu and 1 aryanmen, if we choose to keep the pair and the ryanmen in the aryanmen shape and drop one tile from other taatsus, at that moment there is one surplus tile from the destroyed taatsu, and thus our acceptance is narrowed. In such cases, it is better to simply reduce the aryanmen to a mentsu. Indeed, headless hands come with the risk of tanki wait. With ankos or other shapes from which we can easily get a head, there is no need to fret about such risk.

To make all of our tiles useful in (95), we drop 6p from aryanmen instead of 8m or 8p. Drawing any of 368m also gives us ryanmen tenpai (G・ウザク, 2019, p. 105).

(95) 34568m 6678p 34888s

Since there is a 22s pair and a 344568s ryanmenkanchan in (96), dropping 6m from the aryanmen shape leaves us no surplus tiles. Indeed, dropping 8s and treating 2234s also as an aryanmen leave us no surplus tile as well, and the difference in acceptance is just 1 tile. However, keeping 6678m aryanmen shape means that we must compromise on 9m draw which makes us lose tanyao (G・ウザク, 2019, p. 104).

(96) 6678m 34p 22344568s

There is 2 kanchan blocks in (97). As shown in Table 36, dropping any tiles from the kanchan blocks leaves us a surplus tile and narrowed acceptance. Taking headless 1-away indeed has the widest acceptance but also the risk of tanki wait, and, in the case of (97), there is no shape that gives us a head or serves as a strong wait when our hand is headless. However, notice that for our 23s “ryanmen”, we already have two copies of 1s in our hand, and, if there is another copy of 1s on the table, this 1123s shape is actually almost as weak as a kanchan waiting on 4s. Considering

also the fact that 1123s has no improvement while kanchans at least have ryanmen improvements, dropping 1s is slightly better here (平澤元気, 2024).

(97) 24678m 34579p 1123s

Drop	Acceptance
2m	4m8p14s 13 tiles
1s	234m789p 20 tiles

Table 36: Acceptance of (97).

1.2.4 Confirming a Mentsu or Confirming a Head

In cases where our hand is full of ryanmens and shuntsus and the only pair is buried in nakabukure or ayanmen, we must either drop the sequential part or drop the pair part. While it is true that dropping the pair and taking headless 1-away gives us the widest acceptance and thus higher chance of reaching tenpai, we also face the risk of tanki wait. Tanki waits (except for terminal and honor tile tanki waits) have terrible win rate; considering that our goal is to slam that riichi stick with best outcome possible, we would treat this as undesirable outcome¹⁸ and focus on the rest of our acceptance. The theme of this section is thus to maximize the number of good-wait (≥ 6 tiles) tenpai depending on the actual composition of our hand. According to G・ウザク (2019, pp. 46–55) and ゆうせー (2021g),

- Confirming head is better when
 - It is still early and other players are still dealing with useless terminal tiles.
 - We are in a table situation where we are required to force good-wait or yaku.
 - Taatsus are involved in yakus.
- Confirming mentsu is better when
 - It is late and we would like to reach tenpai as soon as possible or to reach keiten¹⁹.
 - Drawing duplicate copies of tiles in taatsus is more likely than completing those taatsus.
 - The shapes of the rest of our hand makes it salvageable even if we complete the taatsus first.
 - Making it a mentsu enables dora slide.
 - We have an anko and thus not required to confirm a head.

In case of the uncompleted taatsus being unrelated ryanmens, headless 1-away has the widest acceptance of 28 tiles, 16 tiles of which complete one of the ryanmen taatsus and force us to drop one tile from the other taatsu and take tanki wait. On the other hand, confirming mentsu leaves us 12 tiles of good-wait tenpai draws, as listed in Table 37. Thus, it generally is better to drop 13m from (98) and confirm our head.

(98) 1223m 34678p 34578s

¹⁸Practically speaking, even if we unfortunately arrive at tanki wait, we can still choose to dama our hand, regardless of it having yaku or not, and wait for our draws that give us better waits.

¹⁹*Keiten*, or *keishiki tenpai*, refers to a state where we are tenpai without a yaku in late turns. At exhaustive draw, players that are not in tenpai must pay those who are in tenpai, so keiten is something we must keep in mind unless someone already called riichi.

Shape	Confirming head	Confirming mentsu	
	Ryanmen tenpai	Ryanmen tenpai	6-tile tenpai
Ryanmen + ryanmen + shuntsu	16 tiles	12 tiles	0 tile
Ryanmen + sanmenchan	19 tiles	18 tiles	8 tiles
Ryanmen + 56789p	15 tiles	18 tiles	8 tiles
Ryanmen + 66778p	15 tiles	14 tiles	8 tiles
Ryanmen + 77889p	15 tiles	10 tiles	0 tile
Ryanmen (nidouke) + shuntsu	12 tiles	12 tiles	4 tiles

Table 37: The difference between confirming head and confirming mentsu in terms of the number of draws that would give us tenpai with good quality. Bad waits such as shanpon and tanki are omitted here.

For shapes like 66778p in (99), even if we draw 14s first, we can still drop 7p and make the resulting aryanmen shape as our final wait, not too bad. If this shape sits on the edge like 77889p in (100), then we lost the leisure of aryanmen tenpai since 7789p does not accept “10p.” Therefore, we confirm its mentsu in (99) while we confirm its head in (100).

(99) 1223m 66778p 23678s

(100) 1223m 77889p 23678s

In case of sanmenchan, it has more tiles of acceptance than a ryanmen and thus more likely to get completed. However, even then, we can still expect to get good waits when we draw tiles around the completed sanmenchan. For example, after 2m drop from (101), upon drawing 8p, we can first dama²⁰ and expect 235689p draws to give us sanmenchan or irregular three-sided waits.

(101) 1223m 34567p 23678s

While it is true that, even in the case of (98), suppose that we take headless 1-away anyway and end up tanki wait, we can still wait several turns before we get nobetan waits²¹. However, in such cases, we have less draws that give us nobetan waits than the case of (101), and we can barely draw into three-sided waits. Therefore, completed sanmenchan is regarded as salvageable, while unrelated shuntus are not.

In case of two ryanmens with overlapping acceptance, the number of ryanmen tenpai draw after having our head confirmed decreases from 16 tiles to 12 tiles, the same number when we confirm our mentsu. Note that, for shapes like 3467p in (102), if our hand is headless, the nidouke shape actually additionally accepts 5p since we can get nobetan wait out of 34567p shape. Therefore, in terms of good-wait draws, having our mentsu confirmed is actually better. Drop 2m from (102)

(102) 1223m 3467p 123678s

As a rule of thumb, when the ryanmen taatsus are unrelated and do not compound with existing

²⁰Dama refers to a state where we are in menzen tenpai but no riichi is called.

²¹Waits with yonrenkei shape 3456p.

mentsus, having our head confirmed gives us the most probability of having good waits. However, for shapes like 34567p, 34556p, 33445p which can still give us aryanmen or nobetan wait even when our hand is headless, taking headless 1-away is optimal.

Cases where Taatsus are not All Ryanmens

In cases where there are taatsus other than ryanmen, the number of draws that give us ryanmen tenpai decreases, but the number of draws that turn incomplete taatsus into head remains the same, so it becomes clear that confirming mentsu is better than confirming head.

For example, if we confirm our head in (103), the only draw that gives us ryanmen tenpai is 4p 4 tiles. On the other hand, if we confirm our mentsu, our ryanmen tenpai draws become 35p 6 tiles.

(103) 1223m 35789p 23678s

In the case of 67889p in (104), if we drop 9p, we will have either aryanmen wait or ryanmen wait on any draws, as shown in Table 38. On the other hand, if we confirm our mentsu with 2m drop, we will have wider acceptance, and more tiles to reach good-wait tenpai. Therefore, 2m drop is the best choice here. Note that the shape 67889p becomes a mentsu and a head with 5689p draws when our hand is headless.

(104) 1223m 67889p 23678s

Drop	2m	9p
Acceptance	56789p1234s 29 tiles	2m58p14s 16 tiles
Good-wait draws	5689p14s 20 tiles	2m58p14s 16 tiles

Table 38: Acceptance of (104).

In the case of 66788p shape in (105), confirming mentsu here is better since we can either drop 6p or 8p for aryanmen tenpai when 23s ryanmen gets filled; we can dama with 7p draw which gives us iipeiko yaku. Its acceptance is listed in Table 39.

(105) 1223m 66788p 23678s

Drop	2m	6p
Acceptance	56789p1234s 29 tiles	2m58p14s 16 tiles
Good-wait draw	5689p14s 20 tiles	2m58p14s 16 tiles
Kanchan tenpai	23s 6 tiles	0 tiles
Tanki tenpai	7p 3 tiles	0 tiles

Table 39: Acceptance of (105).

However, if this iipeiko kanchan shape sits on the edge, like 77899p shape in (106), confirming 123m mentsu or 7899p aryanmen do not differ in terms of good-wait draws, as listed in Table 40. In terms of hand value, however, for the same menzen-exclusive 1 han, iipeiko gives more fu than pinfu. The number of draws that give us yaku after 2m drop is also greater than the acceptance after dropping 7p. Therefore, dropping 2m is slightly better here.

(106) 1223m 77899p 23678s

Drop	2m	7p
Acceptance	6789p1234s 25 tiles	2m69p14s 16 tiles
Good-wait tenpai	679p14s 16 tiles	2m69p14s 16 tiles
Kanchan tenpai	23s 6 tiles	0 tile
Tanki tenpai	8p 3 tiles	0 tile
Pinfu draw	14s6p 12 tiles	2m69p14s 16 tiles
Iipeiko draw	8p23s 9 tiles	0 tile

Table 40: Acceptance of (106).

When the manzu part becomes 1123m aryanmen, even though dropping 7p gives us less acceptance, as shown in Table 41, we actually end up 1-away that cannot draw into bad wait. Therefore, the best move here is not to confirm mentsu or to confirm head from 1123m aryanmen, but to drop 7p and create two aryanmens.

(107) 1123m 77899p 23678s

Drop	1m	7p
Acceptance	6789p1234s 25 tiles	14m69p14s 20 tiles
Good-wait tenpai	679p14s 16 tiles	14m69p14s 20 tile
Kanchan tenpai	23s 6 tiles	0 tile
Tanki tenpai	8p 3 tiles	0 tile

Table 41: Acceptance of (107).

1.3 6-Tile Shapes

6-tile shapes are a family of shapes that consist of 6 tiles in a suit and, in the context of tile efficiency, compounded with a pair to work with two-pair principle. These particular shapes either are copies of tiles stacked together, like 334456p33z, or have one seemingly useless tile, such as 334568p33z, in a way that often puzzles people and makes them cut tiles that could cost them acceptance and tenpai chance.

Among all the shapes listed in Table 42, the shape 344568p is the most popular one called *ryanmenkanchan* and is the one that is mentioned the most in books and materials on tile efficiency. There are also cases where we have 7-tile shapes such as 3345568p where we can either make it 334556p or 344568p, and the best way of handling this shape depends on many other factors.

Dropping Surplus Taatsus and Forming Ryanmenkanchan

One of the many cases where ryanmenkanchan emerges is when we have surplus taatsus and must drop one. When our taatsus and mentsus are positioned in a particular way, the remaining shape forms a ryanmenkanchan and we get to keep part of the acceptance of the destroyed taatsu, thus minimizing our loss. Examples include:

- Dropping 2s from 2356778s

Shape	Acceptance	Improvement
444556p33z	34567p3z 16 tiles	
334567p33z	2358p3z 15 tiles	
344456p33z	2457p3z 14 tiles	
444566p33z	4567p3z 12 tiles	Draw 3p drop 6p
455566p33z	4567p3z 12 tiles	Draw 3p drop 6p
344568p33z	257p 11 tiles	Draw 4p drop 8p
244456p33z	347p3z 11 tiles	Draw 56p drop 2p
334556p33z	347p3z 11 tiles	Draw 25p drop 3p
334456p33z	235p3z 11 tiles	Draw 47p drop 3p
444557p33z	56p3z 8 tiles	Draw 3p drop 7p
334568p33z	37p3z 8 tiles	Draw 245p drop 8p
		Draw 4p drop 3p
134568p33z	27p 8 tiles	Draw 4p drop 1p
		Draw 5p drop 8p
344457p33z	46p3z 7 tiles	Draw 235p drop 7p
133455p33z	24p 7 tiles	Draw 356p drop 1p
		Draw 6p drop 3p
334566p33z	36p3z 6 tiles	Draw 2457p drop 3p or 6p

Table 42: 6-tile shapes, with a head included (chitist, 2021).

- Dropping 1s from 1356778s
- Dropping 3s from 3356778s

In (108), if we drop either of 67m3578s, at that moment we have a surplus tile in our hand, and we end up 4 kinds 16 tiles of acceptance, as shown in 43. The key here is to identify the 356778s ryanmenkanchan, which makes use of all souze tiles and gives us one more kind of tiles of acceptance. Drop 2s (G・ウザク, 2019, p. 118).

(108) 67m 11456p 2356778s

Drop	2s	578s
Acceptance	58m469s 19 tiles	58m14s 16 tiles

Table 43: Acceptance of (108).

(109) has 6 blocks consisting of mentsus, ryanmens, and pairs. We intend to drop either of the pair. The pinzu shape may look completed, but if we drop 8p, there is actually 344568p ryanmenkanchan in our hand and it gives us more acceptance than dropping 8s.

(109) 345m 3445688p 3488s

(110) is a similar case where dropping 2m gives us the ryanmenkanchan 245667m which makes full use of every tiles (ヨ一テル, 2023f).

(110) East-1, turn 4
2245667m 22678p 45s

Floating Tiles and Ryanmenkanchan Potential

When comparing floating tiles in our hand, we tend to overlook some floating tiles that position near other shapes in a way that has ryanmenkanchan potential. For example,

- 36778p in want of 5p
- 35677p in want of 8p

Both (111) and (112) have floating 8m and 8p, and 8m is more isolated than 8p, so we drop 8m in hope that we can get ryanmenkanchan in pinzu suit (G・ウザク, 2019, p. 122).

(111) 348m 34458p 357788s

(112) 348m 44568p 357788s

Choosing between Pinfu and Tanyao

For the same 1 han, tanyao has a probability of scoring higher than pinfu, and we are ready to make calls and speed up, so we prioritize tanyao over pinfu (G・ウザク, 2019, p. 119). Dropping either of 9p2s from (113) leaves us no surplus tile and gives us the same 19 tiles of acceptance. Dropping 9p gives us tanyao chance, and dropping 2s confirms pinfu. Considering the merits mentioned earlier, we opt to drop 9p and go for tanyao.

(113) 678m 455679p 2238 -8s

Likewise for (114), we drop 9p and choose tanyao (G・ウザク, 2016, p. 10).

(114) 23488m 455679p 556s

Cases where Pinfu is Impossible

Due to the characteristic of ryanmenkanchan that it always produces two shuntsu when completed, it works very well with pinfu. For cases where pinfu is not within sight such as having yakuhai pair in our hand, the importance of maintaining ryanmenkanchan decreases.

Hand (115) has 77z pair and thus cannot have pinfu. Dropping 3m and 8p do not have any difference in terms of acceptance, but if we drop 8p here, we end up with just rii-nomi. Therefore, we drop 3m here. We trade 4m acceptance with 8p7z acceptance (ヨ一テル, 2022c).

(115) 356778m 788p 234s 77z

It does not matter too much which to drop for rii-nomi hands such as (116). Adjust according to actual table situation (ヨ一テル, 2022c).

(116) 356778m 788p 33399s

Choosing between Guaranteed Pinfu and Iipeiko Chance

For shapes like 3556778p or 3567788p, we can either make it ryanmenkanchan, or make it other kinds of 6-tile shapes. Either way, we still have 11 tiles of acceptance with the remaining shape. Therefore, the difference lies in the value the remaining shape could bring us.

Drawing 6m with 55677m or 67788m in (117) or (118) gives us iipeiko, so, in both cases, dropping 3m gives us the most possible value. It is true that, since the remaining shapes after 3m drop all have shanpon acceptance, we still could end up rii-nomi. However, The increase in value with one more han is greater than the decrease in value with one less han, so, in terms of expected value, 3m drop is still optimal (G・ウザク, 2019, pp. 120–121; ヨーテル, 2022c). The cases where we would choose ryanmenkanchan and guaranteed pinfu instead are,

- *Agaritoppu*, a kind of table situation where winning our hand confirms our first place.
- In lead, and we are going to crush others' renchan²² chance
- Already valuable, such as dora 3.

(117) 3556778m 45p 12388s

(118) 3567788m 78p 23499s

Drop	3m	5m
Acceptance	569m36p8s 19 tiles	469m36p 19 tiles
Pinfu drawn	69m36p (78.9%)	469m36p (100%)
Iipeikou draws	6m (15.8%)	None

Table 44: Acceptance of (117).

For hands like (119) that has yakuhai anko, there is no pinfu, so the merit of keeping ryanmenkanchan diminishes. For merits such as 0p acceptance, we opt to drop 3p here (ZERO, 2020, p. 35).

(119) 23m 3556778p 33s 777z

Dropping Ryanmenkanchan for a Better Chance of a Head

The shape 34556m in 134556m ryanmenkanchan produces a head easily. To produce a head out of 134556m ryanmenkanchan, however, the only draw is 1m. In hands such as (120) where we seek to confirm our head, we can drop 1m without losing the opportunity to incorporate 2m, while leveraging the flexibility of 34556m shape.

(120) 134556m 67p 307889s

Drop	1m	9s
Acceptance	1234567m5678p34568s 49 tiles	247m58p46s 27 tiles

Table 45: Acceptance of (120).

²²To win a hand as oya, and thus continue their oya status.

Ryanmen-Toitsu and 6-Tile Shape

When we are choosing between ryanmen-toitsu and 6-tile shape (with also a pair in our hand,) reducing ryanmen-toitsu to a ryanmen is, in most cases, the best choice in terms of acceptance and, when the number of accepted tiles are the same, the number of good-wait draws. Hand has 334568p which may seem weak, but it indeed is a 6-tile shape according to Table 42. Dropping 3p and 5s have the same acceptance of 16 tiles, but, as shown in Table 46, keeping 334568p has the maximal 8 tiles of good-wait draws. By dropping 5s, we are actively denying the possibility of 9m5s draws that destroy our ryanmen.

(121) 99m 334568p 123556s

Drop	Acceptance	Good-wait draws
3p	9m7p457s 16 tiles	7p 4 tiles
8p	9m3p457s 14 tiles	9m3p 4 tiles
5s	9m37p47s 16 tiles	9m37p 8 tiles

Table 46: Acceptance of (121).

(122) too follows the same principle and 7m drop gives us the maximal acceptance. Even though 111223p sits on the edge and loses 4 tiles than its middle counterpart, in terms of total acceptance, 7m drop still has more acceptance than, say, 2p drop (G・ウザク, 2019, p. 134).

(122) 677m 111223p 11567s

Drop	Acceptance
7m	58m1234p1s 20 tiles
2p	578m14p1s 17 tiles
6m	7m1234p2s 14 tiles

Table 47: Acceptance of (122).

From 7 Tiles to 6 Tiles

In the following cases, we will deal with 7-tile shapes that give us possibility of different 6-tile shapes. The point is:

- Always pick the 6-tile shape that gives us the maximal acceptance.
- If the result 6-tile shapes have the same number of acceptance, pick the one with most yaku potential, be it tanyao, pinfu, or iipeiko.

We must drop one tile from pinzu part of (123). We can either drop any of 57p and have what is called “sanmen-toitsu,”²³ or drop any of 48p and have what is called “*niinii* shape.”²⁴ Either way, we have 69m1s acceptance, so the difference is in the acceptance of sanmen-toitsu and *niinii* shape. As is shown in both Table 42 and Table 48, *niinii* shapes have 4 less tiles than sanmen-toitsu, so the correct drop is 57p. If 0p is still live, drop 7p to accept it (G・ウザク, 2019, p. 132).

²³A name I made up since, for , it is a sanmenchan  and a toitsu .

²⁴From Japanese where, for , *ni* “two” refers to  which we have two copies and *i* “one” refers to  which we have one copy.

(123) 78m 4556778p 11345s

Drop	Acceptance of pinzu shape
5p	3679p 13 tiles
7p	3569p 13 tiles
4p	569p 9 tiles
8p	367p 9 tiles

Table 48: Acceptance of (123).

When the shape sits on the edge like in (124), the sanmen-toitsu shape loses 4 tiles of acceptance and there is no difference, in terms of acceptance, between 68p drop and 9p drop. However, by keeping 566788p niinii shape, we have iipeiko on 7p draw and thus the highest expected hand value. Therefore, drop 9p (G・ウザク, 2019, pp. 132–133).

(124) 78m 5667889p 11345s

Shapes like 234445s accepts 1 less tile than sanmen-toitsu but still 3 more tiles than niinii shape. Therefore, we drop 2s from (125) for maximal acceptance. If 234445s becomes 123334s and loses 4 tiles of acceptance as in (126), taking 112334s niinii shapes instead gives us the maximal acceptance and also iipeiko chance (G・ウザク, 2019, p. 133).

(125) 67m 34599p 2234445s

(126) 67m 34599p 1123334s

1.4 Other Shapes

1.4.1 1345 Shape

Keep 1p in 1345p over isolated 28s (noprops, 2022). Drawing 2m gives us 45m ryanmen, and drawing 36m gives us a 13m kanchan which can be a decent wait. Compared to an isolated 2s which produces a penchan if we draw 1s, kanchan if we draw 4s, and a ryanmen if we draw 3s, 1345m has slight advantage. In this regard, we drop 8p from (127).

(127) East-1, West seat, turn 3, 1p
1345m 2338p 5889s 11z

1.4.2 13457 Shape

Keep isolated 28s over 1p in 13457p (noprops, 2022). Drawing 7p with 13457p makes 1p much weaker since, if we drop 1p, we will not lose sanmenchan chance with 6p draw, and we can still incorporate 2p since 23457p is a shape that has 1348p ryanmen-improvement draws. Therefore, drop 1m from (128).

(128) East-1, East seat, turn 4, dora 6p
13457m 2668p 23699s

1.4.3 Entotsu

The shape 22234p22s is called *entotsu* in Japanese whose literal meaning is “chimney.” Why this particular shape is associated with chimneys remains disputed. Since 22234p can be treated as either 22p + 234p and 222p + 34p, with 2s pair in our hand, we end up 25p2s wait.

(129) is an open hand and we draw 7m. Since dora is 8m, we would like to work our shapes closer to dora. On top of that, since we have a kanchan 68p in our hand, it is more likely to draw another copy of 68p (6 tiles) than tsumo our winning tile 7p; in the former case, we can upgrade our hand to entotsu shape and catch dora 8m. Therefore, drop 4m²⁵.

(129) South-2, South seat, turn 4, dora 8m
40556m 68p -7m -1'23p222z

(130) has 6 blocks (seeing pinzu part as 22p + 234p) and dropping any taatsu will not cause too much difference – any of 8m234p drop gives 16 tiles of acceptance and any of 34m78s gives 15 tiles. According to Section 1.5.2, the best course of action is to drop a pair. Although there is no immediate difference between 8m drop and 2p drop, but keep in mind that 22234p can become shapes listed in Table tab:6plus2 with one additional pinzu tile, so 8m drop is the best here (G・ウザク, 2019, p. 130).

(130) 3488m 22234p 12378s

(131) is similar to (130) but with its shapes more to the middle and one additional anko, tanyao and sananko weigh than pure efficiency and the improvements with future draws. Since 78s is now our bottleneck when it comes to scoring tanyao, we can drop it here (G・ウザク, 2019, p. 131).

(131) 3488m 44456p 22278s

1.5 6 Block Theory

We have seen the term “block” here and there previously without defining its exact meaning. In this section, we will cover the term “block” thoroughly and explore the possibility of keeping multiple blocks in our hand, even cases where we have more blocks than needed.

A block is a group of tiles that we expect to become mentsu when we complete our hand. To complete a hand in mahjong using 14 tiles, we need 4 mentsus and 1 head²⁶. In that regard, we may think that we only need to keep 5 blocks in our hand. However, when we are still far away from completion and each block does not take up too much space, we can fit 6 blocks in our hand. The method of developing our hand while keeping all 6 blocks is called “6 block theory” (G・ウザク, 2019, p. 16).

Merits of keeping 6 blocks include (G・ウザク, 2019, p. 17; ゆうせー, 2022g; ゆうせー, 2022m):

- Keeping maximal choices between yakus and waits

²⁵This is from one of my games and in the end I did pick entotsu shape and draw .

²⁶Except for kokushi musou and chiitoitsu.

- Able to adapt to new table situation such as pon or kan from other players
- Dropping tiles and fixing ryanmens make their outside wait come out more easily

Demerits of keeping 6 blocks include (G・ウザク, 2019, p. 18; ゆうせー, 2022g):

- Keeping 6 blocks leaves us no room for follow-up tiles for each block, thus harder to reach perfect 1-away
- The moment we drop 1 block to take 1-away, the destroyed block leaves us 1 surplus tile, thus making our acceptance narrower and lowers our defense

1.5.1 Criteria of Keeping 6 Blocks

Our policy regarding block number in our hand is (G・ウザク, 2019, p. 22–23; ヨーテル, 2023c; ゆうせー, 2022g; ゆうせー, 2022m):

- Keep 6 blocks when
 - Going straightforward leaves us high probability of cheap bad wait
 - Blocks involve hand value and yaku
 - There is no huge difference between blocks
- Keep 5 blocks when
 - There is huge difference between blocks
 - We prioritize the speed of reaching tenpai
 - Our hand is open and we want pairs for pon calls
 - We prioritize defense

Cases where We Keep 6 Blocks

Keep 6 blocks if going straightforward with blocks we currently have leaves us bad waits or cheap hand (G・ウザク, 2019, p. 23). For (132), if there is no difference in player points, drop 3p and take perfect 1-away. However, if we are far behind other players, keeping 3p gives us a slight chance of sanshoku and dropping 7m confirms pinfu. This is the case where even a single tile can be treated as a whole block.

(132) 234 677m 3 88p 234 78s

When there is no discernible difference between taatsus, we can delay our decision by keeping 6 blocks (ヨーテル, 2023c). (133) has 6 taatsus, and dropping any of 78m23p45s leaves us 5 blocks. We can also reduce 334m to a ryanmen and progress with 6 blocks.

(133) East-1, South seat, turn 3, dora 3z
334 78m 23 678p 45 88s

We can drop 4s from (134) and keep 6 blocks, or drop 6z and reduce it to 5 blocks²⁷. If we drop 4s, drawing 7s will not destroy sanshoku, and we keep the possibility of drawing another copy of 6z. If any other ryanmen gets filled up before we complete 66z, we can drop 66z and aim for pinfu. This is the case where keeping 6 blocks gives us more choices between yakus (ゆうせー, 2022g).

²⁷If we drop  we lose pinfu and sanshoku becomes uncertain. Do not drop .

(134) East-1, South seat, turn 3, dora 8p
345m 34 78p 44s 77s 66z

There are 3 possible options in (135),

- Drop 3m and wait for improvement of 3440p
- Drop 4p and follow pure tile efficiency
- Drop 9p and force tanyao

Considering the current turn and the value of our hand,

- It is already in the middle of a round → we cannot afford going back in shanten with 9p drop.
- There are 3 pairs and we could possibly end up rii-nomi without tanyao → dropping 4p is not favorable.
- It is more easily to make another group out of 3440p than out of 233m.

Therefore, dropping 3m is the best choice here (福地誠, 2017, pp. 57–58). If we draw around 3440p, we can drop 9p and go for tanyao. If we draw 9p4s first, we can still drop 4p and riichi.

(135) Turn 8, dora 7p
233678m 344099p 44s

Cases where We Keep 5 Blocks

(136) has a penchan that is clearly weaker than kanchan, so we drop 89s (石橋伸洋, 2015, p. 21).

(136) 11 667m 68p 34 89s 333z

(137) has 6667p shape which can be treated as 2 blocks 66p + 67p. If we reduce it to 666p 1 block, we have the widest acceptance of 7 kinds 24 tiles. However, if we don't draw any of 789m 9 tiles (1 copy of 8m is the dora indicator) first, we end up a weak penchan wait. We therefore drop 8m. If we later draw another copy of dora 9m, we can drop 7p (G・ウザク, 2016, p. 16).

(137) East-1, East seat, turn 7, dora 9m
23489m 6667p 34789s

Being able to pon is crucial for open hands (ヨーテル, 2023c). We just called pon 7z in (138). 6 block theory would require us to drop 3m here and abandon the ability to call 3m7p pon, which defies the purpose of going open in the first place. Open hands also come with the ability to change tiles and waits, so having tiles following taatsus becomes more important and contributes to our flexibility. If (138) develops into (139), calling 1245m chi gives us two ryanmens, drawing 5p also improves our hand into ryanmens. Therefore, drop either of 24p68s depending on table situation.

(138) 33478m 2477p 68s -77'7z

(139) 233478m 2477p -77'7z

At the moment we draw 5p in (140), we have 6 blocks. The only purpose of 12m is to accept 3m; if the other taatsus get completed first, we end up 3m penchan wait rii-nomi. It is therefore close to useless, and we would expect to draw another copy of 16z to give our hand more value; even for defense purposes, 16z is generally more safe than 12m. Thus we drop 12m here (お知らせ, 2020, pp. 154–155).

(140) East 1, East seat, turn 1, dora 7p
12m 44578p 67s 1336z -5p

(141) already has 5 blocks necessary for manzu honitsu. If we keep 23p, at worst our hand becomes haku nomi 1500 points. Therefore we drop 23p. Keep in mind that dora is 7s and keeping 6s makes us able to incorporate dora (お知らせ, 2020, pp. 156–157).

(141) East-1, East seat, turn 4, dora 7s
3357m 23p 6s 114456 -5z

1.5.2 Order of Dropping a Block

Comparing Different Shapes

The order of dropping a block is **penchan** > **kanchan** > **pair** > **ryanmen** (石橋伸洋, 2015, pp. 21–22). Dropping either 6m (keeping 6 blocks) or 1p (reducing to 5 blocks) from (142) gives us the same acceptance of 24 tiles. However, dropping 6m leaves us a kanchan and we lose the chance of drawing 16m to avoid kanchan wait, so there is no reason to keep 6 blocks. Considering that 68p can be upgraded to ryanmen with 5p draw, we drop the weakest block 12p.

(142) 11 668m 12 68p 35s 333z

Even if the weakest taatsu has 1 dora, the order will not change (ゆうせー, 2021e). The 79s kanchan in (143) has a dora, but we have nice shapes in the rest of our hand. Thus, we drop 7s first, and drop 33p pair if later we draw another copy of 9s.

(143) Dora 9s
234788m 3367p 3479s

The reasons why we keep toitsus over kanchans (G・ウザク, 2019, pp. 30–33; ゆうせー, 2020d) are:

- By keeping two pairs, we keep the possibility of their ryanmen improvements and therefore the possibility of perfect 1-away
- If our pairs are outside, they come out more likely, especially for honor tile pairs
- Two pairs means that our waits becomes *atosuji*²⁸ twice likely

²⁸When our discards after riichi makes our wait suji, it is called *atosuji*. The part *ato* means “behind, after” in Japanese.

- Pairs do not use neighboring tiles, thus making them no-chance²⁹ more likely

In terms of ryanmen improvement and complex taatsu improvement, an outside kanchan and one pair are weaker than two pairs of 2 (or 8). When we have two pairs, we keep ryanmen improvement and complex taatsu improvement of each pair. On the contrary, when we only have a pair and a kanchan, we must keep the pair as our head, so we lose its improvement; our only improvement becomes the improvement of kanchan. Table 49 lists acceptance and all improvement of 24p22s (a kanchan and a pair) and 2288s (pairs of 2 and 8.)

Shape	24p22s	2288s
Acceptance	3p 4 tiles	28s 4 tiles
Ryanmen improvement	5p 4 tiles	37s 8 tiles
Ryankanchan improvement	6p 4 tiles	None
Kanchan-toitsu improvement	None	46s 8 tiles

Table 49: Comparing 24p22s and 2288s.

Even for pairs like 1199s which lack any ryanmen improvement, due to them being the outmost number tiles and twice likely becoming atosuji than a kanchan, they are decent as final waits. In this regard, the only cases where we would keep kanchan is when we have yakus such as tanyao or sanshoku involved, or when table situation makes pairs less favorable than kanchan.

When we decide to reduce our hand to 5 blocks, keep safe tiles over the block to drop (お知らせ, 2020, pp. 152–153). At the moment of drawing 6z in (144), we have 6 blocks. We will not touch yakuhai pairs that give us value and speed; in that regard, we have no incentive to keep 68s. Instead of keeping 68s and maintaining 6 blocks until we complete another block, we must drop it immediately to make room for 1z safe tile. Drop 6s.

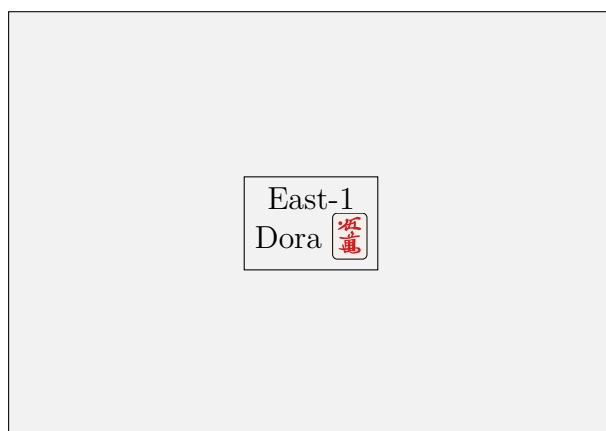
(144) South-2, East seat, turn 4, dora 6p
456m 2245p 68s 1226 -6z

When we are comparing ryanmens and kanchan, when they have the same number of acceptance left, **keep kanchan** (福地誠, 2017, pp.148–149, p. 218). In Situation 1, a lot of manzu tiles has been cut, and both of 34m ryanmen and 79m kanchan in our hand accepts 4 tiles. Here, kanchan is better than ryanmen because,

- We can draw 6m and upgrade it into a ryanmen.
- Even if all copies of 6m are cut, kan 8m becomes no-chance and more likely to come out.
- Occasional increase in value due to the fu given by kanchan.

Even though 234 sanshoku seems plausible, with 3 copies of 2m already cut, and 34p ryanmen still uncertain which side of it is getting filled up, we opt to drop 34m here.

²⁹If there are 4 copies of , it makes  wait or  wait *no-chance* since the ryanmens waiting on  or  requires 1 copy of . It makes  kanchan also impossible. It, however, does not negate the possibility of  or  shanpon wait.



Situation 1

Comparing Similar Shapes

Even for shapes in the same class, since their acceptances are same (without table condition,) their differences lie in their improvement and shapes around them. In particular, we must also take into consideration how bad it backfires when we drop these taatsus. For each class of shapes, we have following merits (お知らせ, 2020, pp. 152–153; ゆうせー, 2022l):

1. (Kanchan, penchan) remaining number of acceptance
2. (Ryanmen, penchan) follow-up of acceptance
3. (Kanchan) follow-up of ryanmen-improvement, ryankanchan-improvement draws
4. (Toitsu) whether or not we can slide towards dora
5. (Toitsu) whether or not, when reduced to a floater, it has improvement

There are two penchans 89m and 12s in (145). To prepare for the moment we pon 5z, we must keep another pair 33m, so we cannot drop 3m. In this hand, after dropping 12s, its acceptance 3s can still be incorporated by its neighboring kanchan, which forms 357s ryankanchan. On the other hand, drawing 7m after dropping 89m is useless; it is just a floater with furiten risk. In other words, dropping 12s loses the least. Drop 12s (ゆうせー, 2022l).

(145) East-1, North seat, turn 3, 7z
33489m 057p 1257s 55z

(146) is also choosing between 89m and 12s. Since we have 55m pair, the acceptance 7m of penchan 89m gives us 557m kanchan-toitsu, and its kanchan-improvement draw 6m gives 556m ryanmen-toitsu. When there is no other information available to us, judging on how we can still incorporate accepted tiles and improvement draws, 89m becomes the weakest penchan. However, when 3s is once cut, or when we immediately draw 7m after dropping 9m, we still keep manzu and drop 12s (ゆうせー, 2022l).

(146) East-1, East seat, turn 2, dora 8s
5589m 123678p 1288s

(147) is choosing between 56m and 67p. With 123m shape, when we drop 56m and draw 4m, we

can still slide towards 4m and drop 1m. Drop 56m (ネマタ, 2014j).

(147) 12356m2267p23456s

(148) is choosing between 99p and 11s. With 11234s shape, when we draw another dora 4s, we can drop 1s and maintaining a head in our hand. Therefore, drop 9p (G・ウザク, 2016, p. 20).

(148) East-1, West seat, turn 7, dora 4s
45m 67899p 1123467s

1.5.3 Dealing with Multiple Pairs

We have discussed situations where we would destroy pairs or keep them in Section 1.1.1. However, the arguments in Section 1.1.1 are made in the context of 5 blocks. In the following section, we will look at pairs in the context of 6 block theory.

When there are three pairs, it is generally a good idea to destroy one (ネマタ, 2014d), **since keeping pair has less acceptance than keeping ryanmen or kanchan** (p. 34 G・ウザク, 2019). We value the strength of a pair according to the following merits (ネマタ, 2014d; G・ウザク, 2019, p. 169; G・ウザク, 2016, p. 8),

- The number of improvement draws
- Involvement in yakus
- Strength as final waits

There are three pairs in (149) and we must drop one. Considering that we can get 11355m shape with 3m draw, 11p becomes the most useless pair. Drop 1p.

(149) 1155m11p678p34789s

In the case of (150), we keep 22p for ittsu chance with 13p draw, and drop 11s for slight chance of tanyao.

(150) 77m 22456789p 1134s

Dropping either 1s or 2s from (151) gives us the same number of 18 tiles of acceptance. In the case when 78p is completed first, 1s7z wait is more likely to win than 2s7z. Therefore, drop 2s.

(151) 06778p 1122345s 77z

Keep pairs that connect to other shapes (福地誠, 2017, pp. 78–79). We are choosing between 7m8p5s from (152). It seems that, since we already have 3 copies of 7m, 77m pair has the least acceptance. However, when we draw 4m, we can turn 7m into anko and incorporate 4m into our hand as 46m kanchan, which is impossible if we drop 7m first. 55s pair does not only has 0s acceptance but also 1s improvement. In this regard, drop 8p pair that has the least improvement.

(152) Turn 7, dora 2z
67778m 2388p 23455s

Likewise, when choosing one from 9m3p6p to drop from (153), we drop 9m pair (福地誠, 2017, pp. 80–81).

(153) East round, non-dealer, dora 1p
0699m 12333668p 67s

2 Building Yaku

Most of Section 1 is about choosing the choice that gives us the most tiles of acceptance with good quality. In cases where choices lead to the same number of acceptance, we then start discussing the expected value of our choices. In this section, we are going to focus more on the value of our hand, and, to some extent, to defy tile efficiency and choose actions that will not give us the most acceptance.

In a ruleset with yakus such as dora, akadora, ippatsu, and uradora, players can easily reach mangan without putting too much thought in building their hand; all they have to do is to be a little bit luckier with how the wall is built. In a ruleset where these opportunistic yakus do not exist, or when we are in dire need of points, making rii-nomi hands seems equal to digging our own grave, and we certainly do not want to sacrifice value potential just to reach rii-nomi faster.

Mahjong is a game of building mangan. For 30 fu hands, we start at 1000; for each additional 1 han we have, our hand value doubles, until it reaches 7700 for 4 han (or 8000 if it rounds up in some rulesets.) Likewise for other hands with different fu numbers, our hand value doubles as han number goes up until it reaches mangan. Therefore, when we have 1 or 2 han confirmed in our hand, spending effort on building hand value becomes worthwhile.

When we already have 3 han or more, yakus are still important – it gives us the opportunity to dama or go open. Since the value of our hand is close to mangan cap, the benefit of riichi (potential ippatsu and uradora) becomes marginal, which then does not justify the decrease in win rate. Only riichi as last resort – no yaku, and must push.

For a concrete example of how much we can sacrifice our efficiency for value, see (154). If we drop 3p, as is prescribed by tile efficiency, we end up 3900 points. Unless we are in huge lead in south game, we can opt to drop 45p, and keep ryanmen ryanmen for 7700 (福地誠, 2017, pp. 98–99).

(154) Turn 8, dora 3p
40m 3345p 22334s -66'6z

2.1 Yakuhai

Yakuhais refer to a family of honor tiles that gives us 1 han when we have 3 copies of them. Yakuhais include,

- Sangenpai: any of haku 5z, hatsu 6z, or chun 7z.
- Bakaze: the same wind as the round. For east games, it would be 1z.
- Jikaze: the same wind as your own seat. For a player in South seat, it would be 2z.

Honor tiles that are not included in the previous categories are called *otakaze*.

When we have a pair of yakuhais in our hand, we can call pon when one copy of it is dropped and we have a yaku confirmed instantly. When to call pon or not will be left for the section of call judgement. In this section, we are going to deal with lone yakuhai tiles and compare it with other floaters.

To put it simply, **it depends**. While we always teach beginners to drop honor tiles first, there are certainly situations where we would keep honor tiles and drop 19 or even 28 floaters first. Efficiency wise, honor tiles can only accept same kinds of tiles, so their acceptance always is less than a floating 1 or 9. However, the value and potential increase in speed it brings us when it pairs up makes it more versatile than a floating 1 or 9, whose job is to either provide a head or to create a penchan or kanchan.

The assessment of pure tile efficiency and need of speed depends on table situation, hand value, hand shape, and of course the person making the assessment. Different people and AIs have their own rules and preferences. The list below shows different kinds of preferences from 6 sources.

- 千羽黒乃 (2020)
 - Hand is in good shape (one pair and two good shape blocks), 19 floater > otakaze > yakuhai
 - Otherwise, yakuhai > otakaze > 19 floaters
- Maru-Jan チャンネル (2014)
 - Hand close to winning, or hand we want to win: 19 floaters > otakaze > yakuhai
 - Hand that does not seem close to winning: otakaze > yakuhai > 28 floaters > 19 floaters
- NAGA's preference according to 美咲 (2020)
 - Pinfu hand: otakaze > yakuhai
 - Otherwise: yakuhai > 19 floaters > otakaze
- 渋川難波 (2023b)
 - The 1 in 1345 > yakuhai > otakaze
 - Otakaze will be your first drop most of the time
- ヨーテル (2021b)
 - One dora: 19 floaters > yakuhai > otakaze
 - No dora: yakuhai > 19 floaters > otakaze
- 稚児 (2023)
 - For headless hands, drop yakuhais first.

To sum these up, it would be three kinds of orders, depending on our hand:

1. It is slow to remain closed, or it would very likely end up valueless rii-nomi: yakuhai > otakaze > 19 floaters
2. Pinfu or tanyao hands, or headless hands: 19 floaters > otakaze > yakuhai

3. Otherwise, hand is not too good nor too bad: 19 floaters > yakuhai > otakaze

Comparing Yakuhais and Floaters

A more detailed comparison between yakuhais and 19 and 28 floaters is as follows (ゆ う せー, 2024a; ゆ う せー, 2023a; ZERO, 2020, pp. 14–15),

- Keep yakuhais over 19 floaters when
 - Hand is too far away from tenpai when it stays closed (the number of ryanmens and mentsus are no more than 2.)
 - The floater is suji to another floater, such as 14p.
 - We don't have enough blocks and we have a pair.
 - We have 5 blocks but one of those is bad shape.
 - Pinfu does not seem doable.
 - Floaters are not related to dora.
- Keep yakuhais over 28 floaters when
 - There are more than 2 bad shapes and no completed mentsu.
 - Terminal tiles are dora.

For hands such as (155) which only has two ryanmens, it would take long to complete the kanchans or to improve them, not to mention 9m1s still could give us penchans; by the time they are filled up, our opponents already tsumo'd their riichi. Keeping yakuhais here not only potentially speeds up when they pair up, but they also are generally safer than number tiles. Therefore, we opt to drop either of 9m1s here (ゆ う せー, 2024a).

(155) East-2, South seat, turn 1, dora 8s
1249m 078p 156s 1557z

Hand (156) has only 4 blocks. Efficiency wise, we could use 789p acceptance of 9p floater. However, drawing a taatsu around 9p means that 0s becomes surplus, and we end up dora 1 kanchan/penchan riichi of 2600 if we cannot improve the taatsu in time. Value wise, 9p is the most useless tile; we would like to see draws near 0s, or drawing another copy of 56z to give us good wait 2000 (ゆ う せー, 2024a).

(156) East-1, East seat, turn 3, dora 5m
45677m 239p 0777s 56z

Since there is only 1 ryanmen and 1 mentsu in (157), it meets the criteria of keeping yakuhai over terminal floaters and the choices are 1p and 1s. When we drop 1s, its acceptance 2s cannot be picked up by its neighboring taatsu 45s. If we drop 1p, 2p forms a ryankanchan with 46p and 3p is the ryanmen improvement of that kanchan. Therefore, we drop 1p which backfires the least (ゆ う せー, 2024a).

(157) East-1, South seat, turn 4, dora 7m
124789m 146p 145s 1z

For once cut yakuhais, we still keep them over terminal floaters when our hand meets the criteria.

Drop 1s from (158).

(158) South-1, East seat, dora 6s, 1z is once cut
34067m 1168p 16s 167z

For the same (159), if dora is 7s, NAGA chooses to drop 6z. Otherwise, it chooses to drop 9s.

(159) East-4, North seat
447m 45p 1109s 44467z

For hands such as (160) which has no mentsu or ryanmen or whatsoever, if we just goes by pure efficiency, we end up rii-nomi, possibly bad wait also. For hopeless hands like this, we can even drop the penchan taatsu and go back in shanten. Either we draw another copy of 167z, or we get rid of terminal tiles and honor tiles altogether and go for tanyao (ZERO, 2020, p. 15).

(160) South seat, turn 1, dora 8p
4688m 22p 35589s 167z

For hands like (161) where there is no completed mentsu or even ryanmens, we can even drop 2m2s (ゆ う せ ー, 2023a).

(161) East-4, West seat, turn 4, dora 1p
268m 455899p 279s 56z

When we have terminal tile doras in our hand, confirming yaku becomes first priority. Therefore, we drop 8s and rely on 56z in (162) (ゆ う せ ー, 2023a).

(162) East-1, West seat, turn 3, dora 1p
2367m 11699p 348s 56z

However, if our hand looks like (163) which has more ryanmens and completed blocks, going full efficiency is better than rely on a single yakuhai. Drop 5z (ZERO, 2020, p. 15).

(163) South seat, turn 1, dora 9s
18m 23667p 557999s 5z

Yakuhai Attack

Yakuhai attack refers to the strategy where we drop yakuhais first before they get to pair up in others hands, or when others hands are still in a state unready to call (ZERO, 2020, pp. 16–17). Dropping yakuhai first means that our hand is blessed with good shapes, or yakuhais contribute little to our hand. In other words,

- Our hand is full of shuntsus or taatsus and there are no pairs; calling pon on yakuhais does not speed up our hand.
- Our hand consists of only tanyao tiles if we drop yakuhai tiles.
- Our hand is “in good shape,” and to be in good shape, it has to

- Have more than 3 ryanmens or mentsus combined (ZERO, 2020, p. 17)
- Have 1 pair and two good shape blocks (including kanchan-toitsus) (千羽黒乃, 2020)

If our hand meets any of the aforementioned conditions, we can drop yakuhais before otakaze. Note that otakaze can serve as our head for pinfu. For example, for hands like (164) where tanyao is very likely, or for hands like (165) which is in good shape according to 千羽黒乃 (2020)³⁰, we can drop yakuhais first.

(164) East round, West seat, turn 1
33468m 22268p 08s 25z

(165) East-1, North seat, turn 1, dora 5m
1156m 23p 1779s 1347z

We already have two yakuhai pairs in (166), more than enough to end the game and take first place. It means that 7z does not contribute to our victory and, if we hold it for too long and drop it too late, we may end up letting others call it. Therefore we drop 7z here first despite an otakaze 3z in our hand (川村晃裕, 2017, pp. 181–182).

(166) All-last, non-dealer, haipai, dora 4z, 500 points to overtake first place
136m 78p 389s 223557z

Hand (167) is the moment we pon dora 5z. The lone yakuhai 7z again contributes very little to our hand, since its value is already mangan. At this rate, the demerit of letting others call and possibly allowing them to win before we could is greater than the merit of value or defense. Drop 7z, even though 9m is close to useless also (川村晃裕, 2017, pp. 183–184).

(167) East-1, non-dealer, turn 1, dora 5z
1369m 247p 389s 7z -55'5z

Lone Dora Yakuhai

If there are other yakus for our hand, drop lone dora yakuhai before anything else. Otherwise, do not drop it easily (ヨーテル, 2022a). As mentioned earlier, if our hand already has strength points such as tanyao or pinfu, we do not particularly need dora yakuhai to pair up, and dropping them early on reduces the probability of them getting called pon. Therefore, for hands like (168) with tanyao and dora and pinfu potential, we drop 5z first.

(168) Dora 5z
34m 468p 334056s 125z

For (169), however, it has no pinfu, no tanyao, and no dora whatsoever, so we keep 5z and other honor tiles. Drop any of 8p125s.

³⁰  would be the first to go according to ゆうせー (2024a): not enough blocks, having one pair. Some see hands like this doable, some see this far away from tenpai. This is why different folks have different preferences when it comes to yakuhai and terminal tiles.

(169) Dora 5z
24668m 8p 12578s 125z

For (170) which is mentanpin³¹ 1-away, we would still drop dora 5z first, even if we draw this scary thing this late.

(170) Turn 7, dora 5z
23467m 34588p 56s 3z -5z

If our hand already has mangan value, it is better to drop lone dora yakuhai. (171) already has honitsu aka and chun, which is 8000, and getting 5z paired up only gives us 4000 extra points. If we cannot get it paired up in time and someone calls riichi, it would be very hard to drop it against riichi. Keep it if you are falling behind and are in dire need of points.

(171) East-1, South seat, turn 7, dora 5z
11340899s 125z -7'77z

2.2 Tanyao

Dropping Terminal Block from 6 Blocks

In most cases, keeping 5 tanyao blocks and drop the non-tanyao one is correct(ネマタ, 2014e; ZERO, 2020, p. 22), even if it is not the choice that gives us the most acceptance. If we drop either 4s or 6s from (172) we will have maximal acceptance of 16 tiles. However, we will only get pinfu with 5s draw; any other tiles will only give us rii-nomi. If we drop 9s, we have tanyao confirmed, and we can open our hand. Drop 9s.

(172) 23467m 345p 446699s

Likewise, drop 9m from (173). Efficiency wise, we would drop 2p. However, with only one ryanmen in our hand, we could end up waiting with kanchan wait. Therefore, 9m is the best course of action here, giving us tanyao and the opportunity to get rid of kanchans by calling or having any of those tiles paired up.

(173) East-1, South seat, turn 3, dora 8s
35799m 2467p 24888s

Drop 9m from (174). Efficiency wise, 3m is correct, which loses acceptance of 2m dora and gives us only riichi pinfu. If we keep it and drop 9m instead, we have tanyao and possibly dora or sanshoku.

(174) East-1, South seat, turn 3, dora 2m
356789m 346788p 45s

Efficiency wise, dropping 7s from (175) gives us the maximal acceptance of 28 tiles and also the risk of ending up shanpon wait rii-nomi if we complete ryanmen first. We drop 2p here first, and when we draw any useful souzu tiles, we can drop 1p pair entirely (福地誠, 2017, pp. 68–69).

³¹Short for menzen (riichi), tanyao, and pinfu.

(175) Turn 4, dora 6z
56m 11256p 2246778s

We already have enough blocks in (176). Going full efficiency means that we end up riichi dora 1. Note that, in this hand, we have strong yonrenkei and a strong floater 7p, and its still early, so we can afford going back in shanten and go for tanyao. Drop 3z (福地誠, 2017, pp. 70–71).

(176) East round, North seat, turn 4, dora 1p
3406m 7p 2346688s 33z

When any of the following conditions is met, do not force tanyao (ZERO, 2020, p. 23),

- We have enough good shapes.
- Forcing tanyao increase the probability of ending up bad wait

If we drop 11m from (177), we most likely end up waiting on kan 3m. Ryanmen + ryanmen is sufficient for us to stay closed, so we drop 2s here and go full efficiency.

(177) East-1, South seat, turn 3, dora 8p
1156m 34678p 24666s

Likewise, dropping kanchan from (178) leaves us 1-away with good shapes, so we do not need to drop the only 99m pair and force tanyao and slow down our hand. We can either drop 4p or 8p; drop 4p if we are in the middle of a round for safety reasons, and drop 8p for possible iipeiko improvement with 3p draw.

(178) East-1, South seat, turn 3, dora 8s
4599m 234468p 3478s

Likewise, for (179), we go full efficiency and drop the entire ryankanchan. Although this hand looks like (173), since (173) is too far away from menzen tenpai, we had to force tanyao. For cases like (179), we are closer to menzen tenpai with 3 ryanmen blocks and not in need to force tanyao.

(179) East-1, South seat, turn 3, dora 8s
35799m 2367p 23888s

Likewise, forcing tanyao in (180) by dropping 13s leaves us pairs and kanchan and makes us lose the acceptance of dora 5s. Efficiency wise, keeping two pairs is better than a kanchan and a pair in terms of possible ryanmen improvement draws, so we drop 68m here (福地誠, 2017, pp. 32–33).

(180) Turn 5, dora 5s
23468m 0577p 12334s

Forcing Tanyao when Agaritoppu

If we are agaritoppu, we must win our hand as soon as possible, and tanyao becomes our most reliable tool. For abysmal hands like (181), we are forced to call, and 89m penchan becomes

completely useless that we can drop 9m first. After that, if we draw any of 678m, drop 9p. We can also call any of 368p257s and drop 9p (in the case of 6p chii, we cannot drop 9p; drop 8p instead.) (川村晃裕, 2017, pp. 11–12).

- (181) All-last, West seat, turn 1, dora 3z, 500 points to overtake first place
489m 247889p 22468s

Hand (182) is hopeless, and we still want to get that first place. As always, tanyao is our most reliable tool; yakuhai is possible too. In this regard, since we already have 4 tanyao blocks, the 4z pair is the most useless. Drop 4z. Same for (183) which has two ryanmens (川村晃裕, 2017, pp. 27–30).

- (182) All-last, South seat, turn 1, dora 3z, 500 points to overtake first place
2488m 358p 46s 44567z

- (183) All-last, South seat, turn 1, dora 3z, 500 points to overtake first place
3488m 568p 46s 44567z

For (184) where 234 sanshoku seems viable, 44z becomes useful, 8p567z less so. Still, we keep 8p and drop 567z first since 8p still accepts 678p 11 tiles for tanyao. If we can't draw around 8p, pon 4z and force sanshoku instead (川村晃裕, 2017, pp. 29–30).

- (184) All-last, South seat, turn 1, dora 3z, 500 points to overtake first place
2488m 348p 24s 44567z

Hand (185) has two penchans and no completed blocks, making the 500 point difference far far away from it seems. Still, our dim hope lies in tanyao. In that regard, both 1m9p are useless. Drop 1m first, since, even if we draw 3m and 23m ryanmen is a furiten taatsu, we can still chi 4m to get rid of this taatsu. Dropping 9p makes 78p ryanmen furiten and it would be very hard to get rid of the furiten taatsu while also incorporating 7p. Do not immediately drop 2m after 1m; 2m draw which gives us pairs backfires a lot (川村晃裕, 2017, pp. 53–54).

- (185) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
12578m 24688s 2489p

Forcing Tanyao when Menzen is Also Close

Hand (186) looks far better than previous hands and menzen tenpai is within reach. Even so, opening our hand with tanyao is still faster. In particular, in (186), there are 2s8p pairs and **2(8) comes out fairly easily**. Thus, drop 1m here and force tanyao (川村晃裕, 2017, pp. 41–42).

- (186) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
123457m 56688p 224s

If our hand is better such as (187) which is perfect 1-away if we drop 7m, dropping 7m is fine for very early turns.

(187) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
123457m 22678p 667s

Hand (188) looks like we can force tanyao by dropping 1p pair, but doing so will lose 178p 8 tiles of acceptance. Notice that, 5s drop not only give us the widest 1-away which accepts 36m1578p 20 tiles, 456 sanshoku is also possible with 6m5p chi and then 1p drop. Therefore, drop 5s here (川村晃裕, 2017, pp. 43–44).

(188) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
45m 11234688p 4556s

Likewise, for (189), 1p drop renders 6m5s useless, and it actually increases our shanten, unlike 1p drop from (188). Widest 1-away we can get from (189) is 16 tiles. In that case, it is better to drop 6m and maintain 1-away with 12 tiles of acceptance, and after we draw or chi any of 3467s, we can opt to drop 1p entirely. This is similar to (175) (川村晃裕, 2017, pp. 43–44).

(189) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
34688m 11234p 4556s

Hand (190) has 4 pairs, and chiitoe seems closer than normal hand, but in agaritoppu, tanyao should still be the first priority. Still, to make chiitoe our backup plan, 9m drop then becomes the best course of action. If we want to keep the possibilities of a normal hand and a chiitoe hand and drop 2m2s, our hand becomes way slower (川村晃裕, 2017, pp. 57–58).

(190) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
247889m 2277p 2488s

Hand (191) is mostly tanyao tiles and forcing tanyao by dropping 9p seems fine. However, the pairs of this hand are all middle tiles, so they would come out harder than usual. Thus, for (191), drop 5s and keep menzen possibility while also not overlooking 345 sanshoku chance (川村晃裕, 2017, pp. 59–60).

(191) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
355m 357789p 22455s

Hand (192) too has middle tile pairs, and even if we drop 9p, the acceptance of 778p overlaps with that of 45p. It is better to simply take the 1-away with 7p drop (川村晃裕, 2017, pp. 59–60).

(192) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
34577m 457789p 468s

If we change the hand a little bit and make it (193) which now has pairs of 2(8) tiles, we can force tanyao with it. Drop 9p (川村晃裕, 2017, pp. 59–60).

(193) All-last, turn 1, non-dealer, dora 3z, 500 points to overtake first place
34588m 347889p 468s

Keeping Tanyao as Plan B

Before turn 7 or until someone opens their hand, we can keep tanyao possibility in our hand instead of safe tile. This particularly works well for cases where we only have one tanyao bottleneck such as 23m ryanmen. Scoring tanyao not only increases our hand value, in cases where we are agaritoppu or we have dora 3 it would also increase our win rate by allowing us to open our hand (川村晃裕, 2017, pp. 155–158).

When we look at hands like (194), we should not only think that 2367m draws give us perfect 1-away, but also keep in mind that any of 345678p345678s gives us tanyao chance. This tanyao chance is particularly helpful when we are agaritoppu or we have dora 3. Suppose that we draw 3p and our hand becomes (195), we can chi 4p and drop 3m. Even with 3s draw as in (196), we can chi 4s and drop 3m.

(194) 2367m 23477p 067s 3z

(195) 2367m 233477p 067s

(196) 2367m 23477p 3067s

If our hand has no dora, the increase in value (1000 or 1300) brought by tanyao does not justify the loss in defense. Thus, for (197), we keep 3z and tsumogiri³² anything other than 2367m7p.

(197) Dora 4z
2367m 23477p 567s 3z

Likewise, hand (198) has a bottleneck 78s, so we keep 4s here in hope for tanyao and sanshoku. For hand (199), however, since its tanyao is already confirmed, and there is already dora 2, there is no need for sanshoku, and 4s becomes useless. Drop 3z in (198) and 4s in (199).

(198) Dora 5p
23467m 23455p 478s 3z

(199) Dora 5p
23467m 23455p 467s 3z

Previously we learned that, in the context of 6 blocks, drop 9m from the skipping pairs 7799m is usually better especially when there is tanyao chance. Even when we compare skipping pairs to floaters, we still must always keep tanyao in mind. For hands from (200) to (202), we are comparing 7799m to a floater 7s. Even though our hand already has 6 blocks and 7s does not contribute to acceptance, with this particular shape 123457s, we can opt to chi 6s and drop 1s, so it contributes to hand value and speed.

If the hand has 3900 or 3900 potential after we open it, drop 9m (川村晃裕, 2017, pp. 165–170). Hand (200) has dora 2 and is already 3900 if we open our hand, and drawing any

³² *Tsumogiri* refers to the action of drawing a tile and immediately discarding it. The word is a compound of *tsumo* “draw” and *giri*, derived from *kiru* “to cut.”

of the aka doras gives us 7700. Hand (201) only has dora 1, but its 45m ryanmen accepts dora 6m and 34p accepts aka dora 0p, so it still has 3900 potential. Hand (202), however, only has one dora and dora 3z is useless, so this hand has no 3900 potential and is thus not fit for calls, and we drop 7s instead.

(200) Dora 7m
457799m 34p 123457s

(201) Dora 6m
457799m 34p 123407s

(202) Dora 3z
457799m 34p 123407s

Hand (203) looks like pinfu hand. Except for 34568m4567p which give us tenpai or perfect 1-away or headless 1-away with anko, keep in mind that 23p draws allow us to pon. In flat situation, 23p is useless, but it is crucial in all-last agaritoppu situations (or dora 3) (川村晃裕, 2017, pp. 17–18).

(203) All-last, non-dealer, haipai, dora 4z, 500 points to overtake first place
4588m 12356p 234s 4z

Deciding between Tanyao and Dabuton

If menzen tenpai is close, drop dabuton (福地誠, 2017, pp. 138–139). If we drop 11z from (204), we still have enough good shapes for tenpai. Dabuton and dora 1 is 5800, but if we stay closed, with riichi tanyao dora 1 we score 7700, and with ippatsu, tsumo, or uradora we can reach 11600. Therefore, value wise, dropping 11z and go for tanyao is better.

(204) East round, East seat, turn 6, dora 3z
22456m 22240p 67s 11z

If we already opened our hand like (205), without opportunistic yakus such as ippatsu, the value of dabuton dora 1 becomes greater than tanyao dora 1. Even if we cannot reach tenpai in time and someone calls riichi, 1z is still generally safer than other tanyao tiles. We drop 3p first. After that, compromise on 2900 if the ryanmens get filled first, or drop 67s when we draw 6p (福地誠, 2017, pp. 143–144, p. 218).

(205) East round, East seat, turn 6, dora 3z
22456m 30p 67s 11z -2'22p

Deciding between Tanyao and Yakuhai

Anything other than ryanmen ryanmen or dora 3 means we keep yakuhai pair (川村晃裕, 2017, pp. 177–180). For hands from (206) to (208), we just called 7s chi on turn 7 to get rid of a kanchan, and we can either drop 7z for tanyao or drop any of the other taatsus for chun.

Some may be thinking that if 7z does not come out this late, *mochimochi*³³ is very likely the case, and we cannot win our hand without dropping 7z. Paradoxically, dropping 7z means that it will get picked by our opponent and make it harder to win our hands.

Therefore, there is little merit of dropping 7z. If it really is *mochimochi*, we are giving others advantage; if it is not, we lose defense when facing riichi. Only drop 7z pair if our taatsus are all ryanmens or we have dora 3 and we must get tanyao confirmed. Drop 6m from (206), 7z from (207), and 7z from (208).

(206) Turn 7, dora 3m
23468m 56p 22s 77z -7'68s

(207) Turn 7, dora 3m
23467m 56p 22s 77z -7'68s

(208) Turn 7, dora 2s
23468m 06p 22s 77z -7'68s

2.3 Pinfu

Forcing Pinfu when Our Hand Has No Dora

Force pinfu when our hand has no dora or other yakus (土井泰昭, 2014, pp. 200–201). Hands like (209) very likely end up rii-nomi with little value, the cheapest 1300. Efficiency wise, the 3z that we just drew is useless. If we drop 3z here, considering that unseen doras are more likely end up in others hands as the game progresses, most likely we will face riichis from others that have values higher than average³⁴. From the perspective of defense and hand value, we drop 3m here.

(209) East-1, West seat, turn 5, dora 8p
334789m 56p 11788s -3z

(210) is a similar hand, but we have 1 han from tanyao. With 1 han in our hand, we can become more aggressive and keep maximal acceptance. Drop 3z.

(210) East-1, West seat, turn 5, dora 8p
334678m 56p 22788s -3z

We must drop any of the ryanmens of (211). In a ruleset with akadoras, 2–5 or 5–8 waits are stronger than 3–6 or 4–7 waits, so we drop 45s here. If our only pair 9m is not dora, drop 4s first, so that when we luckily draw 0s, we can drop 9m instead. If 9m is dora, drop 5s first, so that when we luckily draw 0s, we can choose other ryanmens to drop instead (川村晃裕, 2017, pp. 13–14).

(211) 3499m 12367p 45789s

³³Derived from Japanese *motsu* “to possess,” it refers to a state where two opponents have two copies of one kind of yakuhai and thus they can never call their yakuhais.

³⁴If all 𞄎 are seen, it is another story.

Deciding between Pinfu and Yakuhai

If we have (212) in a flat situation, since it already has good shapes, we can drop 7z and go for menzen riichi with pinfu. If we are agaritoppu, pinfu itself becomes less important and being able to use 7z becomes more important. Our instinct is to drop any of the ryanmens and keep two pairs, but we should drop 2s pair instead – so as to not lose the chance to directly call riichi on any draws of 69m47p (川村晃裕, 2017, pp. 179–180).

(212) Dora 3m
12378m 56p 22678s 77z

2.4 Iipeiko

Since iipeiko worth only one han and is limited to menzen hands only, usually we do not go against efficiency and choose bad shapes to force iipeiko. Shapes such as 33445p only have a fifty-fifty chance of giving us iipeiko. Shapes such as 33455p guarantee us iipeiko when our hand is completed, but it is very hard to get this 35p kanchan filled since we already have one copy of its acceptance. Therefore, we keep ryanmens over 33455p, and we keep other kanchans over 33455p when one more 4p is dropped (ネマタ, 2014f). In this regard, we drop 79m from (213) (多井隆晴・郡道美玲, 2021, pp. 121–122).

(213) North seat, turn 6, dora 8p
77899m 11166p 2378s

When our hand has no dora or has dora 3, we can lean a little bit towards iipeiko chance when choosing which floater to keep. Hand (214) has three floaters 26m8p. In terms of acceptance, 2m is weaker since we already have 13m in our hand and it is less likely to draw them than to draw 79p. However, drawing 13m gives us iipeiko chance and makes our hand worth of fighting when it has no dora and allows us to dama when we have dora 3. Therefore, keep 2m when we have no dora or dora 3. When we have dora 1 or 2, we should just go straight for riichi and drop 2m for maximal acceptance (ZERO, 2020, p. 32).

(214) 1m is once cut
12236m 2348p 35799s

However, for hands like (215) which does not have its head confirmed, aiming iipeiko with 1223m becomes less feasible because, if we arrive at 1-away with ryanmens, we keep 22m and drop 13m to confirm our head. Drop 8p.

(215) 1m is once cut, dora 9s
12237m 2348p 23079s

Three Skipping Pairs

For shapes such as 113355p where we have three pairs, efficiency wise, dropping the middle tile gives us the most acceptance as is shown in Table 50. However, if we keep this 6-tile shape, we have iipeiko upon 24p draw. Therefore, the discussion of this shape will involve how much we want

Shape	Acceptance of 3m drop	Acceptance of 8s drop
113355m78s	1245m69s 20 tile	12345m 14 tile
113355m68s	1245m7s 16 tile	12345m 14 tile
113355m89s	1245m7s 16 tile	12345m 14 tile
113355m88s	1245m8s 14 tile	12345m 14 tile

Table 50: Acceptance of shapes resulting from one tile drop from various combinations of three skipping pairs and taatsus.

iipeiko even if that means sacrificing efficiency (ZERO, 2020, p. 35; ヨーテル, 2021c; G・ウザク, 2019, pp. 114–117).

For example, the drop candidates of (216) are 3m9p. Their merits are,

- Merits of 3m drop
 - 45m draws give us 1m9p shanpon wait which has high win rate
- Merits of 9p drop
 - Iipeiko chance on 24m draw
 - After done dropping this pair, we have space for safe tiles

If our only goal is to riichi, drop 3m and hope that our wait is 1m and/or 9p. If there are one or more 1m9p visible on the table which makes our wait near dead, or we are in a point situation which favors dama, then drop 9p and rely on iipeiko chance (ZERO, 2020, p. 35).

(216) 113355m 123 99p 567s

Hand (217) has its three skipping pairs closer to the middle. Unlike the possible 1m9p shanpon wait in (216), the best shanpon wait we can get from (217) is 2m1p and it already has lower win rate than 1m9p. In such cases, we lean towards dropping 1p (ヨーテル, 2021c).

(217) 224466m 11p 789s 123s

For hands like (218) which has tanyao confirmed, we can still either aim for iipeiko, which has theoretically the highest expected value, or drop 4p and hope for ryanmen improvement of 7s. Since we have tanyao confirmed, we can call and wait for improvements even after we reach tenpai. The only ryanmen improvement of 224466p is 7p, while 22466p77s improves on 7p6s (78s cannot accept 9s if we force tanyao.) Therefore, in situations such as agaritoppu where hand value is not to be maximized, drop 4p; otherwise drop 7s (G・ウザク, 2019, p. 117).

(218) 678m 224466p 23477s

2.5 Honitsu and Chinit-su

In the following cases, we ignore general tile efficiency and force honitsu,

- Acceptance is narrow as there are many bad shapes and thus far from menzen tenpai (渋川 難波, 2023c)

- Hand value is mangan (or near mangan) even after call (平澤元氣, 2021a)
- Hand is still cheap (below 2 han) if we go straight for riichi (平澤元氣, 2021a)
- We have two pairs of yakuhai, which we call *yakuyaku* (ZERO, 2020, pp. 96–97)
- We have one pair of yakuhai and the rest of our hand is abysmal (ZERO, 2020, pp. 97–98)

Honitsu can only use tiles from one suit and honor tiles, so, in terms of pure efficiency, forcing honitsu sacrifices efficiency. Generally, this loss in efficiency is justified by its value (2 han minimum, easily becomes mangan with dora or yakuhai) and its defense (honor tiles generally have lower deal-in rate.) When we have abysmal haipais, forcing honitsu is usually a good option, value wise and defense wise.

Forcing Honitsu in Case of Yakuyaku

Mahjong is a game of building mangan. If we have two yakuhai pairs, we can maximize our hand value and easily reach mangan by forcing honitsu (ZERO, 2020, pp. 96–97). The value of hand (219) is 1000 or 2000 if we just follow pure efficiency. If we force honitsu, it can be 3900 or 7700, way bigger than before. Drop 8s.

(219) East-1, North seat, turn 1, dora 3z
7m 12456p 228s 44557z

Hand (220) has a completed souzu block. If we are super early in a round like before turn 4 and yakuhai comes out, we can call pon and drop the entire souzu block to force honitsu (ZERO, 2020, p. 97). This is the extent we can deny pure efficiency and maximize hand value.

(220) East-1, North seat, haipai, dora 3z
12456p 123s 44557z

Hand (221) has 6 blocks. With 4 pairs in our hand, we can reach mangan with either honitsu or toitoi. In this regard, the weakest tiles are 56m35z. From there, we have two options (独歩他, 2017a, pp. 98–103),

- Drop 6m for 0m acceptance and maximal toitoi and honitsu chance.
- Drop 5z if we are okay with a 2000 hand. Drop 5z before 3z because 3z is jikaze.

(221) East-1, West seat, turn 1, dora 8p
56m 22p 2499s 113566z

Hand (222) is similar to (221), but its manzu part becomes a kanchan. In this case, 46m kanchan becomes even weaker and we can drop it without much doubt (独歩他, 2017a, pp. 98–103).

(222) East-1, West seat, turn 1, dora 8p
46m 22p 24699s 11566z

Think Twice When There is No Yakuhai Pair

When there are no yakuhai pairs in our hand, forcing honitsu will mostly give us 2000 or 3900 with one dora. This holds as well for cases where we have otakaze pairs. Therefore, we are less

incentivized to force honitsu for offense. For defense purposes, though, honitsu still remains an option.

We already have a head and iipeiko in (223). Having iipeiko already confirmed means that if we reach riichi this hand we already have 2 han. Taking the possibility of tanyao and pinfu into consideration, staying menzen has more value than forcing honitsu and opening our hand, especially when there are no yakuhai pairs and the only thing that could increase our value is aka dora. Still, there is a slim chance of chinitsu. Therefore, we drop 37z first and keep 1m (渋川難波, 2023c).

(223) South-1, South seat, turn 1, dora 5p, last place
144556688m 8p 26s 37z

Hand (224) has 5 lone honor tiles. Drawing tiles near floating number tiles is generally faster than drawing duplicate copies of honor tiles. In addition, there are no head confirmed in our hand; even if we draw duplicates of honor tiles, calling pon on them does not speed our hand up. Therefore, drop honor tiles; drop yakuhais first, in particular, since otakaze pairs can still contribute to pinfu (ゆうせー, 2020c).

(224) East-1, West seat, turn 1, dora 1m
48m 23457p 25s 12456z

Hand (225) and (226) are both hands with narrow acceptance and mostly consisting of bad shapes of non-dominant suit. While it is true that we do not have yakuhai pairs and increasing our hand value is difficult, defense wise the otakaze pairs provide us a lot of safety. Therefore, we force honitsu for defense purpose here (渋川難波, 2023c).

(225) East-3, East seat, turn 2, dora 2m, 4m is once cut
114478m 246s 44567z

(226) East-3, East seat, turn 2, dora 8s, 4z is once cut
35m 57p 123449s 2244z

Hand (227) has no yakuhai pair at all. Still, this is agaritoppu, and we must force some yaku to take first place. For this kind of messed up hand, honitsu becomes our hope, with chanta also slightly probably. Therefore, drop 67m, and prepare to call everything possible (川村晃裕, 2017, pp. 37–38).

(227) All-last, West seat, turn 1, dora 3z, 500 points to overtake first place
67m 13p 112279s 1445z

When There is Only One Yakuhai Pair

Having only one yakuhai pair means that we are in the middle ground between the sweetspot of honitsu value increase and menzen riichi. Deciding whether to force honitsu or not will largely depend on how good the rest of our hand is.

With one yakuhai pair and two lone yakuhai tiles, we are in a state referred to as “ready for yakuyaku.” In this state, we can tolerate dropping floating 1289 tiles of lesser suits (ZERO, 2020, p. 97). In this line of thought, we drop 8p from (228) and 8s from (229).

(228) East-1, North seat, turn 1, dora 2s
1126m 8p 56889s 1566z

(229) East-1, North seat, turn 1, dora 2m
34m 1455689p 8s 4556z

Likewise, we drop 28s from (230) and get prepared for pinzu honitsu (平澤元氣, 2021a).

(230) Dora 1p
347m 34479p 28s 1157z

On the contrary, we do not force honitsu for (231). We only have one lone yakuhai tile and it is difficult to make more blocks in manzu suit. Drop 6z (ZERO, 2020, p. 98).

(231) East-1, North seat, turn 1
35999m 379p 48s 5556z

Hand (232) is also not fit for honitsu. It is true that we have one yakuhai pair and two lone yakuhai tiles, but the shapes in lesser suits are all ryanmen. Therefore, drop 67z (ZERO, 2020, p. 98).

(232) East-1, North seat, turn 1, dora 1s
67m 23p 223056s 5567z

Hand (233) has no shapes in suits other than pinzu. Even though we have iipeiko, with 7z pair in our hand, its value is mostly 2600. If we force honitsu, we will have at least 3900, and drawing another copy of jikaze 4z increases our mangan chance. Duplicate copies of 3z can also allow us pivot to toitoi. Considering also the slim chance of 0m, we drop 7s first (独歩他, 2017a, pp. 104–109).

(233) East-1, North seat, turn 1, dora 6z
5m 22333447p 7s 3477z

Drop The Suit with Second Most Tiles First

Hand (234) is the moment when we called pon on the first drop of 4z from oya. All of 6m46p are useless since we are aiming for honitsu here. If we drop 6m here, soon we will start dropping pinzu tiles, making others realize that we are waiting on souzu tiles. Therefore, the trick here is to drop pinzu tiles first, and delay the timing when opponents figure out what the suit we are waiting on is (鈴木大介, 2020, pp. 118–119). If unfortunately we draw manzu tiles, just tsumogiri.

(234) East-1, North seat, turn 1, dora 2m
6m 46p 14467s 336z -444'z

Hand (235) is the moment right after we called pon on dabuton. In cases where there is no difference in the number of pinzu and souzu tiles. drop the outermost tiles to make our river look like we are just following pure efficiency and not forcing honitsu. Drop 8p (鈴木大介, 2020, pp. 120–121).

(235) East round, East seat, turn 2
1179m 68p 36s 447z -111'z

2.6 Yakuyaku

Actively seek ways to make yakuyaku mangan (ヨーテル, 2023g). As mentioned previously, yakuyaku gives us 2 hans that is the sweetspot where, for 2 additional hans, our value increases from 2000 to 7700, a whopping 5700 points of difference. Honitsu is the most common way to achieve mangan, but keep in mind that toitoi is also an option.

Hand (236) has two yakuhai pairs and 34s is our only ryanmen. If we keep 34s and follow pure efficiency, we will mostly end up 2000 (1000 if we do not get to pon one of the yakuhai pairs.) If our hand is slow anyway, it is better to force honitsu to maximize our value. We can drop any of 68p34s.

(236) East-1, South seat, turn 1, no dora
2579m 68p 34s 135577z

Hand (237) has four pairs and toitoi becomes more viable than honitsu. Middle tiles are harder to call than outside tiles, so we drop 3m here first.

(237) East-1, South seat, turn 1, no dora
223m 119p 28s 135577z

Cases Where We Have One Dora

With one dora in our hand, the increase in hand value now becomes limited. Suppose that we have all yakuhai poned, our hand value is already 3900 minimum, and forcing honitsu will only make it 8000, since its value is capped at mangan. Therefore, we do not force 2 han yakus, and for hands like (238) where we also have a ryanmen in lesser suits, it is perfectly fine to proceed with 3900. Drop 13z.

(238) East-1, South seat, turn 1, dora 2m
67m 1330p 46s 135577z

Likewise, we do not force honitsu in (239). However, there is also a great possibility of 123 sanshoku and chanta which are 1 han yakus that brings us mangan. Drop 4s.

(239) East-1, South seat, turn 1, dora 1m
137m 3799p 124s 5577z

Hand (240) indeed has one dora and two yakuhai pairs, but note that we are close to tenpai yet

none of our yakuhai pairs are ponned, which means that, when we reach tenpai, we either have one pair as our head, or wait on yakuhai shanpon, and our hand value is still 2000 minimum. Therefore, we force honitsu here and drop 67s.

(240) East-1, South seat, turn 7, dora 8p
13m 67s 5577z -6'0799'9m

Hand (241) likewise is a hand where we did not get to pon our yakuhai pairs in time. Drop 4s and force toitoi for mangan.

(241) East-1, South seat, turn 7, dora 8p
405s 25577z -1'11p888'm

Cases Where We Advance Yakuyaku Without Forcing Yakus

When our hand is slow, the increase in value from forcing 2 han yaku outweighs loss in speed, but this is not the case when our hand already has enough good shapes like (242). Drop 2z and aim for a fast 2000 to get to next round.

(242) East-1, South seat, turn 1, dora 1m
2467m 223p 56s 25577z

If we are in huge lead and just want to move to the next round, or we are agaritoppu, forcing honitsu or toitoi is also not a good option.

2.7 Sanshoku Doujun

Asada Tetsuya once said, “look for sanshoku³⁵ as soon as you see haipai.” In the early days of mahjong when riichi (in the sense that we declare our tenpai in the middle of a round), ippatsu, uradora and stuff didn’t exist, building your hand with yakus such as sanshoku was important. As riichi and uradora became widespread, the importance of sanshoku dropped, and it is not as popular nowadays. Some would even go so far as to say that sanshoku is a “luck” yaku.

Still, being a 2 han yaku (1 han if our hand is open) and a yaku that only uses 3 blocks, sanshoku is still a useful yaku. The following criteria will help us decide when to force sanshoku (ネマタ, 2014g):

- How easy it is to get sanshoku confirmed.
- Loss of acceptance in the case when it is hard to confirm and we have to force it.
- The merit of increase in value with 2 hans of sanshoku or just the yaku itself.

When our hand has no dora, building value to make it worth fighting becomes crucial. When our hand has a lot of dora but also far away from tenpai, having a yaku to back up our calling is also important. In these two cases, we tend to force sanshoku.

³⁵There is another yaku called *sanshoku doukou* which requires three triples with the same number, i.e., , , and . Since sanshoku doukou is far less common than sanshoku doujun, in most cases, *sanshoku* refers to sanshoku doujun.

If our hand is close to tenpai, or our head is not confirmed, sanshoku becomes harder to score. Hand (243) already has 7 tiles necessary for 123 sanshoku, but there is no confirmed head. The most likely outcome is drawing some good shape around 6778s yonrenkei and drop 1m to confirm 22m as our head. Going for pinfu is more efficient (ZERO, 2020, p. 67).

(243) South seat, turn 5, dora 2p
122m 12357p 236778s

Hand (244) too has two blocks completed. With our hand being headless, we cannot have all 123 sanshoku parts without destroying other blocks (ZERO, 2020, p. 67).

(244) South seat, turn 5, dora 8m
235678m 13567p 123s

Force Yaku to Avoid Rii-Nomi

Hand (245) and (246) would be perfect 1-away if we drop 7p. For (245), since our value is 2600 minimum (riichi tanyao,) we are at that sweetspot of increase in value brought by riichi. Therefore, we prioritize speed over value and take perfect 1-away. For (246), however, our hand value is 1300 minimum and increase in value brought by riichi is smaller. The benefit of 2 han from 567 sanshoku therefore justifies the loss of 16m acceptance after we drop 6m (クリアレイン, 2023b).

(245) Turn 6, no dora
22566m 457p 222567s

(246) Turn 6, no dora
11566m 457p 222567s

Dropping either 4m or 6s from (247) gives us the same number of acceptance. According to Table 3, dropping 6s and confirming ryanmen gives us the most good-wait acceptance. However, dropping 4m confirms 234 sanshoku. In terms of expected value, confirmed sanshoku is greater, so we drop 4m here (ZERO, 2020, p. 65).

(247) South seat, turn 3, dora 3z
244m 234p 234667s 44z

For the same tiles, if we have two doras like (248), we should just go for good waits and drop 6s (ZERO, 2020, p. 65).

(248) South seat, turn 3, 4z
244m 234p 234667s 44z

If our hand has only one dora, there is no big difference in terms of expected return. Adjust your judgement according to actual table situation.

Go Big or Go Home with Bad Haipai

Hand (249) is 4-away and has only one good shape. If our hand is messed up anyway, we can try to force yakus to increase value or move on with quick hands. From this hand, we can see yakuhai and tanyao. Note that with 5 tiles of 123 sanshoku already in our hand, sanshoku is also viable, especially when we do not have enough blocks. Therefore, drop the most useless tile 9s. Even though it is in a 899s block, notice that it is not part of any of the yakus mentioned previously (ZERO, 2020, p. 64).

(249) South seat, turn 1
1356m 124p 15899s 56z

Keep Seeds of Sanshoku Over Bad Blocks

Hand (250) has 6 blocks. According to 6 block theory, we drop floater 8s and take maximal acceptance of 22 tiles. However, dropping 2p and making lower³⁶ pinzu part one block only lose 2 tiles of acceptance, and we get to keep the “seed” of 789 sanshoku. Therefore dropping 2p is good (ZERO, 2020, p. 64). Another way to look at this is to treat 8s as a block, so we are still maintaining 6 blocks in our hand, and we are ready to drop 8s when we are closer to tenpai.

(250) South seat, turn 3, dora 9p
79m 2244789p 458s 33z

Likewise, in the case of (251), it would be very difficult to make two mentsus from 2446m, so we drop 2m and keep 7p with allows us to aim for 789 sanshoku (ネマタ, 2014g).

(251) 2446789m 7p 113578s

6 Block Theory Sanshoku Edition

If you have read 1.3, you will notice that dropping 8m from (252) will give us maximal acceptance as well as guaranteed pinfu. Notice that we also have 7 tiles required for 789 sanshoku as well, but since our shapes in upper parts of suits are ryanmens, we could easily end up not getting sanshoku, and keeping 78m78s makes us los 3 less tiles of acceptance than 8m drop. Therefore, we will keep 78m only when we have no dora and have to chase for value; if we have at least one dora, we should drop 8m (ネマタ, 2014g).

(252) 2334578m 789p 5578s

Hand (253) has 6 blocks $334m + 234p + 46p + 88p + 34s + 88s$. According to 6 block theory, we drop 4p from the weakest block and proceed with 5 blocks which gives us the maximal chance of perfect 1-away. Notice that we have 7 tiles required for 234 sanshoku. By dropping 3m early and confirming 34m, the chance of 234 sanshoku increases³⁷, and it becomes easier for others to drop sotogawa 2m (ZERO, 2020, p. 65).

³⁶The “upper” part of a suit refers to tiles with greater number and the “lower” part refers to tiles with smaller number.

³⁷Because with  shape, drawing  withh ruin this ryanmen. In case where there are no head in the rest of our hand, we must drop  and sacrifice this ryanmen.

(253) South seat, turn 4, dora 4z
334m 2344688p 3488s

Hand (254) has 6 blocks and 6 tiles required for 567 sanshoku. Unlike previous cases where we can reduce taatsus involved in sanshoku to simpler taatsus, we must either drop a sanshoku taatsu entirely or drop 4p which reduced our perfect 1-away chance. On top of that, we already have 0m in our hand, so there is no need to force sanshoku here. Drop nidouke 67p (ネマタ, 2014g).

(254) 06m 34467p 1112367s

Hand (255) looks like it has 567 sanshoku confirmed, but to score 567 sanshoku we have to keep three all kanchans, which requires incredible luck to get them filled. Therefore, unless we absolutely have to force some value out of this hand, follow pure efficiency and drop 5m (ネマタ, 2014g).

(255) 5788m 223457p 3357s

Choosing between Floaters

Hand (256) is at 1-away. Since we already have 789m789p, keeping 9s allows us to draw 78s and score sanshoku. However, compared to other floaters 3m3p, 9s has 8 less tiles of acceptance, and it cannot form a ryanmen. Therefore, we will only keep 9s when (1) we have no dora and must force value or (2) it is still early in a round. When we have at least 2 doras, or it is already in the middle of a round, simply go for maximal efficiency and drop 9s. If we only have one dora, either is fine (ヨーテル, 2022b).

(256) South seat
3789m 3789p 229s 333z

Hand (257) is further away from tenpai than (256). Likewise, keep 9s when we have no dora. When we have dora 3, we can also rely on sanshoku to give us yaku to open our hand and speed it up. In other cases, drop 9s (ヨーテル, 2022b).

(257) South seat
3789m 3379p 3559s 33z

Hand (258) has only 4 blocks and currently does not have any yaku confirmed. Dropping 1s means that, if we don't get 12p penchan filled up first, we will end up rii-nomi. For hands without any dora, drawing around 7s would not be desirable, so we drop it here and keep 1s and aim for 123 sanshoku (ネマタ, 2014g).

(258) 123567m 1255p 17s56z

Choosing between Seeds of Sanshoku

Sanshoku with a floater as its middle tile is weak (ZERO, 2020, pp. 65–66). Take (259) for example. 3p can be the seed of lower part sanshoku and 7p is dora, so we cannot drop either. Thus, it becomes a decision between 1m and 4m, which is equivalent to choosing between 123 and 234 sanshoku. As shown in Table 51, the greatest difference is 1p draw which, if we dropped

1m previously, gives us no yaku at all. Dropping 4m always gives us a yaku, so we drop 4m and aim for 123 sanshoku. We thus conclude that, with 3p floater, 234 sanshoku is weaker than 123 sanshoku.

(259) South seat, turn 5, dora 7p
123405m 37p 122334s

	Drop 1m	Drop 4m
1p draw	No yaku	Sanshoku
2p draw	Pinfu, (sanshoku)	Pinfu, (sanshoku)
4p draw	Pinfu, (sanshoku)	Pinfu

Table 51: Comparing 1m drop and 4m drop from (259). Sanshoku in parentheses means that it is not confirmed due to ryanmen waits.

Hand (260) and (261) are both tenpai if we drop the floater, which gives us dora 1 bad wait. We have learned previously that, ryanmens formed around 7m only gives 678 sanshoku chance, and any kanchan formed from it is undesirable. Just drop 7m from (260) and accept tenpai. On the other hand, 6m accepts 578m all of which give us yaku. Considering also possible 4p draw, we actually have 16 tiles of upgrade if we drop 1p, a reasonable trade-off at turn 3. Therefore, we deny this tenpai in (261). 678 sanshoku is weaker and does not justify for going back in shanten if the floater is the middle one 7m (ZERO, 2020, p. 66).

(260) South seat, turn 3, dora 1m
1237m 13678p 67778s

(261) South seat, turn 3, dora 1m
1236m 13678p 67778s

Ryoutenbin

Ryoutenbin is a Japanese term which generally translates to “sitting on the fence,” and it refers, in the context of mahjong, to a situation where, with 13 tiles, there are possibilities of two yakus that cannot coexist. Which yaku we eventually score depends on the subsequent draws. There are ryoutenbins of sanshoku and ittsu and also ryoutenbins of different sanshokus.

If we drop 4m from (262), with the shape 5678m67p67s, drawing 5p5s gives us 567 sanshoku chance, and drawing 8p8s gives us 678 sanshoku chance. This shape is therefore the ryoutenbin of 567 and 678 sanshoku. While it is true that dropping 67p gives us sanmenchan and maximal acceptance, in a flat situation, we can sacrifice 3 tiles of acceptance and take sanshoku ryoutenbin (Chiba, 2020).

(262) 45678m 3367p 34567s

Hand (263) becomes ryoutenbin of 123 and 234 sanshoku if we drop 3p. While it indeed is 4 tiles less of acceptance than 1s drop which gives us perfect 1-away, our sanshoku chance also increases,

which gives us way higher expected value than 14s drop³⁸. Drop 3p (クリアレイン, 2023b).

(263) Turn 6
23067m 23399p 1234s

Hand (264) also becomes ryoutenbin of 345 and 456 sanshoku if we drop 4p. In this case, we do not have pinfu, thus there is a higher chance of undesirable rii-nomi, which further justifies loss in 4 tiles of acceptance in exchange of increased sanshoku chance (ネマタ, 2014g).

(264) 3456m 445 999p 45 88s

Hand (265) also becomes ryoutenbin of 567 and 678 sanshoku if we drop any of 2m3p. However, since this is a headless hand with anko, dropping 2m3p actually loses 11 tiles of acceptance, a wide gap in acceptance that do not justify increased sanshoku chance. Considering that it is more likely for 8 of 678 sanshoku to come out and complete our hand, we drop 5s here (クリアレイン, 2023b).

(265) Turn 6
22267m 33367p 5678s

If we drop 3p from (266) for 567 and 678 sanshoku ryoutenbin, we lose 4p which gives us mentanpin dora 1 mangan tenpai. In previous cases where we would trade 4 tiles of acceptance for sanshoku chance, those 4 tiles do not contribute to value, but this 4p puts us from 3900 to 7700, an increase too great to lose. Therefore, drop 8m, since 567 sanshoku is more likelier than 678 sanshoku with 067p block (ネマタ, 2014g).

(266) 566678m 306778p 67s

2.8 Ikkituukan

Ikkituukann in Japanese means “to strike through in one breath,” and refers to a yaku we score when we have all nine tiles of a suit. It is often shortened to *ittsu*, or translated into English as *pure straight*. Ittsu has the same value as sanshoku, which is 2 han for menzen and 1 han for open hands. Since ittsu requires all nine tiles from a suit, there are only 3 possible patterns of ittsu (1–9 tiles in manzu, souzu, and pinzu), less than sanshoku (123, 234, 345, 456, 567, 678, 789 tiles), it is a less common yaku.

When Ittsu is Viable

We can go for ittsu if our hand has no dora, it is still far away from tenpai, and we have at least 5 tiles in a suit. In such case, we keep floaters in the dominant suit and drop other floaters, even if they have more acceptance. Take (267) for example where we are choosing one from 7m9p8s. Acceptance wise, 9p is the weakest, but this hand has no tanyao no pinfu no dora whatsoever. Thus, keep 7m for acceptance of 6m dora, and keep 9p for a slim chance of pinzu ittsu. Drop 8s which is the one that contributes the least value (ZERO, 2020, p. 68).

³⁸The difference in expected value justifies loss of acceptance up to dora 2. When we have dora 3, the difference in expected value is very small and we should go for maximal acceptance.

(267) Dora 6m
3457m 2345679p 118s

We must drop either of 1m1s from (268). Manzu ittsu is further than (268) with only 5 manzu tiles, but it is still slightly better than 1s which gives us nothing but the chance of a penchan or a kanchan. Likewise, drop 2p from (269) (ZERO, 2020, p. 69).

(268) Dora 5m
14578m 1136p 1899s 5z

(269) 267899m 2678p 1399s

When comparing 12m penchan and 13s kanchan in (270), efficiency wise, it is better to drop penchan and wait for ryanmen improvement of kanchan. However, dropping 12m means that we lose ittsu chance on 45m draws. Therefore, we drop 1s (ZERO, 2020, pp. 68–69).

(270) No dora
1267899m 678p 1399s

Dropping ryanmen in favor of penchan may sound counterintuitive, but for cases such as (271) where we have no dora and must force ittsu, two penchans that confirm ittsu is better than a penchan and a ryanmen that give us only rii-nomi. Therefore, drop 45s (ネマタ, 2014h). If our hand has 1 dora, follow pure efficiency and drop any of the penchans.

(271) No dora
1245689m 99p 45567s

Cases Where Ittsu is Not Viable

We rarely go for ittsu when we have 1 dora, or we can have tanyao confirm if we remove non-tanyao blocks, even if we have 7 tiles in a suit. While it is true that ittsu has more value than tanyao when stay closed, the shapes in the ittsu suit are fixed and cannot upgrade to ryanmens. If we have at least 1 han, this inflexibility becomes undesirable.

Hand (272) has 7 distinct manzu tiles, but dropping 9m will not make us lose the acceptance of 27m, and 7m allows us to drop 1m and confirm tanyao. It is thus the right choice to drop 9m. (ZERO, 2020, p. 69)。

(272) 1345689m 33468p 06s

Likewise, if there is 1 dora in (273), drop 9m and go straight forward. Only drop 246p ryankanchan when there is no dora (ZERO, 2020, p. 70).

(273) 1345689m 11246p 78s

Dropping 19m from (274) gives us tanyao, so we do not need ittsu here. Drop 19m (ZERO, 2020, p. 70).

(274) 1345679m 3557p 455s

Hand (275) too does not need ittsu if dropping 12m already gives us pinfu (ネマタ, 2014h).

(275) 1256789m 99p 45567s

Golden 1-Away

Golden 1-away is a subset of kuttsuki iishanten that is also a ryoutenbin of sanshoku and ittsu. Take (276) for example. If we drop 6s, 16p draws give us pinzu ittsu chance while 79s draws give us 789 sanshoku chance. Taking the iishanten with ittsu and sanshoku chances has higher expected value than 8m drop which gives us only pinfu, so we drop 9s here (Chiba, 2020).

(276) 8m 2345789p 556789s

Hand (277) too is golden 1-away. Golden 1-away does not always come in shapes like (276) where there is clearly a floating 8m. For the combination of yonrenkei and ryanmen-toitsu, (277) is also a kuttsuki iishanten.

(277) 234m 2346789p 344s

Hand (278) becomes golden 1-away if we drop 6s. Dropping 3m gives us kuttsuki iishanten with two yonrenkei shapes, whose 53 tiles of acceptance is undeniably the most, but its bad wait rii-nomi draws is also the most (渋川難波, 2024a). Since 344m ryanmen implies 345 sanshoku chance, and we need pinzu yonrenkei for ittsu, 6s becomes the most valueless tile.

(278) East-1, East seat, turn 5, dora 2z
344m 2345789p 3456s

Hand (279) becomes golden 1-away if we drop 6m. However, dropping 3p is not too bad either, since it still gives us a wide acceptance of mentanpin (unless with unfortunate 1m19s draws.) Therefore, drop 3p for the best chance of good wait tenpai (ネマタ, 2014h).

(279) 345688m 3p 2345678s

However, if the pair becomes 9m and tanyao is gone, drop 6m for value³⁹.

(280) 345699m 3p 2345678s

Choosing between Sanshoku and Ittsu

Since both sanshoku and ittsu uses 3 blocks, we cannot have both when we are dropping one block from a hand with 6 blocks. In most cases, **keeping sanshoku is better** (ネマタ, 2014h) since sanshoku can coexist with tanyao.

³⁹The result I got from simulator.

Hand (281) looks like 789 sanshoku or pinzu ittsu. However, it has 6 blocks, and we must drop one block from any of 78m, 23p, and 56p. To complete ittsu, we need exactly 14p, which is one more kind of tiles than 789 sanshoku and more difficult to score. Therefore, we drop ittsu here and drop 5p. If we draw 6s and drop 9s9p, we can even have 678 sanshoku and tanyao at best.

(281) East round, North seat, turn 7, dora 8s
3378m 2356789p 789s

Hand (282) already has pinzu ittsu confirmed. If we drop any of the ryanmens and have the other one filled before 79p does, we end up 5200. On the other hand, if we drop 9p, we have 123 and 234 sanshoku ryoutenbin, and 4m4s draws give us tanyao chance. Therefore, we do not go for ittsu here. Drop 9p.

(282) East round, West seat, turn 8, dora 3p
23m 12345679p 2377s

2.9 Toitoiho

Toitoiho, or *toittoi* for short, has 2 hans regardless of hand being open or not, making it highly compatible with other yakus such as yakuhai and easily giving us high value. However, since toittoi hands mostly consist of pairs and triplets, when our opponents call riichi, we have less kinds of tiles and thus less possible safe tiles (ヨーテル, 2023d). In this section, we will look at cases and see if it is worth sacrificing tile efficiency for value; for cases where others drop tiles and we are presented the option to open hand and go for toittoi, see Section 3.

From 1300 to 5200

If we drop 6m from (283), its value is just 1300. If we force toittoi, we have half the tenpai tiles but 4 times the value, a good trade-off. Drop 5m (ZERO, 2020, p. 57)。

(283) Dora 1z
566m 22288p 222s -666'z

Even if we have two open sets and we are more exposed to the danger of riichi, we can still force toittoi. Drop 5m from (284) (ZERO, 2020, p. 57).

(284) Dora 1z
566m 88p 222s -22'2p666'z

If we already have mangan with ryanmen tenpai, there is no need to force toittoi for haneman. If we halve our tenpai tiles, our value only becomes 1.5 times higher. Drop 6m (ZERO, 2020, p. 57).

(285) Dora 2p
566m 88p 222s -22'2p666'z

Hand (286) too has 4 pairs and we can go for toittoi. Drop 3s7p7m (ZERO, 2020, p. 57).

(286) Dora 8s
788m 799p 113s 33z -666'z

Hand (287) is the moment when we draw 2p. Seeing 123p already completed, people would drop 2p without thinking too much. Do keep in mind that, if we keep 2p, there is a chance for toitoi and 5200 points. If it is before turn 4, drop 3p and go get that 5200. If it is later than that, drop 2z, and keep toitoi route as our backup plan (ZERO, 2020, p. 58).

(287) East round, South seat, turn 3, dora 8s
8899m 123p 233z -2p -666'z

Hand (288) has still needs one more pair for toitoi. Efficiency wise, 3z would be the right choice, but we most likely end up 1300. The point here is to keep toitoi possible without sacrificing efficiency too much. In this regard, we cannot drop any of the pairs. 3z would be a good pon candidate when it pairs up or a good tile for defense. 7p then becomes the least useful tile; indeed, it accepts 8p, but dropping it also makes 9p to come out more easily. Therefore, we drop 7p first; if we draw another pair, we can go drop 45s (ZERO, 2020, p. 58).

(288) Dora 8s
223m 11799p 45s 3z -77'7z

Ryanmen 5200 is Sufficient

Ryanmen 5200 is better than bad wait 8000, which is a theory proposed by Fukuchi Makoto (ヨ一テル, 2023d). For hands like (289) which already has dora 2 and chun, and all the fu from terminal tile triplets making it 40 fu, we can just go take ryanmen wait 5200. Halving our wait tiles just for 2800 points increase is not worth. Drop 5s.

(289) 111m 05p 056s -999'm 77'7z

On the other hand, (290) has one less dora and only has 2600. It is perfectly fine to take toitoi for mangan. Drop 6s.

(290) 111m 05p 556s -88'8s 77'7z

Forcing Toitoi

When we do not have all 5 pairs necessary for toitoi, force it anyway if our hand meet any of the following two conditions (ZERO, 2020, p. 59):

- Forcing toitoi gives us 5200 or higher value
- There are 4 pairs and there is at most 1 middle tile pair among them

Hand (291) has 4 pairs and no ryanmen, so it would be very slow even if we follow pure efficiency, and we would end up 1000 from 7z pon. Therefore, we force toitoi with this hand, and drop tiles from the middle. We call pon on any tiles possible.

(291) Dora 6s
2288m 1146p 18s 5677z

Likewise, force toitoi with (292). To deliberately set up sotogawa⁴⁰, drop 7m first.

(292) Dora 6s
22799m 1179p 1s 5677z

We just called 9m with (293) at the premise of toitoi. From opponents' perspective, if someone calls 9m pon and drops 3p, our visible parts would look like manzu honitsu, which makes 1p to come out more likely. Therefore, after we call pon of a suit, we drop a tile from a different suit. If we call 1p pon first, we drop 7m instead.

(293) Dora 6s
1227m 1139p 677z -99'9m

Cases Where We Do Not Go for Toitoi

Even though (294) has 4 pairs, we do not force toitoi with this hand. The greatest hurdle of toitoi, in this case, is that we cannot use double dora 0m. Therefore, toitoi will only give us 2600. Forcing tanyao would be 3900 at the very least and would be a far better option. Since honor tiles are generally safer, we drop 1s pair first (福地誠, 2017, pp. 128–129).

(294) East round, South seat, turn 3, dora 5m
0m 45666p 112246s 44z

2.10 Chiitoitsu

Chiitoitsu, or *chiitoi* for short, is a yaku that was invented in the United States of America and later became a part of Riichi Mahjong ruleset. To get ready for chiitoitsu, our hand must have 6 pairs and 1 lone tile, which also means that chiitoitsu is not compatible with other yakus that consist of sequences and triplets. On top of that, chiitoi 1-away is usually just us waiting on 3 lone tiles which give us 9 tiles of acceptance at most. Except for rare cases where we can still keep both normal hand 1-away and chiitoi 1-away, we usually do not force chiitoi. Even when our hand has 3 or 4 pairs, we still try to grasp the slim chance of a normal hand.

From Three Pairs to Chiitoi

For hands like (295) which has 3 pairs and is pretty much hopeless, chiitoi can be our plan B. Chiitoi 3-away accepts 21 tiles, while dropping any of 3p5z and going for normal hands also accepts 15m8p456s3z 22 tiles. On top of that, going by full efficiency will most likely only leave us bad wait dora 1. Therefore, we will not touch pairs in this hand, and start dropping from the middle tile 3p (ZERO, 2020, pp. 82–83).

⁴⁰*Sotogawa* in Japanese means “outside,” and it refers to the outside of early discards in mahjong context. Sotogawa is generally considered safe because early discards are often not connected or related to other shapes in our hand, and cases where we drop  from  early are rare (but not impossible.) The act of deliberately dropping  from  early is called *sakigiri*.

(295) East round, North seat, turn 3, dora 9p
1146m 379p 3557s 335z

We draw 9p in the next turn and (295) becomes (296). Again, we do not force chiitai here; drop the weakest kanchan 46m (ZERO, 2020, p. 83).

(296) East round, North seat, turn 3, dora 9p
1146m 799p 3557s 335z

Forget about Chiitoitsu When Our Hand is Full of Good Shapes

For both (297) and (298), we *could* drop any of the tiles with a single copy and make our hand ryoutenbin of chiitoitsu 1-away and normal hand 2-away, but we *should not* do that with (297) which has 3 ryanmen-toitsus. With 7z anko, we can open our hand and proceed just fine; we are 2-away from normal hand indeed, but it is faster than chiitai. On the other hand, for (298) which has one kanchan-toitsu, dropping 1p loses 2p acceptance but gains 7m6p acceptance, so it is actually 2 more tiles of acceptance. Therefore, we confirm a ryanmen in (297) and drop 1p from (298) (ZERO, 2020, p. 83).

(297) East seat, turn 5, dora 2s
788m 334677p 05s 777z

(298) East seat, turn 5, dora 2s
788m 133677p 05s 777z

Hand (299) has neighboring pairs and our acceptance is less ideal, but we still exclude chiitai from the possibilities early in a round and drop 4m. When we start discarding second row discards, drop 78m and advance normal hand and chiitai in parallel (ZERO, 2020, p. 84).

(299) Dora 1p
334478m 22388s 777z

Hand (300) has a kanchan and is much less ideal than previous hands. No matter how early it is, drop 7m and advance normal hand and chiitai in parallel (ZERO, 2020, p. 84).

(300) Dora 1p
334479m 22388s 777z

As shown in Table 52, dropping either of 8p6s from (301) will give us the same number of 17 tiles of acceptance. If we drop 8p, the neighboring pairs 2233p is forced to serve as head candidates and 14p draws cannot make it mentsu + ryanmen. Still, 1p draw cancels tanyao out, so it doesn't hurt much when it backfires; the draws around 5667s would hurt more than that. Therefore, we drop 8p here; if we happen to reach chiitai tenpai, take it (G・ウザク, 2016, p. 18).

(301) East-1, East seat, turn 5, dora 4m
2233556678p 5667s

Drop	Acceptance
8p	2347p57s 17 tiles
6s	123457p 17 tiles

Table 52: Acceptance of (301).

Hand (302) has enticing dora dabuton. However, we are already at turn 9, and the later it is the harder it is for dora to come out. Thus, we would take chiitai as our plan B. Dropping either of 4m7p gives us ryoutenbin of normal hand of chiitoitsu, and the difference lies in the shapes after we call 1z pon, as shown in Table 53. If we keep 567p mentsu, it would be difficult for 3344p to function as ryanmens; conversely, and somewhat unexpectedly, dropping 67p in tandem leaves us 34p ryanmen + 55s pair + 344m ryanmen-toitsu. Therefore, drop 7p, which gives us chiitai possibility and the best acceptance after 1z pon (G・ウザク, 2016, p. 20).

(302) East-1, East seat, turn 9, dora 1z
344m 3344567p 55s 11z

44m 3344567p 55s -1'11z		344m 334456p 55s -1'11z	
Drop	Acceptance	Drop	Acceptance
4m	23458p5s 17 tiles	4m	25m235p5s 19 tiles
3p	4m2458p5s 17 tiles	3p	245m25p5s 19 tiles
4p	4m2358p5s 17 tiles	6p	245m25p5s 19 tiles
5s	4m23458p 17 tiles		
(a) With 3m already dropped.		(b) With 7p already dropped.	

Table 53: Shapes of (302) after 1z pon.

Setting Up First Row Discards

Efficiency wise, it should be 8m drop from (303). However, this hand is very likely rii-nomi, so we would like to have chiitai as our backup plan. We can drop either of 59s to keep acceptance in souzu suit, or we can drop 7s, and make 9s look like sotogawa, which increases our win rate (ZERO, 2020, pp. 84–85).

(303) South seat, turn 4, dora 1p
8899m 33445p 579s 33z

Likewise, drop 3m first from (304), even though both 36m are equally bad for a chiitai tanki wait.

(304) East seat, turn 4, dora 1s
136m 88p 6699s 22334z

Note that in both cases we are still at turn 4 where our discards are very likely considered sotogawa. If it is already second row discards, sotogawa trap becomes ineffective, and we can stop thinking about how our river looks.

Passive Chiitai

Chiitai hands, much like toitoi hands, lack defense. When we are in the middle of a round where it is likely that our opponents would call riichi next turn, the floater we keep in our hand would be more defense oriented. Therefore, while 2z is twice cut and gives us furiten if we end up waiting on it, we should still drop 6p so that, even if some one calls riichi, we can still drop safe tiles while maintaining chiitai 1-away (ZERO, 2020, p. 85).

(305) East seat, turn 10, dora 4m, 2z is twice cut and furiten
499m 67799p 11779s 2z

Choosing between Chiitai and Toitoi

According to ネマタ (2014i),

- The following conditions favor toitoi,
 - We have an anko.
 - Yakuhai or honitsu is feasible.
 - We have terminal pairs, or pairs close to terminals.
 - Agaritoppu in all last
- The following conditions favor chitai,
 - We have a pair which is already twice cut
 - We have no anko
 - We have no other yaku
 - We have a lone dora
 - Our pairs are middle tiles
 - We are in the lead and would like to avoid unnecessary risk

Hand (306) looks like 8s drop for 7z pon or 7z drop for pinfu. 8s drop would only leave us a 1000 hand, so unless we are agaritoppu, it would not be ideal. 7z drop is the menzen way, but it could also be slow while none of the menzen yakus such as pinfu and iipeiko are guaranteed. In this regard, the possibilities of chiitai and toitoi would give us maximal expected value.

For chitai and toitoi, 4m34p would be equally useless, but since we still have 7z pair, to not completely confine ourselves to chitai, consider the shapes after we actually call 7z pon, 34p would counterintuitively be the best drop here (ネマタ, 2014i).

(306) Dora 4z
33477889m 34p 88s 77z

Hand (307) has both conditions that favor toitoi and conditions that favor chitai. Considering that, in the case we end up chitai tanki, our win rate plummets, so we drop 8s and go for toitoi unless we are in the lead (ネマタ, 2014i).

(307) Dora 4z
33388m 1155p 8s 4566z

- Conditions that favor toitoi:

- We have an anko
- We have yakuhai pair 6z
- We only have one middle tile pair 55p
- Conditions that favor chitoi:
 - We have a lone dora

For hands with two anko and four pairs, our choice will be based on (1) if there is yakuhai pair and (2) how many ideal chitoi tanki wait tiles are left. If we drop 3m8s from (308), any draws other than 8m15p6z would give us instant chiitoi tenpai; on the other hand, if we drop 5p, we can either ponten⁴¹ or even go for suuanko. Since we already have 66z pair, toitoi would be the best course.

(308) Dora 4z
33388m 1155p 888s 66z

3 Call Judgement

Calls, melds, or *fuuro*, refer to *chi*, *pon*, and *kan*, the methods you claim a tile from other players discards. When we call and open our hands, the state of *menzen* expires, and menzen yakus such as riichi iipeiko and pinfu also become impossible, thus decreasing the expected value of our hand. We cannot touch the open part of our hand, so we will defend against riichi with fewer tiles as well. The merits of calling are thus speed and confirmation of yaku, as summarized in Table 54.

	Open	Menzen
Speed	○	×
Value	×	○
Defense	×	○

Table 54: Comparison between open hands and menzen hands (平澤元氣, 2019, p. 87).

Call judgement is based on the following factors:

- How fast we can reach tenpai by opening our hand
 - Does the hand “look like menzen”?
 - Can we get rid of bad shapes with calls?
 - How good and how far from tenpai is the result shape?
 - Are there any other fast opponents?
- How much defense there is after shortening our hand
- How much the value of our hand decreases (except for cases such as agaritoppu where value is relevant)
- Whether the open part would alarm other players (and whether they care)
- Turn; the later it is, the better it would be for us to open our hand for tenpai

It is very difficult for humans to perform real-time calculations and weigh these factors, and the decision process can be laden with preferences and biases and easily affected by many other things

⁴¹A portmanteau of pon and tenpai, meaning that we reach tenpai after we call pon and drop a tile.

even outside of mahjong table, so often different people have different call judgement. This section is compiled with material from various sources and put in a way as coherent as possible, but there still could be some disagreements between these sources. We could have some leeway in call judgement in very specific conditions, but the general guidelines should stay the same.

As a rule of thumb, if any of the following conditions are met, go ahead and make the call (平澤元氣, 2019, pp. 36–41, p. 45, pp. 58–61):

- The increase in value brought by riichi is limited. That is, there are no menzen yakus such as iipeiko and pinfu.
- The value of our hand is already mangan.
- We can get rid of bad shape or drained up ryanmens.

If all of the following conditions are met, do not call (平澤元氣, 2019, p. 48; Chiba, 2020, p. 216):

- Cheap; if we open our hand, its value is no more than 2000.
- Far away; there is no head, or it has so many bad shapes that it is very likely we have to make 3 calls.
- There are no safe tiles.

3.1 Tanyao

When we open our hand with tanyao as its main yaku, it is called *kuitan*. Since kuitan is completely made of middle tiles, while it is very easy to score, it also comes with a risk of low defense. For players with menzen preference, the criteria for kuitan would be higher, and cheap kuitan are rarely forced except for situations such as agaritoppu. As a rule of thumb, the criteria for kuitan are (ヨーテル, 2024a; 独歩他, 2017a, pp. 170–175),

- Ponten or chiten for hands worth 1000 points
- Call all bad waits for 3900 hands; leave ryanmen calls for mid round.

Only Call for Tenpai for Hands Worth 1000 Points

Except for agaritoppu, regardless of the shapes of the rest of our hand, regardless of the callable part is bad shape or not, refrain from early cheap calls (ヨーテル, 2024a; 独歩他, 2017a, pp. 170–175; 平澤元氣, 2019, pp. 127–128). Therefore, do not open (309) or (310) or (311).

(309) East-1, non-dealer, turn 4, dora 7z
24m 3468p 3358899s

(310) West seat, turn 4, dora 1s
23467m 246p 35889s

(311) East-1, West seat, turn 5, dora 3z
367m 234499p 3377s

If we can get rid of bad shape and also get tenpai, it is fine to call. We can call 3m7p from (312) (ヨ一テル, 2024a).

(312) East-1, non-dealer, turn 4, dora 7z
24m 23468p 34588s 4z

Hands Worth 2000 are Tricky

Generally we can treat hands worth 2000 as if they are worth 1000 only (独歩他, 2017a, p. 179). That is, do not open unless opening our hand gives us tenpai. This 1000 points of gap is not too much of a difference.

For hands with one dora, we are at the sweetspot of reaching mangan with riichi, so we would still be inclined to stay menzen even at 1-away. According to 平澤元氣 (2019, p. 33, p. 38),

- If we have other menzen yakus, stay menzen until the end of second row discards.
- If there are no other menzen yakus and there is no much difference between menzen and open in terms of value, open our hand as soon as we enter second row discards.

Hand (313) is perfect 1-away and has pinfu chance which could make it 7700, or 5200 if there is no pinfu anyway. If we open our hand here, it will only be 2000 or 3900 if we draw any of the remaining aka doras. Thus we should not open this hand until about turn 12, unless we are agaritoppu, there are opponents who already openend their hands, or the ryanmens are drained up.

(313) East-1, West seat, dora 3z
405667m 22667p 34s

If our hand is ryanmen + kanchan 1-away, we can call kanchan part regardless of turn, and leave ryanmen part after about turn 9 (平澤元氣, 2019, p. 40). For (314), we can chi 4s at any turn, or chi 58p when its turn 9 or later.

(314) East-1, North seat, dora 3z
405667m 2267p 35s 2z

Hand (315) has two kanchans, and one may be tempted to call 7m7p chii. However, note that this hand has no head. If we make any call, we have 2 less kinds of head candidate, and it becomes more likely that we have to sacrifice ryanmens for heads. Do not open headless hands easily (平澤元氣, 2019, p. 55).

(315) East-1, West seat, dora 2z
3468m 3468p 067s 34z

Call All Bad Shapes for Hands Worth 3900

Call all bad shapes for hands worth 3900 (独歩他, 2017a, pp. 176–181). For hands like (316) which is laden with bad shapes or hands like (317) which seems close to menzen riichi, 4s is a must call. For (316), we can also call 35p chi. We can also pon 8s to make our hand mangan.

(316) West seat, turn 4, dora 8s
23468m 246p 35889s

(317) West seat, turn 4, dora 8s
23467m 334p 35889s

We can make calls to confirm tanyao even if it does not advance our hand in terms of shanten number (平澤元気, 2019, pp. 127–128). If we pon 7s with (318) and drop 9p, we are still 2-away, but our acceptance is wider with 3m4p floaters instead of 58m9p37s when we stay menzen.

(318) East 1, West seat, turn 5, dora 3s
367m 234499p 3377s

Pon Dora with Hands Already Worth 7700 if Your Opponents Would Not Mind

The merits of dora pon include increase in value as well as wider acceptance when our hand is laden with bad shapes. We have already seen cases such as (316) and (317) where dora pon increase our value from 3900 to 7700. For (316) specifically, if we deliberately get rid of the head, our acceptance becomes 68m246p35s, which is wider than 7m35p4s if we stay closed. For (317), 33p can still serve as our head, so we can pon dora 8s without slowing down our hand.

If our hand is already mangan (or 7700), this merit diminishes, and in a table where your opponents would try to sabotage your visibly expensive hands, the increase in value no longer justifies the decrease in win rate. For hands such as (319) which is already full of bad shapes, a wider acceptance is still favorable. For (320) though, we should only call 4s chi and save us unnecessary attention (独歩他, 2017a, pp. 188–195).

(319) West seat, turn 4, dora 8s
23468m 246p 3088s 4z

(320) West seat, turn 4, dora 8s
23467m 334p 3088s 4z

Calls that Save Us from Bad Wait and Dropping Dora

If we draw 25s into (321) first, we end up awkward tanki. If kamicha drops 4m, we can chi, drop 9p, and make our hand a flexible kuitan 1-away. Any of 4578p25s will give us good wait tenpai (ネマタ, 2016).

(321) Dora 4z
35m 067789p 34678s

If we stay closed with (322), 0s is very likely useless and we end up dora 1 bad wait riichi. Therefore, we should opt to chi 7p6s, drop 89s, and go for kuitan dora 2 (ネマタ, 2016).

(322) Dora 6p
2344588m 68p 0789s

Likewise, double dora 0p is very likely useless for (323) and we could end up dora 1 bad wait riichi. If any of 27m6s comes out, claim it, and drop the entire 1s anko. It may sound bizarre, but considering it could be tanyao dora 3 if we get to keep 0p, it should be worth going for (ネマタ, 2016).

(323) Dora 5p
2234077m 0p 11157s

Hand (324) is already tanyao dora 3, which we want to win no matter what, but is hindered by 12p penchan. Calls that confirm tanyao would help us tremendously even if they don't advance our shanten number. Thus, call any 2457m, and call 3p with 24p kanchan. 58s chi is also fine (平澤元氣, 2019, p. 61).

(324) East-1, West seat, dora 3s
3406m 12456p 3367s

Tanyao Calls in All-Last

When we are in all-last and there is only 500 points left for us to overtake first place or get out of last place, tanyao calls become immensely important. While it is true that each call sacrifices one turn of tsumo, our hand is no longer confined to menzen and is thus faster than menzen. There is no need to consider defense as well since it is better to fight for a slim chance for a better placement than to fold and completely give up the chance of winning. The following hands assume that we absolutely to win 500 points in all-last for a better placement unless otherwise stated.

Chi 6p with 57p and drop 9p from (325). Likewise, chi 36p with 45p and drop 9p from (326). Both hands are perfect 1-away, and we stay menzen in flat situations. In all-last and in dire need to win any hand, value becomes irrelevant and we'd rather get to fastest tenpai with any yaku than wait for menzen riichi which is slow and has lower win-rate. Thus, the point here is to find ways to get rid of non-tanyao tiles, or to slide completed blocks such as 789p for tanyao (マーク 2, 2023).

(325) All-last, dora 1s
34m 567789p 45588s

(326) All-last, dora 1s
34m 456789p 45588s

Keep in mind that dropping 9p after 6'78p chi is *kuikae* which is forbidden in most rulesets. The reason why (325) and (326) are special is because 789p is connected to other completed blocks. Do better see this, we consider shapes minus the terminal tile 9p. The shape 56778p can be either 567p + 78p or 57p + 678p. Likewise, 45678p can be 45p + 678p. However, if we chi 7p with 56p from 567789p and drop 9p, we are left a furiten ryanmen-toitsu instead of a completed block, so this is not a call that really advances our hand. Table 55 lists all possible shapes and calls to allow us pivot to tanyao.

There are indeed cases where we only have an isolated 789p block and must sacrifice one safe tile for tanyao like (327). Chi 6p with 78p and 3z (マーク 2, 2023).

Shape	Combination	Call
123345p	234p + 35p	4p chi
123456p	234p + 56p	47p chi
123444p22s	234p + 44p + 22s	4p2s pon
12456p	24p + 56p	347p

Table 55: Calls that allow us to swap blocks without committing *kuikae* (川村晃裕, 2017, p. 20). In all cases, we can drop 1p after.

(327) All-last, dora 1s
34678m 789p 4588s 3z

Always call shimocha's discards and be cautious about calling kamicha's discards (川村晃裕, 2017, p. 19–20). If we call shimocha's discards, we will only let shimocha draw one more turn. If we call kamicha's discards instead, only our turn of draw is skipped. In other words, calling shimocha's discards benefits our opponents the least. For hands like (328), we can absolutely call 8m2p from shimocha, as well as any other stuff that we'd usually hesitate to call for any other hands. This, of course, is not to say that we cannot call from kamicha; it is a thing to be kept in mind and taken into consideration all the time.

(328) All-last, dora 3z, 9s is twice cut
4588m 1223p 45678s

Chi 47m and pon 2m8p with (329) is a must, but also keep in mind that we can also chi 36p58s ryanmens. Some people may be wary of the possibility of ending up bad waits if we get rid of good shapes, but even then, the bad waits are 2(8) tiles, not too bad as final waits. On the contrary, the bad waits in (330) are middle tile pairs, so do not chi 25p58s ryanmens with (330) unless they are drained up (川村晃裕, 2017, pp. 21–22).

(329) All-last, dora 3z
1223456m 4588p 67s

(330) All-last, dora 3z
1234566m 3466p 67s

37m7p chi is a must for (331). However, besides those, 245s is also callable choices. To get two tanyao blocks out of 2234s, we can see it as 22s + 34s or 23s + 24s. In the former case, 25s chi becomes obvious, but what is better than 2s chi is 2s *pon* which leaves us a ryanmen that accepts 5 copies rather than a pair that only accepts 1 copy. If kamicha drops 5s instead, chi leaves us a 2s pair that still accepts 2 copies of 2s. In the case of 23s + 24s, the true bottleneck is 23s ryanmen which gives us furiten risk when we draw 1s, so 4s is a must and 3s is a no-go. Notice that manzu part is a sankanchan, but calling 5m chi destroys manzu part and leaves us two floating tiles (川村晃裕, 2017, pp. 55–56).

(331) All-last, dora 1p
2468m 68p 2234s 234z

3.2 Yakuhai

In most cases, call yakuhai pon. In any of the following cases, refrain from calling (平澤元氣, 2019, p. 103):

- Our yakuhai pair is the only head and calling it would slow us down (unless our hand is full of bad shapes.)
- Our hand is still cheap and far away after calling

Unlike kuitan, if we skip one yakuhai call, it would be very difficult for the last tile to come out again. To have our yaku confirmed, the criteria for yakuhai pon is generally lower than kuitan.

Yakuhai Pair is The Only Head

For hands like (332) which has only one pair 66z, it does not make our hand faster if we pon it. No matter how many times it is cut, do not pon 6z (平澤元氣, 2019, p. 104).

(332) East 1, West seat, dora 3m
3457m 67p 2346s 266z

For hands such as (333), even though it is close to riichi, its value is 2600 minimum, which is just 600 points higher than open hand. If it is mid round, we must pon 6z; calling pon in early turns would not hurt too much (平澤元氣, 2019, pp. 105–106).

(333) East-1, West seat, dora 3m
345m 667p 23467s 66z

Hand (334) has a bad shape 89s, so getting rid of the only head will also make 89s part of our acceptance. In cases like this where we have bad shapes in our hand, deliberately getting rid of the head actually speeds our hand up.

(334) East-1, West seat, turn 5, dora 4p
123m 23467p 89s 277z

The same principle applies to shapes after yakuhai pon. If we pon 2m with (335), our acceptance changes from 6m3p to 567m234p. This is thus a must call. If our hand has one ryanmen like (336), or if our hand has an anko like (337), calling pon is still better (平澤元氣, 2019, p. 199).

(335) East-1, West seat, dora 5z
12257m 24p 789s -5'55z

(336) East-1, West seat, dora 5z
12257m 45p 789s -5'55z

(337) East-1, West seat, dora 5z
12257m 45p 888s -5'55z

Hand (338) is already worth 7700 points. We should pon 7z first to have our yaku confirmed (独歩他, 2017a, pp. 164–167).

(338) East-1, West seat, turn 4, dora 8p
307m 4789p 207s 277z

Cheap and Far Away

If our hand is 3-away, is cheap, has no good shape, and has no pair other than yakuhai, refrain from calling (平澤元氣, 2019, p. 110). For hands like (339), 7z pon is a big no-go.

(339) East-1, West seat, turn 3, dora 7m
135m 4789p 268s 277z

Atozuke

Atozuke refers to the state where only a part of our waits give us a yaku. For example, (340) waits on 7s7z, but only 7z tsumo or ron allows us to win this hand. In a ruleset called *ariari*⁴², atozuke is allowed and all fine and dandy.

The issue would be how and why we approach atozuke. For example, is there anything we can call other than 7z in (341)? Do we call any of 35m2p from kamicha? Would the number of dora in our hand change how much we want to go for atozuke?

(340) 234m 340p 77s 77z -2'13p

(341) 246m 13340p 77s 577z

Some people may shun atozuke, afraid of the unstable state and attention from other opponents. However, know that even if your hand looks like anything but kuitan, your opponents do not always try to strangle you for that. The criteria for atozuke is as follows (独歩他, 2017a, pp. 146–151),

- Always call bad shapes (unless someone calls riichi first) in any of the following situations,
 - Agaritoppu
 - First place in south round
 - Hands with mangan value
- If we are not in the lead and our hand is not mangan either,
 - Call only when it gives us 2000 1-away
 - Call only when it gives us 3900 2-away

For hands like (342), its value does not justify loss in defense after 3m3p call except for agaritoppu.

(342) Turn 4, dora 7s
24779m 1267p 5s 477z

⁴²The first *ari* refers to kuitan, and the second *ari* refers to atozuke. Some rulesets ban both, and those rulesets are called *nashinashi*. By the way, there is no such thing as *arinashi* or *nashiari* which only bans one of kuitan and atozuke.

For hands like (343) which has 1 dora, opinions may vary from person to person. Since calling 37m2p only gives us 2000 2-away and thus does not meet the criteria shown previously, skipping those should be fine.

(343) Turn 4, dora 2m
24779m 2267p 5s 477z

If we have 2 doras like (344), calling any of 37m2p is fine.

(344) Turn 4, dora 2m
24779m 2206p 5s 477z

If opened parts such as 2'13s from (345) would alarm your opponents and make them hesitate to drop live dora 7z, 2s chi would not be wise. If your opponents do not care even a little bit, 2s chi is fine.

(345) East-1, West seat, turn 4, dora 7z
23899m 45p 136s 477z

Do not call 9p chi from (346). Even though we have dora 3 and we would like to move forward as fast as possible, 9p chi does not get rid of bad shapes, and it reduces the number of tiles that could give us another pair. Calling any of the kanchan is acceptable (福地誠, 2017, pp. 116–117).

(346) Non-dealer, turn 2, dora 6z
268m 3078p 1357s 66z

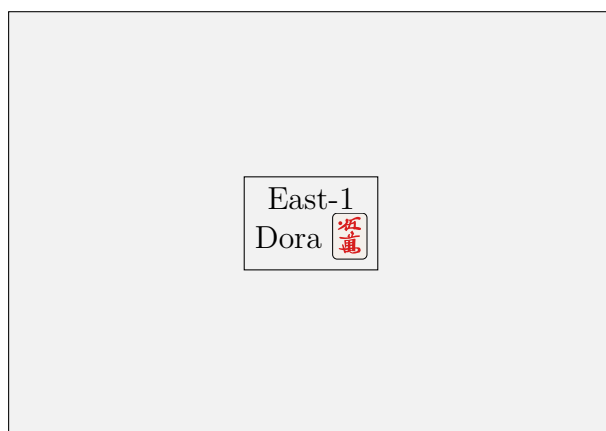
If we have 3 pairs other than yakuhai pair, we can call any of them. With 4 pairs in our hand, toitoi is also within reach; atozuke with yakuhai even if we do not get to score toitoi is also fine. Call 2p pon with (347) (ゆ ー せ ー, 2020e).

(347) East-2, turn 1, dora 4z
37m 22378p 1122s 55z

Situation 2 shows two copies of 1p already seen. Kamicha drops a tile; it could be 1p or 4p (or any other things.) If it is 4p, since 23p is not critical, we can skip it just fine, and 14p are likely useless for kamichi who will drop those later. However, if kamicha drops 1p instead, it is another story. If we choose to skip it, there is only one copy of 1p left somewhere in the wall, and our 23p ryanmen becomes almost as weak as a kan 4p. While it is true that a visible 1'23p may alarm others and make 7z less likely to come out, skipping 1p also weakens our waits. Likewise, we should call outside tiles such as 2 and 3 more compared to their inside suji counterparts (5 and 6) (川村晃裕, 2017, pp. 131–132).

3.3 Honitsu

Calling 1p pon with (348) is fine, while some people may choose not to. We already have 4 blocks necessary for honitsu. Another draw of 7z gives us huge increase in value. If someone calls riichi first, we can drop 5z and buy us two turns (福地誠, 2017, pp. 112–123, p 217).



Situation 2: Call kamicha's 1p, but do not call 4p

- (348) East round, East seat, turn 6, dora 3z
116m 1123468p 557z

Hand (349) is worse than (348). The dealer's first drop is 3z. We can either pon 3z as a bluff and force honitsu, or skip it and hope for *kyuushukyuuhai*⁴³ or at least closed honitsu chiitai. For rulesets which require more defense, skipping 3z pon is better (福地誠, 2017, pp. 124–125, p. 217).

- (349) Non-dealer, haipai, dora 2z
7m 14p 114s 1233467z

We can pretty much call anything with (350). The visible pinzu taatsus of this hand is 34p and 79p, so it is clear that we can call 258p. We can also call dora 3z. However, 3p chi is also a viable option in early turns. Even though 1334p is a little bit awkward, if kamicha drops 3p this early, it is very likely that 2p is useless to him and will drop it later. With 3 copies of 3p visible, it would also be difficult for other opponents to make use of 2p, making it a good wait (川村晃裕, 2017, pp. 121–123).

- (350) East-1, south seat, early turns, dora 3z
24m 13344579p 336z

3.4 Toitohi

If we are chiitai 1-away and also have a yakuhai pair, generally calling is better. The more 1289 pairs we have in our hand, the better is to call. Therefore, with (351) if someone drops 4p, we must call it and call any other pairs (独歩他, 2017a, pp. 204–211).

- (351) East-1, West seat, turn 4, dora 8p
1167m 2244p 788s 77z

If we force toitohi with (352), we will not be able to make use of dora 3m and our hand will only be 2600. Unless we want to confirm a yaku as soon as possible, do not open this hand.

⁴³If we have 9 or more kinds of terminal and honor tiles in our hand with our first draw, without any other calls interrupting us, we can declare *kyuushukyuuhai* and force abortive draw.

(352) East-1, West seat, turn 5, dora 3m
3m 3399p 11889s 224z

If we are given 4 pairs in the haipai (353) and oya drops 1m first, calling 1m pon is fine. If your opponents get alarmed easily, skipping pon is also fine. For some people, 1 yakuhai pair does not qualify as a toitoi hand (福地誠, 2017, pp. 132–133, p 217).

(353) Non-dealer, dora 6s
117m 1266p 224s 557z

Hand (354) has 4 pairs and one dora anko, but 3 of its pairs are middle tiles and hard to call. There are people who wants to call 9m pon on turn 1, and there are also people who wants chitoi even at turn 6 (福地誠, 2017, pp. 134–135, p. 217).

(354) East round, non-dealer, dora 2p
24499m 222577p 55s

3.5 Kan

Any of the following calls are collectively referred to as kan:

- Daiminkan: claiming others discards by opening an anko in our hand.
- Kakan: when we draw the 4th copy of a tile which we already pon'd. This is sometimes referred to as *shouminkan*.
- Ankan: when we draw the 4th copy of a tile which we already have an anko of.

An opened quadruplet is called *minkantsu* or *minkan* for short, and a closed quadruplet is called *ankantsu* or *ankan* for short.

The main benefits of kan is to increase hand value, either through the kan dora that is revealed after the kan call is completed, or through the additional fu from the quadruplet. For example, a terminal anko gives 8 fu, and a terminal ankan gives 32 tu. Another benefit of kan is an additional draw, which is why when we are tenpai with ryanmen kan is usually better. Kan dora also benefits our opponents, and it is three times likely that someone will get kandora than we getting it. Therefore, kan far away from tenpai is not wise and often advised against.

Daiminkan

When 3 out of the following 4 criteria is met, we can commit daiminkan (川村晃裕, 2017, p. 140):

1. We are in last place and in dire need of points
2. It is unlikely for us to stay menzen (for example, only one completed block and we'd call everything else)
3. By calling, we can get faster than others
4. At least 2 hans

The reason why 2 hans would be the sweetspot of kan is due to the value brought by the kan dora.

Lets say we only have a chun ankou and call daiminkan with it, with 1 kan dora our hand value jumps from 1300 to 2600 (or 2000 to 3900 for dealer.) Considering the value increase of others who likely have doras, this 1300 points of different is easily outnumbered.

For hands like (355), since we have 1 dora, in last place, and all other pairs are perfect for calls, we can commit 7z daiminkan (川村晃裕, 2017, pp. 139–140).

(355) South-3, dealer, early to mid turn, last place, dora 3z
11688m 99p 06s 6777z

Hand (356) has no dora, but we can force honitsu with the hand to make it at least 3 han, and any kan dora increase our hand value by a lot. Therefore, we can kan 7z. However, if any of 19p7z becomes new dora, we don't need to force honitsu anymore and we still have mangan, in which case just follow pure efficiency (川村晃裕, 2017, pp. 141–142).

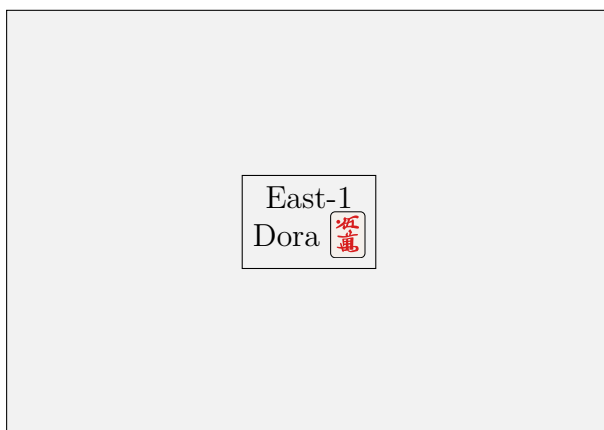
(356) Last place, dora 3z
34m 11246899p 777z

Hand (357) has 2 dora, but our hand is also close to menzen tenpai, so we should not kan 7z unless someone calls or it is mid turn. Even so, only kan shimocha's 7z and skip toimen and kamicha's draws. If we only have one aka dora, the value brought by riichi is bigger than kan, so do not kan at all (川村晃裕, 2017, pp. 141–142).

(357) Last place, dora 3z
11m 34078p 40s 4777z

Kakan

Calling kakan when we are in ryanmen tenpai is perfectly fine, but it would be awkward for tanki wait in Situation 3. Opinions vary (福地誠, 2017, pp. 198–199, p. 222).



Situation 3

Ankan

Call ankan when our hand is 1-away (多井隆晴・郡道美玲, 2021, pp.153–157). Hand (358) is still 2-away, so we don't call kan yet here; drop 5z and see how things goes. If it becomes (359) which is 1-away, we can kan 4p.

(358) South seat, turn 7, dora 7m
66778m 444478p 56s 5z

(359) South seat, turn 9, dora 7m
667789m 444478p 56s

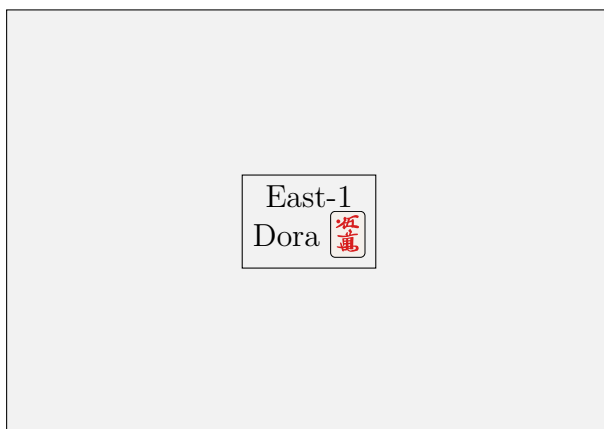
However, for hands like (360) which is also chitoi 1-away, kan makes chitoi impossible. Therefore, unless chitoi is completely out of the question, refrain from kan and maintain normal hand 1-away and chitoi 1-away (?, pp. 163–164).

(360) West seat, turn 9, dora 2s
11688m 2222p 44566s

3.6 Canceling Ippatsu and Shifting Haitei

The only effective ippatsu canceling is when someone can overtake us with haneman tsumo (平澤元気, 2019, p. 188). In other words, when we are in first place, have a lot of safe tiles, and second place who is 14000 below us calls riichi, we can go make any calls to cancel their ippatsu. In other cases, deliberately shortening our hand and giving up our tenpai chance would not worth it.

Always shift haitei (平澤元気, 2019, p. 190), especially when our hand is hopeless. However, shifting haitei only makes a difference of 100 points in terms of expected value (みーにん・梶本琢程, 2019, p. 188). In Situation 4 where we have enough genbutsu, if we drop 1m we won't be able to chi 3m, and dropping any of 27s makes us unable to call genbutsu from kamicha. Therefore, drop 2m here. Since 2 copies of 2m are already on the table, there is no way we can pon it.



Situation 4: When choosing a genbutsu to drop, choose the one that gives us the most chances to call and shift haitei. Drop 2m.

4 Bibliography

Chiba, Daina (2020) *Riichi Book I*.

chiitist (2021) 「2 + 6 枚複合形講座 â | 「にいいに型」「にいいにカン型」」, URL : <https://mahjong.nekoeigo.net/fukugo-7/> (アクセス日: 2024).

chirno (2008) 「牌効率について (リャンカントイツやサンカンチャン等)」, URL : <https://chirno.hatenadiary.org/entries/2008/05/18> (アクセス日: 2023 年 7 月 29 日).

chitist (2021) 「2 + 6 枚複合形講座」, URL : <https://mahjong.nekoeigo.net/category/fukugo/> (アクセス日: 2024).

David (2018a) “[Houou Lab] Looking at Tenhou ASAPIN’s Calling Part 2,” URL: <https://osamuko.com/hououlab-asapin2/>, accessed on 2024.

—— (2018b) “[Houou Lab] Looking at Tenhou ASAPIN’s Calling Part I,” URL: <https://osamuko.com/hououlab-asapin1/>, accessed on 2024.

Equip-chan (2023) “mjai-reviewer/faq.md at master · Equip-chan/mjai-reviewer,” URL: <https://github.com/Equip-chan/mjai-reviewer/blob/master/faq.md>, accessed on 2023.

Feng (2018) *Intermediate Riichi Strategy*.

G・ウザク (2016) 『麻雀 傑作「何切る」300 選』, 三オブックス.

—— (2019) 『ウザク式 麻雀学習 牌効率』, 三オブックス.

manboo (2013) 「イーシャンテンの牌理 (1) | 無料ネット麻雀戦術書」, URL : <http://beginners.biz/pairi/pairi13.html> (アクセス日: 2023 年 5 月 6 日).

Maru-Jan チャンネル (2014) 「土田浩翔が解説 | 土田のデジタル | 23. 不要牌の切り順」, URL : <https://www.youtube.com/watch?v=SQnf6AkDAuE> (アクセス日: 2024).

nisi (2014a) 「とりあえず麻雀研究始めてみました ノーチャンスワンチャンス」, URL : <http://epsilon69399.blog20.fc2.com/blog-entry-117.html> (アクセス日: 2023).

—— (2014b) 「真ん中無筋カンチャンリーチのみの是非」, URL : <http://epsilon69399.blog20.fc2.com/blog-entry-164.html> (アクセス日: 2023 年 5 月 6 日).

noprops (2022) 「麻雀 一択何切る」, URL : <https://play.google.com/store/apps/details?id=info.mygames888.nanikiru&hl=ja> (アクセス日: 2024).

tomo sann (2022) 「麻雀の攻守ともに重要!!〜一向聴 (イーシャンテン) の種類を考える〜 麻雀大好きともさんブログ」, URL : https://majann-daisuki.com/majan-ishanten/?utm_source=rss&utm_medium=rss&utm_campaign=majan-ishanten (アクセス日: 2023 年 5 月 6 日).

ZERO (2020) 『麻雀手役大全』, 竹書房.

お知らせ (2020) 『世界最強麻雀 AI Suphx の衝撃』, マイナビ出版.

- ずんだもんの麻雀何切るチャンネル (2023) 「3 ヘッド最弱理論を『使わない場面』、もちろん知った上で使っていますよね？【初級～中級向け問題】」, URL : <https://www.youtube.com/watch?v=oLfyQACRqcg> (アクセス日: 2024).
- とつげき東北 (2021) 「新 科学する麻雀」, URL : <https://hobbyjapan.co.jp/mahjong/data.html> (アクセス日: 2024).
- ひ いい (2005) 「2 - 0 4 牌 効 率」 , URL : <http://www.ix3.jp/hiii/02mahken/2-04paikouritsu.htm> (アクセス日: 2023 年 5 月 28 日).
- みーにん (2017) 『「統計学」のマーじゃん戦術』, 竹書房.
- みーにん・梶本琢程 (2019) 『知るだけで強くなる麻雀の 2 択』, 竹書房.
- ゆうせー (2020a) 「どのターツが不要かを【的確に】選ぶ基準～“配牌からの”牌効率 第 2 回～リーチを目指せ！麻雀初心者向け講座！」, URL : <https://www.youtube.com/watch?v=EWIZuxUXovo> (アクセス日: 2024).
- (2020b) 「孤立浮き牌とペンチャンターツ、どちらを切るか!? 比較&解説！【ライバルに差をつけよう！】～明日から使える麻雀戦術講義！～」, URL : <https://www.youtube.com/watch?v=Pjf0RhPbo4Y> (アクセス日: 2024).
- (2020c) 「実戦で使える手役紹介シリーズ!! 第 1 回【ホンイツ】 ～初級者から上級者まで！～」, URL : <https://www.youtube.com/watch?v=IErfBmtXMro> (アクセス日: 2024).
- (2020d) 「数牌トイツと愚形ターツの、明日から使える【法則】紹介！ “配牌からの”牌効率 第 3 回～リーチを目指せ！麻雀初心者向け講座！」, URL : <https://www.youtube.com/watch?v=VCrMiHnieTc> (アクセス日: 2024).
- (2020e) 「役牌バックを発進するかどうかの【わかりやすい】目安について！～明日から使える麻雀戦術講義！～」, URL : <https://www.youtube.com/watch?v=RHFHRnLSbwQ> (アクセス日: 2024).
- (2021a) 「24667 などの連続形の強さと【意外と知られていない】捌き方の極意」, URL : <https://www.youtube.com/watch?v=rm4kzJuW6B8> (アクセス日: 2024).
- (2021b) 「【#M リーグ】令和麻雀の必須事項！完全イーシャンテン先切りマニュアル～明日から使える麻雀戦術講義！22 回目～」, URL : <https://www.youtube.com/watch?v=fvKnLKqewpM> (アクセス日: 2023 年 11 月 4 日).
- (2021c) 「もう二度と迷わない！158 から一枚外すなら、1 か 8 かどっち?!～初心者から上級者まで！リーチを目指せ！ “配牌からの”牌効率講座！第 17 回～」, URL : <https://www.youtube.com/watch?v=jtPG0g6J2wM> (アクセス日: 2024).
- (2021d) 「データの衝撃があなたを襲う！今すぐに取り入れたい 1335 の捌き方徹底講座！明日から使える麻雀戦術講義！24 回目～」, URL : <https://www.youtube.com/watch?v=VAyF-DODGcg> (アクセス日: 2024).
- (2021e) 「初・中級者が間違いやすい何切るを解いて【重要戦術】を 3 つ身につけよう！～初心者から上級者まで！リーチを目指せ！ “配牌からの”牌効率講座！第 32 回～」, URL : <https://www.youtube.com/watch?v=8BdKtyk195o> (アクセス日: 2024).

- (2021f) 「初心者の大半が知らない 11355 をさばくコツを 10 分強でマスター！～初心者から上級者まで！～リーチを目指せ！ “配牌からの” 牌効率講座！第 14 回」, URL : <https://www.youtube.com/watch?v=cR4bJxaShX0> (アクセス日: 2023 年 7 月 29 日).
- (2021g) 「分かってスッキリ！皆が一度は悩んだことのある、アタマがないイーシャンテンでのなかぶくれのさばき方をわかりやすく解説！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 31 回」, URL : <https://www.youtube.com/watch?v=-BAh4W5xNW8> (アクセス日: 2024).
- (2022a) 「99%の初心者が間違える 2357 何切る全 4 問！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 41 回～」, URL : <https://www.youtube.com/watch?v=631L6QcUEAw> (アクセス日: 2024).
- (2022b) 「あまりにも簡単すぎてガチで驚きます。3568 マスター講座！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 46 回～」, URL : <https://www.youtube.com/watch?v=Gfksmav7wEs> (アクセス日: 2024).
- (2022c) 「こういうのを分からないままにするのが一番ダメです。34468 に 3 を引いたときは？～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 45 回～」, URL : <https://www.youtube.com/watch?v=Ad5mgfoGfJ0> (アクセス日: 2024).
- (2022d) 「まだ間違った牌を切ってるんですか...？112246 のさばき方徹底講座！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 47 回～」, URL : <https://www.youtube.com/watch?v=iEtFXGbAHE8> (アクセス日: 2023 年 5 月 6 日).
- (2022e) 「ガチで全問正解出来たら神。リャンカン + 四連形問題集！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 40 回～」, URL : <https://www.youtube.com/watch?v=Yo-R3w2tsNw> (アクセス日: 2024).
- (2022f) 「永久保存版 誰でも赤 578 がさばけるコツを説明！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 43 回～」, URL : <https://www.youtube.com/watch?v=5Lx03nDGk0w> (アクセス日: 2023 年 5 月 6 日).
- (2022g) 「覚えたら麻雀が変わる【ブロック使い分け】完全マスター講座！～明日から使える麻雀戦術講義！32 回目」, URL : <https://www.youtube.com/watch?v=oq2ntvWUI1M> (アクセス日: 2024).
- (2022h) 「苦節 20 年。やっと発見した 6889 の法則を紹介します。～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 48 回～」, URL : <https://www.youtube.com/watch?v=4jeri6yDIkA> (アクセス日: 2023 年 5 月 6 日).
- (2022i) 「今日から悩まずに切れる！35689 のさばき方徹底講座！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 35 回～」, URL : https://www.youtube.com/watch?v=_V000khUCw0 (アクセス日: 2024).
- (2022j) 「衝撃の結果が?! 一生使える基準付き！134 ドラ 1 は何を切る？～役牌ポン編～明日から使える麻雀戦術講義！36 回目」, URL : <https://www.youtube.com/watch?v=VxUirDseXPw> (アクセス日: 2023 年 5 月 6 日).
- (2022k) 「絶対、覚えた方がいい 1334 イーシャンテン攻略法！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 39 回～」, URL : <https://www.youtube.com/watch?v=160N1JZcuv4> (アクセス日: 2024).

- (2022l) 「多くの上級者が使いこなす「裏目フォロー打法」をマスターしよう！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 36 回～」, URL : <https://www.youtube.com/watch?v=cB7k0t0N4lY> (アクセス日: 2024).
- (2022m) 「誰も教えてくれない 6 ブロック打法の弊害とは？～明日から使える麻雀戦術講義！35 回目」, URL : <https://www.youtube.com/watch?v=Dn29q4DfHaE> (アクセス日: 2024).
- (2022n) 「土田浩翔は正しかった！極悪形 2468 の攻略法！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 42 回～」, URL : <https://www.youtube.com/watch?v=2chbGHCzpnY> (アクセス日: 2023 年 5 月 6 日).
- (2023a) 「こういうのを分からないままにするのが一番ダメです。孤立 28 と役牌を選ぶ基準！(手なり編) 初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 54 回～」, URL : <https://www.youtube.com/watch?v=yi6-g0ws6Zk> (アクセス日: 2024).
- (2023b) 「ほぼ 100%の初心者が間違える 2 赤 58 何切る 7 問&マスター講座！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 50 回～」, URL : <https://www.youtube.com/watch?v=A48Feg06SEs> (アクセス日: 2024).
- (2023c) 「知らないと絶対に損している 126 から 1 枚選ぶコツ～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 53 回～」, URL : <https://www.youtube.com/watch?v=Go3CHktsAV0> (アクセス日: 2024).
- (2024a) 「99%の初心者が間違えている 19 牌と役牌のガチで効果的な切り順マスター講座！初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 56 回」, URL : <https://www.youtube.com/watch?v=c2bGmxHWKH0> (アクセス日: 2024).
- (2024b) 「なぜ中級者のミスが多発しているのか、動画で説明します。13r5 のコツとは？！初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 57 回」, URL : <https://www.youtube.com/watch?v=0zLZJMZO1ro> (アクセス日: 2024).
- クリアレイン (2023a) 「【麻雀】あがり率が低い人の 9 割が軽視している「浮き牌」の残し方」, URL : <https://www.youtube.com/watch?v=juqw3F6FJ14> (アクセス日: 2024).
- (2023b) 「【麻雀】三色同順を狙っているのかダメなのかがわかる何切る問題」, URL : <https://www.youtube.com/watch?v=PuY0BSBAwsU> (アクセス日: 2024).
- (2023c) 「【麻雀】知らないで絶対にあがり率が下がる「2457」の扱い方【牌効率】」, URL : <https://www.youtube.com/watch?v=N6eK0Ht8WNo> (アクセス日: 2024).
- (2023d) 「【麻雀】知らないで絶対にあがり率が下がる「3456・4556」の扱い方【牌効率】」, URL : <https://www.youtube.com/watch?v=4QtG0mRrarE> (アクセス日: 2024).
- (2024) 「【麻雀】知らないで絶対に損する「34468」の扱い方【牌効率】」, URL : <https://www.youtube.com/watch?v=CqVrnEQStdg> (アクセス日: 2024).
- ツケマイせんせい (2014) 「シャボかカンチャン論争 | ツケマイせんせいの哲学する麻雀」, URL : <https://ameblo.jp/nyanyami2011/entry-11764202490.html> (アクセス日: 2023 年 5 月 25 日).

- ネマタ (2014a) 「現代麻雀技術論 - 押し引き論 1. 絞り」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter02/section001.html> (アクセス日: 2024).
- (2014b) 「現代麻雀技術論 - 牌効率論 10・外カンチャンターツ」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section010.html> (アクセス日: 2023 年 5 月 6 日).
- (2014c) 「現代麻雀技術論 - 牌効率論 11・内カンチャンターツ」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section011.html> (アクセス日: 2023 年 5 月 6 日).
- (2014d) 「現代麻雀技術論 - 牌効率論 13・トイツ落とし」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section013.html> (アクセス日: 2023 年 7 月 28 日).
- (2014e) 「現代麻雀技術論 - 牌効率論 26. 断ヤオ九」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section026.html> (アクセス日: 2024).
- (2014f) 「現代麻雀技術論 - 牌効率論 28. 一盃口」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section028.html> (アクセス日: 2024).
- (2014g) 「現代麻雀技術論 - 牌効率論 31. 三色同順」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section031.html> (アクセス日: 2024).
- (2014h) 「現代麻雀技術論 - 牌効率論 32. 一气通貫」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section032.html> (アクセス日: 2024).
- (2014i) 「現代麻雀技術論 - 牌効率論 33. 対々和」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section033.html> (アクセス日: 2024).
- (2014j) 「現代麻雀技術論 - 牌効率論 45・2 面子余剰牌形」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section045.html> (アクセス日: 2023 年 5 月 20 日).
- (2014k) 「現代麻雀技術論 - 牌効率論 47・2 面子余剰牌形か完全形の選択 1」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section047.html> (アクセス日: 2023 年 5 月 20 日).
- (2014l) 「現代麻雀技術論 - 牌効率論 56・2 シャンテン戻し」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section056.html> (アクセス日: 2023 年 10 月 28 日).
- (2014m) 「現代麻雀技術論 - 牌効率論 5・複合ターツ 1」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section005.html> (アクセス日: 2023 年 5 月 6 日).
- (2014n) 「現代麻雀技術論 - 牌効率論 6・複合ターツ 2」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section006.html> (アクセス日: 2023 年 5 月 6 日).
- (2014o) 「現代麻雀技術論 - 牌効率論 9・ペンチャンターツ」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section009.html> (アクセス日: 2023 年 5 月 6 日).

- (2015a) 「もっと勝つための現代麻雀技術論 第 12 回「2 メンツ形の選択」」, URL : <https://mj-news.net/column/method/2015101623923> (アクセス日: 2024).
- (2015b) 「もっと勝つための現代麻雀技術論 第 13 回「2 メンツ形の選択」」, URL : <https://mj-news.net/column/method/2015101724009> (アクセス日: 2024).
- (2015c) 「もっと勝つための現代麻雀技術論 第 16 回「面子候補固定と雀頭固定」」, URL : <https://mj-news.net/column/method/2015102024317> (アクセス日: 2024).
- (2016) 「もっと勝つための現代麻雀技術論 第 133 回「別の面子、雀頭、面子候補を崩す鳴き」」, URL : <https://mj-news.net/column/method/2016021434639> (アクセス日: 2024).
- マーク 2 (2023) 「和了率が 5」 (2024).
- (2024) 「一部の上級者だけが知っている、手が悪い時に役牌を絞らないという戦術【魂天&天鳳位の解説】【麻雀講座】」, URL : <https://www.youtube.com/watch?v=JnSZH6wzmUk> (アクセス日: 2024).
- ヨーテル (2021a) 「【麻雀】孤立 1・9 の切る順番を決めるたった一つの考え方【天鳳位】」, URL : <https://www.youtube.com/watch?v=Cr6jQJeJ2nw> (アクセス日: 2024).
- (2021b) 「【麻雀実況】役牌と孤立 9、結局どちらを切るべきなの？【雀魂/天鳳位】」, URL : <https://www.youtube.com/watch?v=az2VYGMkMvs&t=307> (アクセス日: 2024).
- (2021c) 「【麻雀牌効率講座】地味だけど頻出する飛び対子の扱い方【ヨーテル/天鳳位】」, URL : <https://www.youtube.com/watch?v=0sWOG84vVXs>.
- (2022a) 「【麻雀講座】もう迷わない！孤立ドラ役牌の扱い方【天鳳位】」, URL : <https://www.youtube.com/watch?v=HQphpEWog4s> (アクセス日: 2024).
- (2022b) 「【麻雀講座】令和最新版！三色の扱い方【天鳳位】」, URL : <https://www.youtube.com/watch?v=nu4uTz0WaPY> (アクセス日: 2024).
- (2022c) 「【麻雀牌効率講座】実戦で使える両面カンチャンの扱い方【天鳳位】」, URL : <https://www.youtube.com/watch?v=vUjvPzPb5r0> (アクセス日: 2024).
- (2023a) 「【麻雀講座】2 枚切れの愚形でリーチするか？【天鳳位】」, URL : <https://www.youtube.com/watch?v=Z7QmW6Rn9Eo> (アクセス日: 2024).
- (2023b) 「【麻雀講座】3 4 5 6 の 3 と 6、適当に切ると痛い目見ます【牌効率/天鳳位】」, URL : https://www.youtube.com/watch?v=6AA8f_vZ2pM (アクセス日: 2024).
- (2023c) 「【麻雀講座】5 ブロックと 6 ブロック、2 つの道の選択基準【天鳳位】」, URL : https://www.youtube.com/watch?v=F0_KVS1oMV4 (アクセス日: 2024).
- (2023d) 「【麻雀講座】トイトイは強い？弱い？癖のある役を攻略しよう【天鳳位】」, URL : <https://www.youtube.com/watch?v=vWdh5y2Dcuc> (アクセス日: 2024).
- (2023e) 「【麻雀講座】一瞬でカンチャンを選択するための判断基準を 3 つ紹介【天鳳位】」, URL : <https://www.youtube.com/watch?v=exCAr8K9ypc> (アクセス日: 2024).

- (2023f) 「【麻雀講座】下手に見られないために、上位互換を選ぼう【天鳳位】」, URL : https://www.youtube.com/watch?v=mPJNhh_8NcU (アクセス日: 2024).
- (2023g) 「【麻雀講座】配牌で役牌対子が2つあるときの進め方【天鳳位】」, URL : <https://www.youtube.com/watch?v=nVnd5xBs-po> (アクセス日: 2024).
- (2024a) 「【タコ鳴き確定】上級者が絶対にしない鳴き3選【天鳳位】」, URL : https://www.youtube.com/watch?v=Nu_eTkgVo5c (アクセス日: 2024).
- (2024b) 「【麻雀講座】134の1を雑に切ってはいけません【牌効率/天鳳位】」, URL : <https://www.youtube.com/watch?v=ai-nHr4Eg14> (アクセス日: 2024).
- 綾鷹 (2020) 「愚形リーチのみ」, URL : https://note.com/ayataka_tell/n/n277c956cf9de (アクセス日: 2023年5月8日).
- 因幡はねる (2023) 「【切り抜き】超絶クソ配牌！これで絶対切っちゃいけない牌はこれです！by 松本吉弘プロ #ぐみひゃく /空星きらめ【因幡はねる / ななしいんく】」, URL : <https://www.youtube.com/watch?v=plpoJAdh50U> (アクセス日: 2024).
- 筋トレプレーヤー (2024) 「麻雀の対リーチ放銃点の期待値に関する考察」, URL : <https://www.withdrawal-civilservice.com/entry/2024/03/01/203018> (アクセス日: 2024).
- 国税庁長官官房企画課 (2023) 「民間給与実態統計調査」.
- 渋谷難波 (2023a) 「【渋谷ハル】守備力を上げる！手出し・ツモ切りを覚えるより大事！ペンチャン・カンチャンなど愚形リーチへの放銃を減らす方法！【雀魂/Mリーグ/KADOKAWA サクラナイツ/渋谷難波切り抜き】」, URL : <https://www.youtube.com/watch?v=H6fR3tYGg0w> (アクセス日: 2024).
- (2023b) 「【渋谷ハル】序盤の1・9・字牌の細かい切り順「練習編」【雀魂/渋谷難波切り抜き・】」, URL : <https://www.youtube.com/watch?v=jkjG73Lc7DE> (アクセス日: 2024).
- (2023c) 「【渋谷ハル魂天計画】染める？染めない？混一色？リーチ？【雀魂/渋谷難波切り抜き】」, URL : https://www.youtube.com/watch?v=JzpS59a_--Q (アクセス日: 2024).
- (2023d) 「【雀聖計画番外編！】この配牌、メリで打つのハリで打つのそれとも様子見！？【教えて渋谷監督】」, URL : <https://www.youtube.com/watch?v=bu2xtQGuFn8> (アクセス日: 2024).
- (2024a) 「【ウザク式何切る？】四連形を残す？一気通貫・三色をみる？両面を固定する？【神域リーグ2024/因幡はねる/渋谷難波切り抜き】」, URL : <https://www.youtube.com/watch?v=cizpHPP7Hug> (アクセス日: 2024).
- (2024b) 「ノータ임で を切れたらウザク式卒業！？最終形の強さを求める問題！【神域リーグ2024/渋谷難波切り抜き】」, URL : <https://www.youtube.com/watch?v=Pqnd1iJK3Mc> (アクセス日: 2024).
- 石橋伸洋 (2015) 『黒いデジタル麻雀～現代流データ戦術を斬る～』, マイナビ出版.
- 千羽黒乃 (2020) 「字牌？端牌？第一打は何を切るのじゃ？VTuber 千羽黒乃の麻雀講座【第25回】」, URL : <https://kinmaweb.jp/archives/123234> (アクセス日: 2024).

- 川村晃裕 (2017) 『麻雀勝ち組の鳴きテクニック』, 竹書房.
- 多井隆晴 (2021) 『麻雀無敗の手筋』, 竹書房.
- (2023) 「有料級！多井隆晴の天才的手順！悪配牌が跳満 18000 点「見てる人は気持ち悪くなると思うけど... このバランスが大事！」咲乃もこ ランカー計画切り抜き # さっきのもの」, URL : <https://www.youtube.com/watch?v=IqHxJvZpAZM> (アクセス日: 2024).
- 多井隆晴・郡道美玲 (2021) 『初心者から ぐんぐん強くなる 麻雀何切る』, 竹書房.
- 滝沢和典 (2015) 『滝沢和典の麻雀読みの公式』, マイナビ出版.
- 稚児 (2023) 「【麻雀】【雀魂】ヘッドレスは字牌を切る【講座】」, URL : https://www.youtube.com/watch?v=aEvKo_Uhp0U (アクセス日: 2024).
- 土井泰昭 (2014) 『最強 確率論で勝つ麻雀』, マイナビ出版.
- (2021) 『シンプルにして最重要! 麻雀 押し引きを決める手牌価値』, マイナビ麻雀 BOOKS.
- 独歩・かにマジン・しゅかつ (2017a) 『麻雀・序盤の鉄戦略: 3 人の天鳳位が出す究極の結論』, マイナビ出版.
- (2017b) 『麻雀・鉄押しの条件: 3 人の天鳳位が出す究極の結論』, マイナビ出版.
- 美咲 (2020) 「麻雀強者の AI から学ぶ字牌の切り方」, URL : <https://note.com/misaki11223344/n/n40ea439ac1e0> (アクセス日: 2024).
- 福地誠 (2017) 『手作りと押し引きの鉄戦術』, 三才ブックス.
- 平澤元気 (2019) 『鳴きの教科書』, マイナビ出版.
- (2021a) 「【麻雀講座】初心者が勝つために必要なホンイツ狙いの基準を解説」, URL : <https://www.youtube.com/watch?v=ykr15yb8QYM> (アクセス日: 2024).
- (2021b) 『通せる牌が倍増! 麻雀鳴き読みドリル』, マイナビ出版.
- (2023) 「上級者は全員やってるけど当然すぎてあまり説明してくれない食い伸ばしのテクニック」, URL : <https://www.youtube.com/watch?v=MAV2AgoMU-w> (アクセス日: 2024).
- (2024) 「勝てない初級者の方がよく間違えてしまう「亜リャンメン」の扱い方を解説します【麻雀講座】」, URL : <https://www.youtube.com/watch?v=S0bj-Cpilaw> (アクセス日: 2024).
- 鳳東のうさぎさん (2020) 「邪魔ポンを使いこなす」, URL : <https://note.com/houtooooooooon/n/n06b7869e19f3> (アクセス日: 2024).
- 堀内正人 (2015) 『神速の麻雀 堀内システム 51』, 洋泉社.
- 鈴木大介 (2020) 『麻雀強者の流儀』, 竹書房.